Caspit SmartRetail Solution

Specification

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Contents

Document History

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Ran Ullmann	0.2	08/02/2016	• Continued to write spec.
Ran Ullmann	0.3	15/02/2016	• Continued to write spec.
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Ran Ullmann	0.6	29/05/2016	
Taras Epikhin	0.7	30/05/2016	• Add Xml Tags into EMV_Output
Ran Ullmann	0.8	14/06/2016	
Ran Ullmann	0.9	28/06/2016	• ResultCode table
Ran Ullmann	1.0	03/11/2016	• New tags in INT_IN: ○ Addendum2 ○ Addendum1Settl ○ Addendum2Settl ○ Addendum3Settl ○ Addendum4Settl ○ Addendum5Settl ○ Zdata ○ RequestId ○ New usage for tag Addendum1. It will be used as the voucher number (like field X from Ashrait 96) • Changes in tag PanEntryMode • TimeoutInSeconds tag must be implemented by the Pinpad. • Transaction Query command
Ran Ullmann	1.1	23/11/2016	• Continue of Transaction Query description • Parts, which describe the inner working and inner flow of the transaction between the SmartRetail and AshraitEMV, are colored with a light blue shadow. I also added the words "Caspit Internal". Integrators can ignore that "Caspit Internal" parts. Unless they are curious to learn the inner stuff. • Swipe request description • We will use <Xfield> instead of <Addendum1> to hold the voucher number. AshraitEMV will keep the <Xfield> value in one of the ISO fields.
Ran	1.2	28/11/2016	• The transaction response XML will include the full text of the receipt in

Ullmann			Tags <ReceiptMerchant> and <ReceiptCustomer>. See response XML for more details. • Retrieve File • Retrieve Deposit Report
Ran Ullmann	1.3	30/11/2016	• More paragraphs marked as "Caspit Internal" • Changed document name to "Caspit SmartRetail Solution" • Making the document more "Integrator" friendly. • Generic command 006 – Call Shva without sending transactions. • Appendix A – Result Codes • Appendix B – PTL Header • <UnsentTransactions> tag in Pinpad Configuration command response. • Appendix C – Table of commands • Appendix D – Glossary • <Session-Number> and <File-Number> in Transmit Transactions response
Ran Ullmann	1.4	08/12/2016	• Example for <ReceiptMerchant> and <ReceiptCustomer> • Example for <DepositReport> • <EnableFullPAN> in Pinpad Configuration command • Chapter 9.2 – XML examples of transactions. Use the examples to learn how to do various transaction types.
Ran Ullmann	1.5	15/12/2016	• <CardInside> tag in Communication Test response
Ran Ullmann	1.6	19/12/2016	• <CaspitInternalError> tag in transaction response. • More examples in chapter 9.2 • Appendix E – SmartRetail Tester • <CardInside> tag also in Pinpad Configuration response • Appendix F – Caspit Payment Windows Service
Ran Ullmann	1.7	25/12/2016	• Appendix G – Idle Swipe • Appendix H – Shva <Status> codes • Appendix I – Shva <AshStatus> codes • New tag <AllowCardInside> in transaction request
Ran Ullmann	1.8	02/01/2017	• Appendix J – Pinpad steps table • Transaction Query – LAST
Ran Ullmann	1.9	05/01/2017	• I started adding some "Switch Support" notes.
Ran Ullmann	1.10	08/01/2017	• I expanded the Retrieve TRAN command description. • <Track2> tag in transaction request
Ran Ullmann	1.11	08/01/2017	• Attribute "name" in <Line> tag in receipt
Ran Ullmann	1.12	09/01/2017	• I added more examples in chapter 9.2 • New tag <AllowCardInsideBefore> in transaction request (it replaces tag <AllowCardInside>) • New tag <AllowCardInsideAfter> in transaction request
Ran Ullmann	1.13	18/01/2017	• Appendix L – Beginner's Guide
Ran Ullmann	1.14	19/01/2017	• Appendix M – Troubleshooting • I expanded the example about J2 (see chapter 9.2.11)
Ran Ullmann	1.15	24/01/2017	• New Appendix N – FAQ
Ran	1.16	29/01/2017	• More examples in chapter 9.2

Ullmann			
Ran Ullmann	1.17	05/02/2017	• I expanded Appendix J – Transaction Steps
Ran Ullmann	1.18	20/02/2017	• <MustTranSteps> tag in Pinpad Configuration command
Ran Ullmann	1.19	21/02/2017	• <CaspitInternalError> possible values. See Appendix K • Examples for transactions with benefits (הטבות). See chapter 9.2 • Config.xml file in Appendix F • New Appendix O – CaspitPayment.dll
Ran Ullmann	1.20	27/02/2017	• <SwitchEnabled> tag in Pinpad Configuration **response** • Generic Command 012 • Generic Command 013
Ran Ullmann	1.21	12/03/2017	• Generic Command 014 • New steps: 500 – 545
Ran Ullmann	1.22	26/03/2017	• Abort transaction using the Transaction Steps mechanism. See Appendix J.
Ran Ullmann	1.23	17/05/2017	• Minor changes
Ran Ullmann	1.24	15/06/2017	• Changes in UI command • <Field55> in transaction response • Mti 420 • New example related to Cancel J5 transaction • New example related to Complete J5 transaction
Ran Ullmann	1.25	02/10/2017	• Typos
Ran Ullmann	1.26	06/12/2017	• LifeStyle card support • <ExtraInfo> - a new tag in transaction response. This tag will hold the LifeStyle response and more stuff. • <CardHash> - a new tag in transaction response • <SwitchName> - a new tag in Pinpad Configuration response • <SwitchId> - tag in Pinpad Configuration response • <ShowErrorsOnPinpad> - a new tag in transaction request • <PossibleCreditTerms> - tag in transaction response, relevant to J2.
Ran Ullmann	1.27	19/12/2017	• I updated the J2 example in chapter 9.2 to include the <PossibleCreditTerms> • I updated all examples in chapter 9.2 to include the new tags like <ExtraInfo> and <CardHash>.
Ran Ullmann	1.28	26/12/2017	• Typos
Ran Ullmann	1.29	22/01/2018	• I added English translation for parts written in Hebrew (I left the Hebrew text and added English after the Hebrew text) • New tag <SwitchSend> • New tag <SwitchStatus> • Tag <CardHash> will return also in Swipe command and in Idle Swipe
Ran Ullmann	1.30	28/01/2018	• The RRN, in receipt text, has been moved to separate line (chapter 9.4) • Hebrew and English description of result codes (see Appendix A)
Ran Ullmann	1.31	31/01/2018	• New tag <SlipLang> - The language of the slip
Michael	1.32	05/02/2018	• Appendix O was rewritten

Lobak Ran Ullmann			• <Xfield> was missing in the response. I added it.
Ran Ullmann	1.33	13/02/2018	• New result code: RETVAL_SWITCH_BATCH_IS_TOO_OLD • tag <BlockAutoSwitchBatchDelete> in Pinpad Configuration command.
Ran Ullmann	1.34	26/02/2018	• New tag <PinpadLock> in Pinpad Configuration command.
Ran Ullmann	1.35	28/05/2018	• I added documentation of <Mti> tag to transaction response. It was missing in the document. • In Pinpad Configuration command, I updated the correct default value of <BlockDoubleTrans>. The correct default value in a new Pinpad is – 1 (Enabled)
Ran Ullmann	1.36	14/06/2018	• New result codes: RETVAL_AMOUNT_TOO_BIG RETVAL_AMOUNT_TOO_SMALL RETVAL_ERROR_GETTING_TOKEN • New tag <GetToken> in transaction request
Arie Gleizer	1.37	26/06/2018	• Application version format updated • The TimeoutInSeconds tag description updated.
Michael Lobak	1.38	03/07/2018	Add to Appendix J a new chapter "**Enable Transaction Steps vs FileSystemWatcher**"
Michael Lobak	1.38	09/09/2018	Continue J2 values range: "0", "1", **"2"**
Michael Lobak	1.38	09/09/2018	Following **Response** Class fields name changed: from 'EnableTranSteps' to 'EnableTran**s**Steps'; from 'TranStepsAddress' to 'TransStep**s**Address'; from 'MustTranSteps' to 'MustTran**s**Steps'; from 'EnableFullPAN' to 'EnableFullP**an**'.
Michael Lobak	1.39	09/12/2018	Added 3 public `RequestTreatment` **deal** functions: 1) `<summary>Get Signature Server (Cardo) Transaction ID</summary>` `<param name="transactionDetails">DEE302292;…</param>` `<param name="signatureVectorBlockData">` `1023,1023,99;2;…</param>` `<returns> return int status.` `if status > 0 then It is the transaction id. It means everything is OK.` `if status = 0 it is some kind of unknown error and we should try again.` `if status < 0 CollectSignature returned with error </returns>` `int GetSigServerTransactionId(string transactionDetails, string signatureVectorBlockData);` 2) `<summary>This function desirable to run before turning off the ECR application for dispose, releasing all resources associated with the CaspitPayment.DLL in order to quickly and properly start communication with the next ECR activation.</summary>` `void OnEcrClosing();` 3) `<summary>Make Jpeg File from Signature Vector BlockData</summary>` `<param name="signatureVectorBlockData"></param>` `1023,1023,592;101,561,65538,0;98,559,65538,0;100,558,65538,0;100,555,...>`

			<param name="signatureJpjFilename"></param> in case signatureJpjFilename is empty string the output Filename will be C:\Users\USER_NAME\AppData\Roaming\CASPIT\CPS\Signature.jpg <returns> Ok = 0, Error codes: ErrVectorDataIsEmpty = 21002, ErrNoVectorPointsInVectorData = 21003, ErrInvalidHeaderOfVectorData = 21004, ErrMaximumXResolutionOfSignatureIsInvalid = 21005, ErrMaximumYResolutionOfSignatureIsInvalid = 21006, ErrInvalidNumberOfPointsInVectorDataHeader = 21007, ErrInvalidPointInVectorData = 21008, ErrExceptionWhileSavingJpegFile = 21009. </returns> int **MakeJpegFileFromSignatureVectorBlockData**(string signatureJpjFilename, string signatureVectorBlockData);
Arie Gleizer	1.39	25.12.2018	New Caspit TAG 260 PanEntryModeExt added
Arie Gleizer	1.40	4.2.2019	Possible Currency for each credit term added in J2 response
Arie Gleizer	1.41	3.3.2019	1. DCC support added 2. Delayed payment
Arie Gleizer	1.41	26.11.2019	QR Code Display Command added
Arie Gleizer	1.42	27.4.2020	• <ReceiptPrint> Tag added in reponse to CR • <DCCAuthorization> Tag added in PINPad configuration to declare DCC support by CR • <Commission> Tag added in response for DCC transaction • <PosHash> upto 32 chars prefix to be used in SHA256 for MAX cards • <BListTime> Tag added in PINPad configuration • <DCC> Tag added in transaction response if DCC done
Arie Gleizer	1.50	25.5.2021	• <PosHash>upto 32 chars suffix to be used in MD5 for CAL cards • Ashrait Delek Support Added • Connection over Broker added
Arie Gleizer	1.51	8.5.2022	Result code added: RETVAL_FILE_BAD_RECORD
David Guetta	1.52	18.12.2022	Support XML Session-Number value LAST in retrieve Deposit report command
David Guetta	1.52	18.12.2022	Support XML terminal Id value 0000000 in send switch and delete tran file commands to be done for all trminal Id of RAV-SAPAK
David Guetta	1.52	1.2.2023	Command 20, add option to scane QR by camera on supported device
David Guetta	1.52	22. 2.2023	New pinpad configuration command 30 for RAV-SAPAK

David Guetta	1.52	22.2.2023	Pinpad STEP with Http
David Guetta	1.52	22.2.2023	Updating DELEK XML parametrs
David Guetta	1.53	1.6.2023	New XML for VAS
Arie Gleizer	1.53	19.7.2023	Result code list defined for VAS
David Guetta	1.53	10.12.2023	Free text for swipe, command 23

1. Introduction

Caspit SmartRetail Solution is a complete set of advanced technologies that enables the retailer to accept all types of credit cards, including EMV, Contactless and magnetic. The solution consists of a Pinpad device, applications for the Pinpad and a software (API) for the PC.

The Pinpad software is made of a couple of applications and the two most important applications are the Ashrait EMV application and the SmartRetail application. Ashrait EMV application is the core element that does all the payment stuff and implements the full Ashrait EMV protocol and specification. But Ashrait EMV application doesn't know how to "speak" with ECRs.

The SmartRetail application on the Pinpad is the application that will bridge between a PC-based-ECR and the Ashrait EMV application. "Speaking" with the outside world will be the main feature of the SmartRetail application whether the outside world will be the PC ECR, on one side, or the Switch, on the other side.

The SmartRetail application will provide services for two types of solutions:

1. In the first solution, the SmartRetail application will be used by standalone ECRs that require neither Switch nor P2PE (Point to Point Encryption). In that solution, the Pinpad communicates directly with Shva for transaction authorization. This will be similar to the Ashrait 96 Pinpad solution we currently have in Caspit. In this solution the SmartRetail application role is to bridge between the PC ECR and Ashrait EMV application. Ashrait EMV will get the transaction details from SmartRetail and then Ashrait EMV will perform the full transaction and will return the result to SmartRetail.

2. In the second solution, the SmartRetail application will do much more. This solution is intended for the big retail market and this solution includes a dedicated Switch and a P2PE solution (Point-to-Point Encryption). So the SmartRetail application still needs to bridge between the PC ECR and the Ashrait EMV application but it will also need to support the Switch, meaning it will need to encrypt the authorization packet coming from Ashrait EMV, sends it to the Switch, gets an answer from the Switch and sends the answer to Ashrait EMV (and then of course get the final answer from Ashrait and sends it back to the PC ECR. The fine details will be described later). In this solution, the transaction authorization is done using the Switch and not using Shva.

It should be noted that the SmartRetail application will not do the low level communication with the PC ECR. The actual listening and communication with the external ECR will be done by the Router Application. The Router application is another application on the same device and it will get the packet from the ECR and will route it to the SmartRetail application.

The SmartRetail Application will run on all the Telium 2 family of devices and also on Telium 3 (Tetra).

The SmartRetail application will give the following features to the PC ECR:

- Perform all kind of credit card transactions (Contact, MSR, Contactless)
- Perform J2 and J4/J5 transactions
- Retrieve STATIS file (a summary file with all deposits done. Up to 99 deposits)
- Retrieve TOTAL file (like STATIS but only the last deposit)
- Retrieve TRAN file (a detailed file with all transactions in a specific deposit/batch)
- Retrieve JENR file
- Retrieve DATA file
- Retrieve deposit report (a detailed report with all transactions in a deposit/batch)
- Check status of transactions
- Check connection to Shva
- Retrieve Pinpad status and configuration

2. Audience

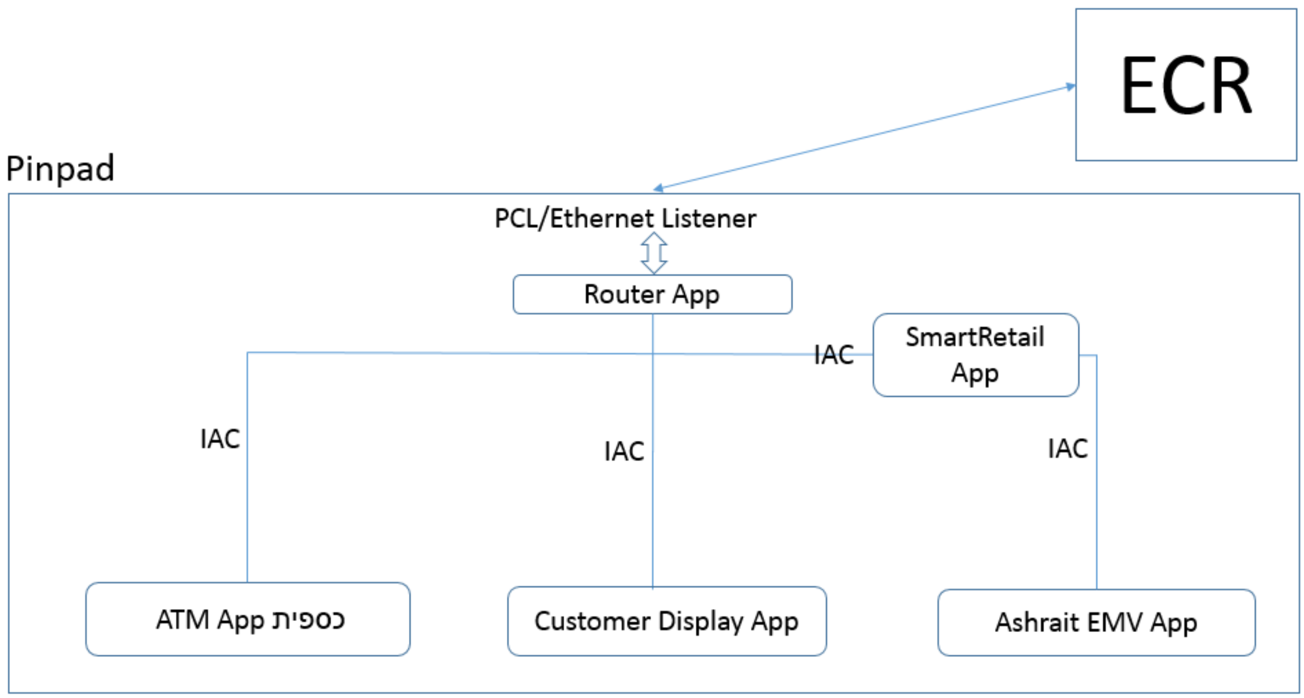
This document is intended for the integrator of the ECR.

However, this document contains also some internal information for Caspit developers. The parts that are intended for the Caspit developer are marked with the words "Caspit Internal" and I also added a blue light shadow (like this paragraph) to those parts. The integrator of the PC ECR can ignore those parts.

3. Topology (Caspit Internal)

The following figure illustrates the internal configuration of a Pinpad which has the Router application, a SmartRetail application, an Ashrait EMV application and other applications as well. As you can see, the internal communication between the Router app and the other apps is made using the IAC protocol (The IAC protocol is a **secure** mechanism for direct communication between Telium applications inside the Pinpad). The Pinpad is connected to a PC ECR (a PC which runs an ECR software) and that PC ECR may send various requests to the Pinpad. For example, if the PC ECR wants to accept a credit card, it will send a request to the Pinpad, a request that will be targeted to the SmartRetail application. The request will arrive to the Router application that will route it to the SmartRetail application. Once the SmartRetail got the request, it will start handling the request as needed. At the end of the transaction, the SmartRetail will return the response to the Router application which in turn will return the response to the PC ECR.

Of course, the SmartRetail application itself is using the help of Ashrait EMV application to perform the actual credit transaction.



The flow of data looks like that:

PC ECR → Router App → SmartRetail App → AshraitEMV → SmartRetail App → Router App → PC ECR

And if you want to go even further, we can include also the EPA (EMV App) and the CPA (Contactless App) to show the flow of a contactless transaction:

PC ECR → Router → SmartRetail → AshraitEMV → EPA → CPA → EPA → AshraitEMV → SmartRetail → Router → PC ECR.

And here is the flow of data using the Switch:

PC ECR → Router App → SmartRetail App → Ashrait EMV → SmartRetail → Switch → SmartRetail → Ashrait EMV → SmartRetail → Router → PC ECR.

4. Telium Summary for the SmartRetail application (Caspit Internal)

Telium Application Type: 51042 (0xC762)
Full binary name: 8510420010 (for version 0.01.0)
Project name in IngeDev: SmartRetail
Name in Telium Manager F menu: Retail V.RR.D (for example Retail 0.01.0)
Telium SDK: Telium Plus with latest version of SDK.
Diagnostics App ID: T23
TID filename: 51042.TID
The SmartRetail application will support the following features:
It will not receive MSR events when in idle.
It will not get keyboard events when in idle.
It will support PCL and/or Ethernet communication.
It will have a menu that will be described later.
It will keep a log file of all packets going through it and of all transactions.
It will support P2PE
It will support the Switch

5. Protocol

The protocol between the PC ECR and the SmartRetail application will be based on XML. To perform a payment transaction we will use almost the same XML that is described in Shva Mefitzim-EMV document with some extra XML tags that will be required by the special features that the SmartRetail application provides.

The ECR should use a simple XML. The following XML features **SHOULD NOT BE USED:**

- Don't use empty elements like <TranType/>, instead use <TranType></TranType>
- Don't use attributes

Before the XML string, the ECR must send the PTL header (see Appendix B), which is a bunch of characters that tells the Pinpad the destination of the XML. But if the ECR is using the CaspitPayment.dll from Caspit then the ECR doesn't need to deal with the PTL header because the CaspitPaymet.dll will take care of it. Only if the ECR is working directly with the Caspit Payment Windows Service, without using the CaspitPayment.dll, then the ECR will have to add the PTL header by itself.

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The SmartRetail will parse the XML and will convert the XML tags into TLV tags that Ashrait EMV understands. Ashrait EMV will handle the transaction and will return a response, in TLV tags, back to SmartRetail. SmartRetail will convert the TLV tags back to XML tags and will return the response back to the PC ECR.

I will mention once more that between the PC ECR and the SmartRetail application there is another application, the Router application, which is making it possible for the PC ECR and the SmartRetail to exchange data. But I will not mention the Router application every time. Just remember that it is there.

I will give two examples of a transaction flow just to give a sense of how it will look.

Here is the flow of a standard transaction between the PC ECR and the Pinpad (**without** Switch and P2PE support)

1. The PC ECR builds an XML string with the transaction details.
2. The PC ECR sends the XML string to the Pinpad (probably with the help of a Dll or with the help of a Windows Service provided by Caspit)
3. The XML string arrives to the Pinpad and is intercepted by the Router application.
4. The Router application routes the XML string (using IAC) to the SmartRetail application.
5. The SmartRetail application parses the XML.
6. The SmartRetail makes some checks on the XML data
7. The SmartRetail converts the XML data to TLV data.
8. The SmartRetail sends the TLV data to Ashrait EMV application (using IAC)
9. Ashrait EMV application handles the transaction and communicates with Shva if needed.
10. Ashrait EMV application returns a TLV response back to SmartRetail application (using IAC)
11. SmartRetail converts the TLV to XML.
12. SmartRetail returns the XML to the Router application (using IAC)
13. The Router application returns the XML response to the PC RCR
14. End of transaction

And here is the flow of a standard **P2PE** transaction **with the Switch**.

1. The PC ECR builds an XML string with the transaction details.
2. The PC ECR sends the XML string to the Pinpad (probably with the help of a Dll or a Windows Service provided by Caspit)
3. The XML string arrives to the Pinpad and is caught by the Router application.
4. The Router application routes the XML string (using IAC) to the SmartRetail application.
5. The SmartRetail makes some checks on the XML data
6. The SmartRetail converts the XML data to TLV data

7. The SmartRetail sends the TLV data to Ashrait EMV application
8. Ashrait EMV starts the transaction and builds the authorization packet (with encrypted track2).
9. Ashrait EMV sends the authorization packet to SmartRetail
10. SmartRetail builds the ISO8583 packet required for the Switch
11. SmartRetail sends the ISO8583 packet to the Switch
12. The Switch returns an answer to SmartRetail.
13. SmartRetail sends the answer back to Ashrait EMV application.
14. Ashrait EMV finishes the transaction and returns a response to SmartRetail
15. SmartRetail converts the response to XML and send it back to the PC ECR (through the Router application).

6. Supported Devices (some examples)

1. iPP350

2. Lane5000/Lane8000

3. Link2500

4. iCMP

7. SmartRetail with a Switch

Some retailers may choose to use a Switch to act as a gateway between the Pinpad and Shva and to aggregate the transactions. Throughout this document, I marked with Switch Support the parts (or the differences) that are relevant to a Switch. Therefore, you can search for "Switch Support" to find all those parts.

The main points of difference between the Switch mode and the normal mode (non-switch mode) are:

- With the Switch, transactions are not kept in the Pinpad. They are kept in the Switch.
- The Switch may return a token in the transaction response.
- Using a Switch, the Transmit Transactions command has a different meaning.

The following commands are irrelevant when using a Switch

- Retrieve TOTAL file
- Retrieve STATIS file
- Retrieve Deposit Report
- Retrieve TRAN File

8. The PC Side

Let's take the PC side for a moment and have a look at what we need to have on the PC in order for this solution to work.

The supported Windows versions are 7 and 10 (both 32bits and 64bits).

It depends on how the Pinpad is connected to the PC.

<u>Using a USB</u>

First, the Ingenico driver must be installed

And then a Windows Service will be installed on Windows (this Windows Service is developed by Caspit).

A Pinpad with USB cable must be connected to the PC.

The Windows Service must be configured to the correct USB port. The Windows Service must run. It listens to TCP connections coming from the ECR software.

Caspit will provide a DLL (CaspitPayment.dll) to the ECR, so the ECR may use this DLL to communicate with the Windows Service.

<u>Using RS232 (serial)</u>

For RS232, it is the same as USB, except you don’t need to install the Ingenico driver.

A Pinpad with RS232 must be connected to the PC.

Note that the ECR may ignore the CaspitPayment.dll and "speaks" directly with the Windows Service. It depends on how the ECR developer prefers to work. If the ECR doesn't use the DLL from Caspit then the ECR must add the PTL header at the beginning of the XML sent to the Windows Service. See Appendix B for explanation about the PTL Header.

<u>Ethernet Connection</u>

The Caspit Windows Service must be installed.

The ECR software will use a DLL to send and receive the XML to and from the Pinpad.

However, the ECR may ignore the DLL and speaks directly with the Windows Service. It depends on how the ECR developer prefers to work. If the ECR doesn't use the DLL from Caspit then the ECR must add the PTL Header at the beginning of the XML sent to the Windows Service. See Appendix B for explanation about the PTL Header.

9. Performing Transactions

The most used feature of the Pinpad is obviously to perform credit card transactions. Every transaction always starts with the ECR and if the selected payment method is a credit card then the ECR will turn to the Pinpad to perform the credit card transaction. The ECR can use the Pinpad to do as many credit card transactions as needed during the day.

At the end of each day, the ECR usually does an end-of-day procedure, as any ECR should do. One of the steps that the ECR must do during the end-of-day procedure is to command the Pinpad to transmit its transactions to Shva. See chapter 15 for the command to transmit transaction to Shva.

This chapter will explain about performing transactions.

See chapter 9.2 for XML examples for various types of transactions. You can learn from these examples.

9.1 The Request XML (INT_IN)

The following XML shows all the possible XML tags that can exist in a transaction request XML. Note that in real life there is no transaction that actually uses all these tags at once and most of the transactions will use only a few tags. We will see a few examples below.

To actually send the XML to the Pinpad, the PC must be prepared with the installation of a Windows Service developed by Caspit (unless the ECR is android based, which requires a different method). And the XML must be prefixed by a PTL header (see Appendix B) and then the XML string will be sent to the Windows Service by TCP socket. The Windows service is responsible to send the XML to the Pinpad and to get the response back from the Pinpad.

```
<Request>
<Command>001</Command>
<RequestId>29482980983</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>30</TimeoutInSeconds>
<SapakMutavNo></SapakMutavNo>
<Mti>100</Mti>
<Eci></Eci>
<CavvUcaf></CavvUcaf>
<Xid></Xid>
<CapDpa></CapDpa>
<CashbackAmount></CashbackAmount>
<Tip></Tip>
<IpayCode>-1</IpayCode>
<IpayAmount></IpayAmount>
<IpayNumber></IpayNumber>
<IpayPercent></IpayPercent>
<OfferCode></OfferCode>
<GiftCode></GiftCode>
<ProductCode></ProductCode>
<ConversionAmount></ConversionAmount>
<ConversionRate></ConversionRate>
<ConversionCurrency></ConversionCurrency>
<GasolineType>0</GasolineType>
<GasolineLiter></GasolineLiter>
<OilLiter></OilLiter>
<OilAmount></OilAmount>
<Speedometer></Speedometer>
<CarNumber></CarNumber>
<ServiceAmount></ServiceAmount>
```

<Zip></Zip>
<Address></Address>
<City></City>
<StndOrdrNo></StndOrdrNo>
<StndOrdrTotalNo></StndOrdrTotalNo>
<StndOrdrTotalSum></StndOrdrTotalSum>
<StndOrdrUniqueRef></StndOrdrUniqueRef>
<Commission></Commission>
<StndOrdrFreq></StndOrdrFreq>
<Cellular></Cellular>
<CardSeqNumber></CardSeqNumber>
<Otp></Otp>
<FirstPayment></FirstPayment>
<NotFirstPayment></NotFirstPayment>
<NoPayments></NoPayments>
<IndexPayment>0</IndexPayment>
<OriginalUid></OriginalUid>
<OriginalTranDate></OriginalTranDate>
<OriginalTranTime></OriginalTranTime>
<OriginalAmount></OriginalAmount>
<OriginalAuthorizedAmount></OriginalAuthorizedAmount>
<OriginalAuthNum></OriginalAuthNum>
<OriginalAuthSolekNum></OriginalAuthSolekNum>
<OriginalAuthorizationCodeManpik>0</OriginalAuthorizationCodeManpik>
<OriginalAuthorizationCodeSolek>0</OriginalAuthorizationCodeSolek>
<OriginalLinkIncrAuth></OriginalLinkIncrAuth>
<CreditTerms>1</CreditTerms>
<TranType>1</TranType>
<AshReasonCredit>0</AshReasonCredit>
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<AuthorizationCodeManpik>0</AuthorizationCodeManpik>
<Addendum1></Addendum1>
<Addendum2></Addendum2>
<AddendumSettl></AddendumSettl>
<Amount>987</Amount>
<AuthorizationNo></AuthorizationNo>
<Currency>376</Currency>
<ClubNumber></ClubNumber>
<ParameterJ>0</ParameterJ>
<Cvv2></Cvv2>
<Id></Id>
<DeferMonths></DeferMonths>
<DueDate></DueDate>
<Rrn></Rrn>
<SpecialProjectCode></SpecialProjectCode>
<SpecialProjectInfo1></SpecialProjectInfo1>
<SelfServiceTrans>0</SelfServiceTrans>
<PanEntryMode>PinPad</PanEntryMode>
<Zdata></Zdata>
<Xfield>209420394</Xfield>
<AllowCardInsideBefore></AllowCardInsideBefore>
<AllowCardInsideAfter></AllowCardInsideAfter>
<Track2></Track2>
<ContinueJ2></ContinueJ2>

<SkipDoubleTranCheck></SkipDoubleTranCheck>
<ShowErrorsOnPinpad></ShowErrorsOnPinpad>
<SwitchSend></SwitchSend>
<GetToken></GetToken>
<PosHash></PosHash>
<Tidluk>1</Tidluk>
</Request>

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
<Request>	The opening tag of the request XML. This tag will exists always.	N/A
Command	פקודה. תגית זו תמיד תופיע. לביצוע עסקה, קוד הפקודה הוא 001. Command. This tag is mandatory. To perform transaction use Command 001.	001
RequestId	מזהה כלשהו של הבקשה. הוא יחזור גם בתשובה. הקופה יכולה להשתמש בו על מנת לוודא שהתשובה מתאימה לבקשה. Request Id. The same Id should return in the response. The ECR should use it to verify that the response belongs to the correct request.	051
TerminalId	מספר מסוף. תגית זו תמיד תופיע. מספר המסוף נבדק בכל שלבי העסקה. הוא ייבדק גם בSmartRetail וגם בAshrait EMV. Terminal Id. Mandatory tag. The terminal id is checked during the transaction.	006
TermNo	מספר קופה. עד 3 ספרות אשר יועברו לקובץ חיובים כחלק של מספר עסקה. כעיקרון משמש לעבודה ברשת על מנת להבטיח ייחודיות של מספר עסקה בקובץ חיובים, אשר נוצר מתחנות עבודה (קופות\עמדות) שונות. Station number. 3 digits. This number is usefull when you have multiple pinpads with same terminal id and you need to distinguish between the devices.	040

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
TimeoutInSeconds	טיימאאוט (בשניות). משך הזמן המירבי שבו המסוף ימתין לתגובה של הקונה. זה יכול להיות בהמתנה לכרטיס (קרבה, חכם או פס מגנטי) או הקלדה (מספר כרטיס או CVV). אם חלף הזמן הזה אז המסוף יחזיר את השליטה לקופה והעסקה תופסק. Timeout in seconds. The maximum timeout that the Pinpad will wait for human activity.	005
SapakMutavNo	מספר רב ספק \ רב מוטב This tag is currently not supported in the request.	086
Mti	קוד המסר: 100 – עסקה רגילה 400 – עסקת ביטול 420 – ביטול בקשה לאישור ללא עסקה (ביטול J5). 100 – Regular transation 400 – Void transaction 420 – Void Pre authorized transaction (void J5)	087
Eci	רמת האבטחה של עסקת אינטרנט 1 – עסקת אינטרנט ללא שדה חתימה 2 – עסקת אינטרנט. לספק יש תוכנה לCAVV/UCAF אבל המנפיק לא רשום לתוכנית 3- עסקת אינטרנט עם שדה חתימה CAVV/UCAF שנבדקה בהצלחה This tag is currently not supported in the request.	088
CavvUcaf	Secure Code של מסטרכרד\ויזה בעסקאות בסחר אלקטרוני בהתאם ליכולת בית העסק לספק את הנתון. This tag is currently not supported in the request.	089
Xid	לכרטיסי ויזה בשיטה 3 This tag is currently not supported in the request.	090
CapDpa	קוד אימות cap/dpa בעסקה טלפונית או עסקת אינטרנט This tag is currently not supported in the request.	091
CashbackAmount	סכום המזומן The cashback amount when performing a cashback transaction	092

Tag name	Description	Matching TLV tag (Caspit Internal)
Tip	סכום התשר This tag is currently not supported in the request.	093
IpayCode	קוד אמצעי התשלום 00 – מטבע 01 – כוכבים 02 – נקודות 03 – מועדון For "Benefits" support. This is payment method 00 – Currency 01 – Stars (miles) 02 – Points 03 - Club	094
IpayAmount	סכום הנחה \ הטענה For "Benefits" support. This is the amount of charge/discount	095
IpayNumber	מספר יחידות For "Benefits" support. This is the number of units	096
IpayPercent	אחוז הנחה בהטבה For "Benefits" support. This is the discount percent.	097
OfferCode	קוד מבצע This tag is currently not supported in the request.	098
GiftCode	קוד מתנה This tag is currently not supported in the request.	099
ProductCode	קוד מוצר This tag is currently not supported in the request.	100
ConversionAmount	סכום לאחר המרה, במאיות, אפסים משמאל, ללא נקודה עשרונית This tag is currently not supported in the request.	101
ConversionRate	שער ההמרה של סכום העסקה This tag is currently not supported in the request.	102
ConversionCurrency	מטבע ההמרה (קוד מטבע ISO שאליו בוצעה ההמרה) This tag is currently not supported in the request.	103

Tag name	Description	Matching TLV tag (Caspit Internal)
GasolineType	סוג דלק 01 – אוקטן 91 02 – אוקטן 96 03 – אוקטן 95 נטול עופרת 04 – סולר 05 – אופנועים 06 – בנזין צבאי 07 – סולר צבאי 08 – נפט 09 – סוגי דלק אחרים 10 – אוקטן 91 נטול עופרת 11 – אוקטן 98 12- אוקטן 98 נטול עופרת 14 – סולר נטול עופרת 36 – גז פחממי 37 – גז אוראה This tag is currently not support in the request.	104
GasolineLiter	כמות הדלק במאיות ליטרים, ללא נקודה עשרונית This tag is currently not supported in the request.	105
OilLiter	כמות שמן בתדלוק במאיות ליטרים, ללא נקודה עשרונית This tag is currently not supported in the request.	106
OilAmount	סכום באגורות בגין שמן, ללא נקודה עשרונית This tag is currently not supported in the request.	107
Speedometer	ספידומטר (מד מרחק) This tag is currently not supported in the request.	108
CarNumber	מספר רכב This tag is currently not supported in the request.	109
ServiceAmount	סה"כ דמי שירות לתדלוק, באגורות, ללא נקודה עשרונית This tag is currently not supported in the request.	110
Zip	מיקוד Zip code	111

Tag name	Description	Matching TLV tag (Caspit Internal)
Address	רחוב ומספר בית This tag is currently not supported in the request.	112
City	עיר This tag is currently not supported in the request.	113
StndOrdrNo	מספר תשלום תורן בעסקה עם הוראת קבע This tag is currently not supported in the request.	114
StndOrdrTotalNo	סה"כ מספר תשלומים בעסקה עם הוראת קבע This tag is currently not supported in the request.	115
StndOrdrTotalSum	סה"כ סכום התשלומים במאיות, אפסים מובילים, ללא נקודה עשרונית בעסקה עם הוראת קבע This tag is currently not supported in the request.	116
StndOrdrUniqueRef	אסמכתא פנימית של ב"ע בעסקה עם הוראת קבע This tag is currently not supported in the request.	117
Commission	עמלת בית עסק במאיות, אפסים מובילים, ללא נקודה עשרונית, בעסקה עם הוראת קבע This tag is currently not supported in the request.	118
StndOrdrFreq	תדירות התשלומים, בעסקה עם הוראת קבע This tag is currently not supported in the request.	119
Cellular	מספר סלולארי This tag is currently not supported in the request.	120
CardSeqNumber	מספר סידורי של הכרטיס, מפריד בין כרטיסים בעלי PAN זהה This tag is currently not supported in the request.	121
Otp	סיסמא חד פעמית בעסקה סלולארית This tag is currently not supported in the request.	122
FirstPayment	סכום תשלום ראשון במאיות, אפסים מובילים, ללא נקודה עשרונית For installments support. This is the amount of the first installment.	123
NotFirstPayment	סכום תשלום קבוע בעסקת תשלומים, במאיות, אפסים מובילים, ללא נקודה עשרונית. For installments support. This is the amount of each of the rest installments.	124

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
IndexPayment	הצמדה לדולר\מדד 1 – צמוד למדד 2 – צמוד לדולר 0 – No index 1 – Madad index 2 – Dollar index	125
NoPayments	מספר תשלומים בעסקאות תשלומים ללא תשלום ראשון (ערכים: 0-999) For installments support. This is number of installments, not including the first installment.	126
OriginalUid	מספר מזהה עסקת מקור The Original Uid. To be used when making a void transaction (cancellation).	127
OriginalTranDate	תאריך עסקת מקור This tag is currently not supported in the request.	128
OriginalTranTime	זמן עסקת המקור This tag is currently not supported in the request.	129
OriginalAmount	סכום עסקת המקור The Original Amount. To be used when cancelling a J5 authorization.	130
OriginalAuthorizedAmount	סכום מאושר בתשובה לבקשה לאישור This tag is currently not supported in the request.	131
OriginalAuthNum	מספר אישור מנפיק של עסקת מקור This tag is currently not supported in the request.	132
OriginalAuthSolekNum	מספר אישור סולק של עסקת מקור This tag is currently not supported in the request.	133

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
OriginalAuthorizationCodeManpik	מאשר עסקת מקור מנפיק 0 – עסקה ללא מספר אישור מנפיק 1 – אושר ע"י המנפיק 2 – דחייה ע"י המנפיק 3 – אושר ע"י שב"א בשם המנפיק 4 – דחייה ע"י שב"א בשם המנפיק 5 – אושר ע" המענה הקולי 6 – דחייה ע"י המענה הקולי 7 – אישור offline ע"י כרטיס חכם 8 – אושר ע"י כרטיס חכם – לא ניתן לבצע התקשרות This tag is currently not supported in the request.	134
OriginalAuthorizationCodeSolek	מאשר עסקת מקור סולק 0 – עסקה ללא מספר אישור סולק 1 – אישור ע"י הסולק 2 – דחייה ע"י הסולק 3 – אישור ע"י שבא בשירות STIP לסולק 4 – דחייה ע"י שבא בשירות STIP לסולק 5 – אושר ע"י המענה הקולי 6 – דחייה ע"י המענה הקולי 7 – אושר ע"י החברה הסולקת באמצעות בקשה לאישור ללא עסקה This tag is currently not supported in the request.	135
OriginalLinkIncrAuth	קישור לבקשת מקור באישור מצטבר This tag is currently not supported in the request.	136

Tag name	Description	Matching TLV tag (Caspit Internal)
PanEntryMode	התג הזה מודיע לפינפד איך רוצים לקבל את נתוני הכרטיס. הערך של התג הזה תלוי אם העסקה היא: • עסקת J2 • עסקת J4 רגילה • עסקת J4 לאחר J2 <u>עסקת J2 או עסקת J4 רגילה</u> אם מדובר על עסקת J2 **או** על עסקת J4 רגילה (כלומר **לא** בוצע J2 לפני הJ4) אז קיימות 3 אפשרויות: "PinPad" - מגנטי, חכם או קירבה. הביצוע בפועל של העסקה הוא זה שיקבע את צורת הקלט. זאת ברירת המחדל. 50 – עסקה טלפונית. הפינפד יבקש לקבל את פרטי הכרטיס ויהיה צורך להקליד את מספר כרטיס האשראי ואת התוקף של הכרטיס בעזרת מקשי הפינפד. 51 – עסקת חתימה בלבד. הפינפד יבקש לקבל את פרטי הכרטיס ויהיה צורך להקליד את מספר כרטיס האשראי ואת התוקף של הכרטיס בעזרת מקשי הפינפד. (זה דומה לעסקה טלפונית אבל בעסקת חתימה הכרטיס צריך להיות נוכח). **כל ערך אחר, או אם התג לא קיים, אז זה כאילו כתוב PinPad** שימו לב שבתשובה לעסקה, לתג זה יש יותר ערכים אפשריים, מאחר שבתשובתו הפינפד מדווח מה בפועל היתה צורת הביצוע. אם מבקשים עסקה טלפונית (50) או עסקת חתימה (51) אז בתשובה יחזור 50 או 51, בהתאמה, כי הפינפד לא ישנה את אופן הביצוע עבור עסקה טלפונית ועסקת חתימה. אבל אם מבקשים "PinPad" אז יחזור ערך אחר. למשל, אם בבקשת העסקה ביקשנו "Pinpad" והעסקה בוצעה עם כרטיס חכם, אז בתשובה בשדה זה הערך 40 <u>עסקת השלמה: J4 לאחר J2</u> אם מדובר בעסקה שהיא עסקה השלמה לאחר ביצוע J2 אז השדה הזה חייב להכיל את הערך שחזר בתשובה של J2, אחרת הפינפד ידחה את ביצוע עסקת ההשלמה. This tag tells the Pinpad how we want to accept the payment. The value of this tag depends on the type of transaction: • A J2 transaction • A regular J4 transaction • A J4 after J2 <u>A J2 or regular J4</u> If we are doing a J2 or a regular J4 transaction (i.e. there was not J2 before the J4) then there are 3 possible values for the Pan Entry Mode: "**Pinpad**" – this means that the Pinpad is responsible to the payment method. All three payment methods are possible: MSR, Chip and Contactless. The "**Pinpad"** is the default value for this tag. **50 –** Card not present transaction (CNP). The Pinpad will ask the cardholder to enter the PAN (card number) and the expiration date of the card. **51** – Signature transaction. Card is present in the transaction but the transaction is done by entering the PAN (card number) and the expiration date of the card. Pay attention, that in the transaction response, this tag has move possible values, because in the response, the Pinpad tells us what was the actual payment method that was used. For example, if the PanEntryMode in the request was "**Pinpad**", and the cardholder used the chip, then the PanEntryMode in the response will be "**40**". <u>A completion transaction: J4 after J2</u> If the transaction is a completion transaction after doing J2 then this tag must contain the value that returned in the J2 response.	138

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
CreditTerms	סוג אשראי 1 – אשראי רגיל 2 – ישראקרדיט\אמקסקרדיט\עדיף\30+ 3 – חיוב מיידי 6 – קרדיט 8 – תשלומים Credit Terms **1** - Regular **2** – Special: IsraCredit, AmexCredit, Adif **3** – Immediate **6** – Credit **8** - Installments	139

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
TranType	סוג עסקה 01 – עסקת חיוב 02 – עסקת פריקה 03 – עסקת חיוב מאולצת 06 – עסקת cashback 07 – עסקת מזומן 11 – עסקה עם הוראת קבע 30 – בירור יתרה 53 – עסקת זיכוי 55 – עסקת טעינה 99 – קבלת מידע Tran Type 01 – Regular 02 – Discharge 03 – Forced 06 – Cashback 07 – Cash 11 – Recurring 30 – Check balance 53 – Refund 55 – Topup (charge) 99 – Get info	140
AshReasonCredit	סיבה לעסקת זכות 41 – זיכוי מיוחד 42 – זיכוי עסקת מימון 43 – זיכוי בגין העברה מחשבון לחשבון This tag is currently not supported in the request.	141

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
AuthorizationCodeSolek	מקור אישור סולק 0 – עסקה ללא מספר אישור סולק 1- אישור ע"י הסולק 2 – דחייה ע"י הסולק 3 – אישור ע"י שבא בשירות STIP לסולק 4- דחייה ע"י שבא בשירות STIP לסולק 5 – אושר ע"י המענה הקולי 6 – דחייה ע"י מענה קולי 7 – אושר ע"י החברה הסולקת באמצעות בקשה לאישור ללא עסקה This tag is currently not supported in the request.	142
AuthorizationCodeManpik	מקור אישור מנפיק 0 – עסקה ללא מספר אישור מנפיק 1 – אושר ע"י המנפיק 2 – דחייה ע"י המנפיק 3 – אושר ע"י שב"א בשירות STIP 4 – דחייה ע"י שב"א בשירות STIP 5 – אושר ע"י המענה הקולי 6 – דחייה ע"י המענה הקולי 7 – אישור אופליין ע"י כרטיס חכם Authorization code from Issuer 0 – Transaction without authorization number 1 – Authorized by issuer 2 – Declined by issuer 3 – Authorized by Shva using STIP 4 – Declined by Shva using STIP 5 – Authorized by voice 6 – Declined by voice 7 – Offline authorization using chip	143
Addendum1	נתונים מיוחדים. שדה להוספת טקסט חופשי Special data	144

Tag name	Description	Matching TLV tag (Caspit Internal)
Addendum2	נתונים מיוחדים. שדה להוספת טקסט חופשי Special data	204
AddendumSettl	טקסט חופשי מהמסוף הנרשם בעסקה (נתוני צי רכב ותדלוק/רכישות שהתבצעו/נסיעות/תחום הלינה/השכרת רכב/בידור/בילוי/נתוני עסקה שיסוכמו עם בית העסק/משלוחים/פוליסת ביטוח) אורך תוכן התג יכול להיות מקסימום 1180 תווים. (בהתאם לאורך התוכן, התוכן יחולק ל4 שדות ISO. חלוקה זו תבוצע ע"י הפינפד. הקופה תשלח את כל התוכן ביחד). Free text that is saved with the transaction record. The maximum lenth is 1180. (Depending on the value length, the content may be split into 4 separate ISO field. This split is done by the Pinpad. The ECR sends the value in a single part).	205 (ISO field 104) 206 (ISO field 118) 207 (ISO field 122) 208 (ISO field 123)
ExpirationDate	תאריך תוקף הכרטיס במבנה YYMM. חיוב להכניס את השדה בעסקה טלפונית. This tag is currently not supported in the request.	145
Amount	סכום באגורות. אורך השדה משתנה עד 12 פוזיציות. אין לרשום נקודה עשרונית לאחר השלמים. The amount in cents. Without decimal dot. Maximum 12 digits.	146
AuthorizationNo	מספר אישור (אופציונלי) השדה מופיע כאשר בוצעה התקשרות אוטומטית או באמצעות שיחה טלפונית רגילה לחברות האשראי, והתקבל מספר אישור לעסקה. השדה אלפא נומרי (רווח אפשרי). השדה יכיל מספר אישור באורך מירבי של 7 תווים, אותיות, מספרים ותווים מיוחדים. השדה יועבר כשהוא מיושר לימין. Authorization number If the merchant did a voice authorization then this tag will hold the authorization number that was given by the issuer. Maximum 7 characters. Alphanumeric.	147

Tag name	Description	Matching TLV tag (Caspit Internal)
Currency	קוד מטבע. קוד נומרי בהתאם לטבלת קודי מטבע ISO8583. קודים שכיחים: 376 – שקל 840 – דולר 978 – יורו Currency code 376 – NIS (New Israeli Shekel) 840 – USD 978 - EUR	148
ClubNumber	קוד תחום מועדון. מספר מועדון – לעסקאות טלפוניות או חתימה בלבד. לישראכרט: מספר המועדון (באורך 4 פוז') בכרטיס נמצא בערוץ 2. לכאל וללאומיקארד: מספר המועדון מוגדר בפוז' 6 -1 משמאל למספר הכרטיס. This tag is currently not supported in the request.	149

Tag name	Description	Matching TLV tag (Caspit Internal)
ParameterJ	אופן הפעלה – 2 – בדיקה בלבד של נתוני העסקה. הבדיקה תתבצע ברמת הפינפד, ללא יציאה לבקשה לאישור וללא רישום עסקה בקובץ התנועות. נוצר XML תשובה בלבד. במיוחד כשמדובר באפליקציה של קופה, מומלץ לבצע את ההפעלה הראשונה של הממשק עם פרמטר J2 – לצורך קביעת חברת האשראי מנפיקת הכרטיס, כך שניתן יהיה להקרין את סוגי האשראי הקיימים של אותה חברה בלבד, ואז ההפעלה השניה תהיה לצורך חיוב (ללא הפרמטר J). **ראו דוגמא והסבר מפורט בסעיף 9.2.11** 4 – זאת ברירת המחדל. אם התג <ParameterJ> לא נשלח אז הכוונה לערך 4 (גם ערך אפס בParameterJ יטופל כמו 4) 5 – בקשה לאישור ללא עסקה. אופציה זו נועדה לעסקים כגון: חברות להשכרת רכב, בתי מלון וכדומה. בשלב הראשון מבצעים בקשה לאישור לצורך בדיקת תקפות הכרטיס והעסקה מול חברת האשראי ולקבלת מספר אישור (גם במקרים שהתשובה לבקשה הינה חיובית, העסקה לא תרשם בקובץ TRAN). העסקה תבוצע בשלב מאוחר יותר כאשר ידוע סכום העסקה המדוייק. 6 – בקשה לאישור ביוזמת הקמעונאי. זאת אופציה נוספת שבה המשתמש יכול ליזום התקשרות לחברות האשראי לצורך קבלת מספר אישור. במקרה של תשובה חיובית, העסקה תרשם בקובץ הTRAN. **אפשרות זו לא נתמכת עדיין.** 49 – השלמת עסקת J5 Mode of operation **2** – Get card details (J2). This mode just checks the card and doesn't perform a real transaction and doesn't write the transaction in the transactions batch file. After getting the J2 response, the ECR may complete the transaction by sending a J4 request. Refer to chapter 9.2.11 to see a sample and an extended explanation. **4** – Real transaction (J4). This is the real transaction. If successful it will be written in the transactions batch file. **5** – Pre authorization (J5). This mode of operation is used mainly in hotels, card rental and petrol stations. The transaction goes online to get an authorization number but the transaction is not a real transaction and it is not kept in the transactions batch. The real transaction can be completed in future time using the authorization number that has been received. **6** – Not supported yet. **49** – Complete J5	150

Tag name	Description	Matching TLV tag (Caspit Internal)
Cvv2	תוכן שדה Cvv2 This tag is currently not supported in the request.	151
Id	מספר תעודת הזהות, מיושר לימין עם אפסים מובילים This tag is currently not supported in the request.	152
DeferMonths	מספר חודשי דחייה (1-12) This tag is currently not supported in the request.	153
DueDate	מועד חיוב הלקוח YYMM This tag is currently not supported in the request.	154
Rrn	מזהה עסקה ייחודי שמוקצה ע"י החברה המאשרת את הבקשה לאישור This tag is currently not supported in the request.	155
SpecialProjectCode	מספר פרוייקט מיוחד This tag is currently not supported in the request.	156
SpecialProjectInfo1	נתונים לפרוייקט מיוחד This tag is currently not supported in the request.	157
SelfServiceTrans	ציון לעסקה בשירות עצמי 0 – עסקה שאינה בשירות עצמי 1 – עסקה בשירות עצמי This tag is currently not supported in the request.	158
Xfield	מספר השובר (או מספר החשבונית) של הקופה למקרה שיש רק תשלום אשראי אחד. במקרה של יותר מתשלום אחד, יש להוסיף למספר חשבונית ספרה נוספת שונה לכל תשלום. זהה לשדה X מאשראית 96. זהו שדה חובה. **הקופה חייבת להעביר ערך ייחודי לכל עסקת תשלום**. הפינפד בודק אם הערך שייך לעסקה קודמת, ואם כן אזי יחזיר הודעה של עסקה כפולה/נעשתה. חיפוש עסקאות, ע"י Transaction Query, מתבסס על הערך של Xfield. כך שאם הערך לא יהיה ייחודי, חיפוש עסקאות לא יעבוד תקין. This tag is mandatory. This is the ECR voucher number (or ECR invoice number). The ECR must send a unique value for each transaction. Quering transaction, using the Transaction Query command, is based on the Xfield value.	212 The X value will eventually be put in ISO field 126 B2

Tag name	Description	Matching TLV tag (Caspit Internal)
AllowCardInsideBefore	0 – **Default**. Right before starting a new transaction, the Pinpad will check if there is a card inserted in the smart card reader. If there is a card inserted, the Pinpad will ask the cardholder to remove the card, and only after removal of the card, the Pinpad will start the transaction and will ask the cardholder to insert a card. This is done to prevent the case of starting a transaction with a card that was forgotten in the card reader from previous transaction. 1 – When starting a new transaction, the Pinpad will start the transaction even if there is a card already in the smart card reader. This can be useful in the case of a J2 transaction. After a J2 usually comes J4 and we would like the cardholder to leave the card inside the card reader. We do not want that the cardholder will need to insert his card twice.	216
AllowCardInsideAfter	0 – **Default**. At the end of transaction, **before returning the response XML to the ECR**, the Pinpad will wait until the cardholder remove his card. Only after removal of the card, the Pinpad will return the response XML to the ECR. It means that while the card is inside, the "control" of the transaction is still in the hands of the Pinpad. 1 – At the end of the transaction, the Pinpad will return the response XML to the ECR, not waiting for the removal of the card. It is up to the ECR to check if there is a card still inside and to do something about it. The ECR can check if there is a card inside by using the Communication Test command and then use the UI Command to ask the cardholder to remove the card.	221
Track2	The Pinpad will never send out the full track2 data of a credit card. But the ECR can send **to** the Pinpad a track2 data to be used in the transaction. If this tag <Track2> exists then the Pinpad will not ask to swipe/present/tap card. The Pinpad will use the <Track2> value and will perform a magnetic transaction. **Only magnetic transaction can be done like that.** Of course, if the <Track2> value is invalid or if the service code in the track2 data requires that the smart card will be used, then the Pinpad will reject this transaction.	65

Tag name	Description	Matching TLV tag (Caspit Internal)
ContinueJ2	This tag should be used only after a J2 transaction. Only if the ECR wants to continue J4/J5 transaction that has started with J2. **0** – Default. If ContinueJ2 is 0 or if ContinueJ2 tag does not exist at all, then there is nothing to do. **1** – This transaction is a continuation of a J2 transaction. **2** – This transaction is a continuation of a J2 transaction without Amount checking.	224
SkipDoubleTranCheck	Using the Pinpad Configuration command, you can configure the Pinpad to block double transaction (see chapter 17 and <BlockDoubleTrans> tag). This tag can be used to override that configuration, but only for this specific transaction. The possible values are: **0** – No override will be done. If Pinpad is configured to block double transactions, then the check for a double transaction will be done. If the Pinpad is NOT configured to block double transactions, then no check will be done **1** – The Pinpad will not check for double transactions, even if the Pinpad is configured to check for double transactions.	237
ShowErrorsOnPinpad	Default is 0. 0 – In case of an error during transaction (e.g. "Invalid Credit Card"), the Pinpad will not show the error on the Pinpad screen. Instead the Pinpad will just return the error code in the response to the ECR. 1 – The Pinpad will display the error on the Pinpad screen, for a few seconds, before returning the response to the ECR.	203
SwitchSend	This tag is relevant only when using **Switch** **0** – Default. If this tag is missing then this is the default. Pinpad will **NOT** send the transaction record after performing the transaction. To send the record, the ECR must use Generic command with sub command 012. **1** – The Pinpad will send the transaction record to the Switch immediately after performing the transaction. This will happen before the Pinpad returns the response to the ECR. It is like the Pinpad is calling Generic command 012 after every transaction. If there are some records that haven't been sent to the Switch, they will all be sent to the Switch, not just the current transaction.	254

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
GetToken	**<u>Switch Support</u>** The <GetToken> tag goes together with <ParameterJ>**2**</ParameterJ>. Usually, a J2 command just checks the card in offline mode. But using <GetToken> the ECR can ask the Pinpad to get a token, from the Switch, even when doing J2. The <GetToken> tag will have no effect if used with a <ParameterJ> different than **2**. 0 – Default. The Pinpad will NOT ask for token. The J2 will be the regular, offine mode, J2. 1 – The Pinpad will go online during J2 command to get a token. The token will return, as usuall, in tag <AddendumSettl>. <u>Notes</u> The first thing the Pinpad will always do is the standard J2 flow, no matter what is the value of <GetToken>. A <GetToken>1</GetToken> just adds a token request at the end of the standard J2. Because we added another step to the J2, the ECR should carefully check the result code of the J2. It is possible that the J2 completed successfully, except for the token request. So, the ECR will have to decide if it wants to continue the transaction (J4/J5) without token or if it wants to abort the transaction and try again the J2. Relevant result codes in response: RETVAL_ERROR_GETTING_TOKEN - It means the J2 completed successfully but the token request failed. Probably because of communication error. <ResultCode> **>=** 900000. This is a Switch error code. It means the J2 completed successfully and the token request has arrived to the Switch, but the Switch returned an error.	258

Tag name	Description	Matching TLV tag (Caspit Internal)
PanEntryModeExt	התג הזה מודיע לפינפד איך רוצים לקבל את נתוני הכרטיס , כאשר כל שיטה מיוצגת ע"י ביט אחד מסויים: **00001** – עסקה טלפונית. הפינפד יבקש לקבל את פרטי הכרטיס ויהיה צורך להקליד את מספר כרטיס האשראי ואת התוקף של הכרטיס בעזרת מקשי הפינפד. **00010** – עסקת חתימה בלבד. הפינפד יבקש לקבל את פרטי הכרטיס ויהיה צורך להקליד את מספר כרטיס האשראי ואת התוקף של הכרטיס בעזרת מקשי הפינפד. (זה דומה לעסקה טלפונית אבל בעסקת חתימה הכרטיס צריך להיות נוכח). **00100** – עסקת פס מגנטי בלבד. **01000** – עסקת כרטיס חכם (צ'יפ) בלבד. **10000** – עסקת כרטיס מגע בלבד. הערך של התג הזה תלוי אם העסקה היא: • עסקת J2 • עסקת J4 רגילה • עסקת J4 לאחר J2 <u>עסקת J2 או עסקת J4 רגילה</u> אם מדובר על עסקת J2 או על עסקת J4 רגילה (כלומר לא בוצע J2 לפני הJ4) אז קיימות 3 אפשרויות: 11100 - מגנטי, חכם או קירבה. הביצוע בפועל של העסקה הוא זה שיקבע את צורת הקלט. זאת ברירת המחדל. 00001 – עסקה טלפונית 00010 – עסקת חתימה בלבד. **כל ערך אחר, או אם התג לא קיים, אז זה כאילו כתוב** 11100 שימו לב שבתשובתו הפינפד מדווח מה בפועל היתה צורת הביצוע. <u>עסקת השלמה: J4 לאחר J2</u> אם מדובר בעסקה שהיא עסקה השלמה לאחר ביצוע J2 אז השדה הזה חייב להכיל את הערך שחזר בתשובה של J2, אחרת הפינפד ידחה את ביצוע עסקת ההשלמה. This tag tells the Pinpad how we want to accept the payment. Each type of transaction is represented by one bit only: **00001 –** Card not present transaction (CNP). The Pinpad will ask the cardholder to enter the PAN (card number) and the expiration date of the card. **00010** – Signature transaction. Card is present in the transaction but the transaction is done by entering the PAN (card number) and the expiration date of the card. **00100** – Magnetic swipe transaction. **01000** – Smart Card transaction (Chip). **10000** – Smart Card transaction (Contactless). The value of this tag depends on the type of transaction: • A J2 transaction • A regular J4 transaction • A J4 after J2 <u>A J2 or regular J4</u> If we are doing a J2 or a regular J4 transaction (i.e. there was not J2 before the J4) then there are 3 possible values for the Pan Entry Mode: 11100 – this means that the Pinpad is responsible to the payment method. All three payment methods are possible: MSR, Chip and Contactless. This is the default value for this tag. 00001 – Card not present transaction (CNP). 00010 – Signature transaction. Pay attention, that in the transaction response, the Pinpad tells us what was the actual payment method that was used. <u>A completion transaction: J4 after J2</u> If the transaction is a completion transaction after doing J2 then this tag must contain the value that returned in the J2 response.	260
PosHash	<PosHash> upto 32 chars prefix to be used in SHA256 for MAX cards, or upto 32 chars suffix to be used in MD5 for CAL cards	
Tidluk	Parameter that indicates fuelling transaction	

Tag name	Description	Matching TLV tag (Caspit Internal)
</Request>		N/A

9.2 XML Examples of transactions

You should use the SmartRetail tester and the following examples to learn how to perform various transactions.

1. Regular Transaction. The most basic transaction.

Amount is 100 NIS (10000 Agorot).

The request XML (INT_IN):

```
<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>1</TranType>
<Amount>10000</Amount>  (Amount is in agorot)
<Currency>376</Currency> (Currency is in ISO currency code)
<PanEntryMode>PinPad</PanEntryMode>
<Xfield>20398049823</Xfield>
</Request>
```

A successful response will look similar to that:

```
<EMV_Output>
<Command>1</Command>
<TerminalId>0880264</TerminalId>
<ResultCode>0</ResultCode>
<Status>0</Status>
<AshStatus>0</AshStatus>
<Mti>100</Mti>
<Pan>0004580289999999709</Pan>
<PanEntryMode>40</PanEntryMode>
<CardName>ויזה זהב     </CardName>
<Manpik>2</Manpik>
<Brand>2</Brand>
<Solek>1</Solek>
<CardType>001</CardType>
```

<spType>0</spType>
<Amount>10000</Amount>
<TranType>1</TranType>
<CreditTerms>1</CreditTerms>
<Currency>376</Currency>
<FileNo>02</FileNo>
<TermNo>001</TermNo>
<TermSeq>003</TermSeq>
<TerminalName>EMV TEST CASPIT</TerminalName>
<Retailer>0880264012</Retailer>
<ComRetailerNum>0071506</ComRetailerNum>
<DateTime>0109130753</DateTime>
<ResponseId>0</ResponseId>
<ResponseCvv2>0</ResponseCvv2>
<ResponseAvs>0</ResponseAvs>
<AuthCodeManpik>0</AuthCodeManpik>
<AuthCodeSolek>0</AuthCodeSolek>
<Uid>17010913075308802640011</Uid>
<parameterJ>0</parameterJ>
<appVersion>8510330084</appVersion>
<CardSeqNumber>01</CardSeqNumber>
<AID>A0000000031010</AID>
<TSI>E800</TSI>
<VerifiedByPIN>1</VerifiedByPIN>
<ATC>001B</ATC>
<TVR>0080000000</TVR>
<ExtraInfo></ExtraInfo>
<CardHash></CardHash>
<RequestId>20170109130703</RequestId>

Here will come the text of the receipts. I do not show it here in this example. See chapter 9.4 for an example of the receipt text.

</EMV_Output>

If all you do is a simple regular transaction then your XML will always be like that, and you will need to change only the <Amount>.

<Request>
<Command>001</Command>

<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>

<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>1</TranType>
<Amount> </Amount>
<Currency>376</Currency>
<PanEntryMode>PinPad</PanEntryMode>

<Xfield>20398049823</Xfield>

</Request>

2. Payments Transaction. 100 NIS in 4 payments.

<Request>
<Command>001</Command>

<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>

<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>8</CreditTerms>
<TranType>1</TranType>
<Amount>10000</Amount>
<Currency>376</Currency>
<PanEntryMode>PinPad</PanEntryMode>

<NoPayments>3</NoPayments> (this number doesn't include the first payment)

<FirstPayment>2500</FirstPayment>

<NotFirstPayment>2500</NotFirstPayment>

<Xfield>3409583598TTTT</Xfield>

</Request>

3. Cashback Transaction (עסקה שבתמורתה מחויב בעל הכרטיס באשראי ומקבל סכום נוסף במזומן)

Here we just add the tag <CashbackAmount>

<Request>
<Command>001</Command>

<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>

<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>06</TranType>
<Amount>10000</Amount>

<Currency>376</Currency>
<PanEntryMode>PinPad</PanEntryMode>

<CashbackAmount>5000</CashbackAmount>

<Xfield>GGG23942304938</Xfield>

</Request>

4. Cash Transaction (עסקת חיוב המתבצעת בכרטיס ישראלי או תייר ובתמורתה מקבל בעל הכרטיס מזומן בלבד)

Note the difference between Cash and Cashback.

<Request>
<Command>001</Command>

<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>

<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>07</TranType>
<Amount>10000</Amount>
<Currency>376</Currency>
<PanEntryMode>PinPad</PanEntryMode>

<Xfield>GGG23942304938</Xfield>

</Request>

5. Cancel Transaction, aka void transaction (עסקה המבטלת עסקה שנתוניה קיימים בקובץ התנועות במסוף)

For each transaction, the Pinpad keeps the Uid of the transaction. That Uid is used to identify the transaction we wish to cancel.

The ECR, on its side, must keep the Uid of the original transaction and to use it when the merchant would like to cancel the transaction.

It is possible to cancel transactions only if they were not transmitted to Shva.

It is important to understand the difference between a "Cancel" transaction and a "Refund" transaction. A "Refund" transaction (sometimes called a "credit" transaction) is a new transaction that is not related to the original transaction. If you did a regular debit transaction and then you made a refund transaction to the same card number and the same transaction details then at the end, the terminal will have two transactions: the first is the regular debit transaction and the second is the refund transaction. The cardholder will not be charged, though in some cases he will see the two transactions (debit and refund) in his monthly statement. In most cases you can do a refund transaction even if there is no original debit transaction. A refund transaction is NOT limited to the same day of the original transaction; it can be done on another day.

A "Cancel" transaction (also valled a "void" transaction) must have an original transaction that is the "target" of the cancelation. If you did a regular debit transaction and the transaction has not been sent to Shva yet, then you can perform a "Cancel" transaction that will actually cancel the original transaction. The original transaction and the cancelation transaction will both exist in the TRAN file and will both be transmitted to Shva.

In Ashrait EMV (as opposed to Ashrait 96) you can cancel any of the transactions that exist in the TRAN file, not just the last transaction.

<Request>
<Command>001</Command>

<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>

<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>**400**</Mti>
<CreditTerms>1</CreditTerms> (must be the same as in the original transaction)
<TranType>01</TranType> (must be the same as in the original transaction)
<Amount>10000</Amount> (must be the same as in the original transaction)
<Currency>376</Currency> (must be the same as in the original transaction)

<OriginalUid>23929384738828374</OriginalUid> (must be the same Uid from the original transaction).

<Xfield>GGG23942304938</Xfield> (must be a new value of Xfield. **Not** the original Xfield from the original transaction)

</Request>

The Pinpad will search the <OriginalUid> in the transaction batch. If found, the Pinpad will compare also the following tags: <CreditTerms>, <TranType>, <Amount>, <Currency>.

If the <OriginalUid> wasn't found in the transaction batch or if there is no perfect match between the original transaction and the XML values then the cancellation cannot continue. In that case AshStatus will be 443.

Note that if the original transaction was made by online authorization then the cancelation will also require an online authorization. And if there is a communication problem then the cancellation cannot be done.

The response will include the tags: <OriginalTranDate>, <OriginalTranTime>, <OriginalAmount>

And depending on the entity that authorized the original transaction, issuer (manpik) or acquirer (solek):

<OriginalAuthNum>, <OriginalAuthorizationCodeManpik>

Or

<OriginalAuthSolekNum>, <OriginalAuthorizationCodeSolek>

6. Authorized transaction, using an authorization number (עסקה עם מספר אישור)

If a transaction is declined with AshStatus 003, it is possible to call the credit company to get an authorization number. The merchant is supposed to call to the credit company support center to get an authorization number for the transaction.

After getting an authorization number, the ECR can redo the transaction with an authorization number.

In this example you can see that we get authorization number 1234567, which we put in tag <AuthorizationNo>.

You also need to send tag <AuthorizationCodeManpik> with value 5.

```
<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>01</TranType>
<Amount>10000</Amount>
<Currency>376</Currency>
<PanEntryMode>PinPad</PanEntryMode>
<Xfield>GGG23942304938</Xfield>
<AuthorizationCodeManpik>5</AuthorizationCodeManpik>
<AuthorizationNo>1234567</AuthorizationNo>
</Request>
```

7. Regular Refund transaction (עסקת זיכוי\זכות)

Tran Type is 53

```
<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>53</TranType>
<Amount>10000</Amount>
<Currency>376</Currency>
<PanEntryMode>PinPad</PanEntryMode>
<Xfield>GGG23942304938</Xfield>
</Request>
```

8. Get Balance (בירור יתרה)

Amount must be 100.

TranType is 30

```
<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>30</TranType>
<Amount>100</Amount>
<Currency>376</Currency>
<PanEntryMode>PinPad</PanEntryMode>
<Xfield>GGG23942304938</Xfield>
</Request>
```

The response will include the tag <DspBalance> and <AddDspBalance>

9. Forced transaction (עסקה מאולצת)

In a forced transaction, the merchant takes responsibility for the transaction. The Pinpad will not go online for authorization.

```
<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>03</TranType>
<Amount>10000</Amount>
<Currency>376</Currency>
<PanEntryMode>PinPad</PanEntryMode>
<Xfield>GGG23942304938</Xfield>
</Request>
```

10. Topup transaction (עסקת טעינה)

A topup transaction is loading funds into a giftcard.
Tran Type should be 55.

```
<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>55</TranType>
<Amount>10000</Amount>
<Currency>376</Currency>
<PanEntryMode>PinPad</PanEntryMode>
<Xfield>GGG23942304938</Xfield>
</Request>
```

Beside the regular tags in the response, you might get two other tags: <DspBalance> and <AddDspBalance>.
See the response XML tags table to learn more about these tags.
Also, in the customer receipt text XML you will two lines with a "Balance" attribute.
Like this:

```
<Line name="Balance">    סיטרכה תרתי םוכס</Line>
<Line name="Balance">           748.88</Line>
```

11. J2 – Get Card Details (פרטי כרטיס)

Some integrators like to perform a J2 before every actual transaction.
J2 is also called Get Card Details, and you do it by adding the tag <ParameterJ>2</ParameterJ> to the payment request XML.

The J2 main features are:

- The Pinpad will not perform a real transaction.
- The cardholder will not be charged.
- Transaction is not saved in the Pinpad
- No communication to Shva is done.

The Pinpad will just read the card details and return the response to the ECR (if a smart card is used, the cardholder will **not** have to enter the PIN code in this stage. The PIN will be asked later, if and when the transaction is completed with J4).
The Pinpad can decline the J2 transaction from various reasons. The regular Ashrait logic is performed during the J2 transaction so any of Ashrait errors may return. If the Pinpad declined the J2 transaction then there is no need to continue with the completion transaction (the J4/J5).
If the Pinpad approved the J2 transaction then the ECR can examine the card details from the response and decide if it wants to proceed with the completion transaction. If yes, the ECR will make the completion transaction with <ParameterJ>4</ParameterJ>.
The completion transaction must be done with the same payment method (PanEntryMode) as the J2 transaction response.

Here is the request for the J2 transaction. This example is for a regular transaction, but any type of transaction and credit terms can be used in the J2 transaction.

<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>1</TranType>
<Amount>10000</Amount> (**Amount is in agorot)**
<Currency>376</Currency> **(Currency is in ISO currency code)**
<PanEntryMode>PinPad</PanEntryMode>
<Xfield>20398049823</Xfield>
<ParameterJ>2</ParameterJ>
<AllowCardInsideAfter>1</AllowCardInsideAfter>
</Request>

Notice the **<ParameterJ>2</ParameterJ>** tag that is used to tell the Pinpad that this is a J2 transaction.

Notice the **<AllowCardInsideAfter>1</AllowCardInsideAfter>** tag. This tag tells the Pinpad to **NOT** ask the cardholder to remove the card after J2 is performed. If the ECR will not send the <AllowCardInsideAfter> tag or if the value of the tag will be zero then the Pinpad will ask the cardholder to remove the card at the end of the J2 transaction. Of course, all this card removal issue is relevant only to J2 that is done using a smart card (chip card).

The response XML of the J2 transaction will be similar to the regular payment transaction response, but without the receipt text parts.

The response will also include the tag <PossibleCreditTerms>. The ECR can parse this tag a display to the cashier the available credit terms to complete the transaction.
The response will also include the tag <PossibleCurrency>. The ECR can parse this tag and display to the cashier the available currency to complete the transaction.

Normally, after J2, the ECR will send the actual transaction. This time with <ParameterJ>4</ParameterJ>.
The continuation of the J2 to the actual transaction varies according to the payment method (pan entry mode).
The ECR should parse the <PanEntryMode> tag that came back in the J2 response and act accordingly.

<u>Continue J2 after using a smart card</u>
The response of the J2 transaction that was done by a smart card (by the chip) will look like that

```
<EMV_Output>
<Command>001</Command>
<AshStatus>0</AshStatus>
<Status>0</Status>
<ResultCode>0</ResultCode>
…
<PanEntryMode>40</PanEntryMode>  (smart card)
<Pan>XXXXXXXX1234</Pan>
<ParameterJ>2</ParameterJ>
<PossibleCreditTerms></PossibleCreditTerms>
<CardHash></CardHash>
<ExtraInfo></ExtraInfo>
……
</EMV_Output>
```

If the J2 was done with a smart card then the actual transaction (the completion of the transaction, the J4) still needs the card inside because during the actual transaction, the PIN (secret code) is checked against the card. The request of the J4 will be like that:

```
<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>1</TranType>
<Amount>10000</Amount>  (Amount is in agorot)
<Currency>376</Currency> (Currency is in ISO currency code)
<PanEntryMode>40</PanEntryMode>
<Xfield>20398049823</Xfield>
<AllowCardInsideBefore>1</AllowCardInsideBefore>
<Pan>XXXXXXXXX1234</Pan>
<ContinueJ2>1</ContinueJ2>
<ParameterJ>4</ParameterJ>
</Request>
```

Notice the **<AllowCardInsideBefore>1</AllowCardInsideBefore>** tag. This tag tells the Pinpad to allow starting the transaction when there is a card inserted. Without this tag, if a card is already inserted, the Pinpad will ask to remove it before starting the transactions. The usage of <AllowCardInsideBefore> tag and <AllowCardInsideAfter> tag gives us the option to perfrom the J2 and then the actual transaction without the need to remove and insert the card between the two phases. Thus, making the payment experience easier for the cardholder.

Notice also the <PanEntryMode>40</PanEntryMode> in the J4 transaction request. Because the J2 was a smart card transaction, the Pinpad returned <PanEntryMode>40</PanEntryMode> in the J2 response XML. If the ECR wants to continue the transaction, it must send also <PanEntryMode>40</PanEntryMode> in the J4 transaction request.

Notice also the <Pan> tag. It must have the same value of the <Pan> tag from the J2 response XML. That way the Pinpad will make sure that the card from the J4 transaction is the same card from the J2 transaction.

Notice also the <ContinueJ2> tag, that tell the Pinpad that this is a continuation of a J2 transaction.

If after the J2, the ECR decides to NOT continue with the transaction, and if the card is still inside the Pinpad then the ECR can check if the card is still inside using the Pinpad Configuration command and then use the UI command to ask the cardholder to remove the card.

The continuation of the J2 is valid only for the last J2. You cannot do other transactions with the Pinpad after doing J2 and then decide to do the J4 transaction. The J4 transaction must come directly after the J2.

If the Pinpad can not continue the J2 transaction, the Pinpad will return with Result code 10028 (RETVAL_CANT_CONTINUE_J2)

<u>Continue J2 after using a magnetic card</u>

The response of the J2 transaction will look like that:

<EMV_Output>
<Command>001</Command>
<AshStatus>0</AshStatus>
<Status>0</Status>
<ResultCode>0</ResultCode>
…
<PanEntryMode>00</PanEntryMode> **(Magnetic card)**
<Pan>XXXXXXXX1234</Pan>
<ParameterJ>2</ParameterJ>
<PossibleCreditTerms></PossibleCreditTerms>
<CardHash></CardHash>
<ExtraInfo></ExtraInfo>
……
</EMV_Output>

If the J2 was done with a magnetic card, we do not want to ask the cardholder to swipe again his credit card, so the actual transaction request will be like that:

<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>1</TranType>
<Amount>10000</Amount> **(Amount is in agorot)**

<Currency>376</Currency> **(Currency is in ISO currency code)**
<PanEntryMode>00</PanEntryMode> (Magnetic card)
<Pan>XXXXXXXXX1234</Pan>
<ContinueJ2>1</ContinueJ2>
<ParameterJ>4</ParameterJ>
<Xfield>20398049823</Xfield>
</Request>

Notice the <PanEntryMode>00</PanEntryMode> tag which tells the Pinpad to do the J4 transaction using a magnetic card. Notice also the <Pan> tag which must be with the same value of the <Pan> tag that the ECR got from the J2 response XML. That way the Pinpad will know that the J4 transaction must be done with the same magnetic stripe data that was swiped during the J2 phase. This will prevent the cardholder from swiping his credit card twice.

Notice also the <ContinueJ2> tag that tells the Pinpad that this is a continuation of a J2 transaction.

The continuation of the J2 is valid only for the last J2. You cannot do other transactions with the Pinpad after the J2 and then decide to do the actual transaction. The J4 transaction must come directly after the J2.

<u>Continue J2 after using a contactless card</u>
The response of the J2 transaction will look like that:
<EMV_Output>
<Command>001</Command>
<AshStatus>0</AshStatus>
<Status>0</Status>
<ResultCode>0</ResultCode>
…
<PanEntryMode>05</PanEntryMode> **(Contactless)**
<Pan>XXXXXXXX1234</Pan>
<ParameterJ>2</ParameterJ>
<PossibleCreditTerms></PossibleCreditTerms>
<CardHash></CardHash>
<ExtraInfo></ExtraInfo>
……
</EMV_Output>

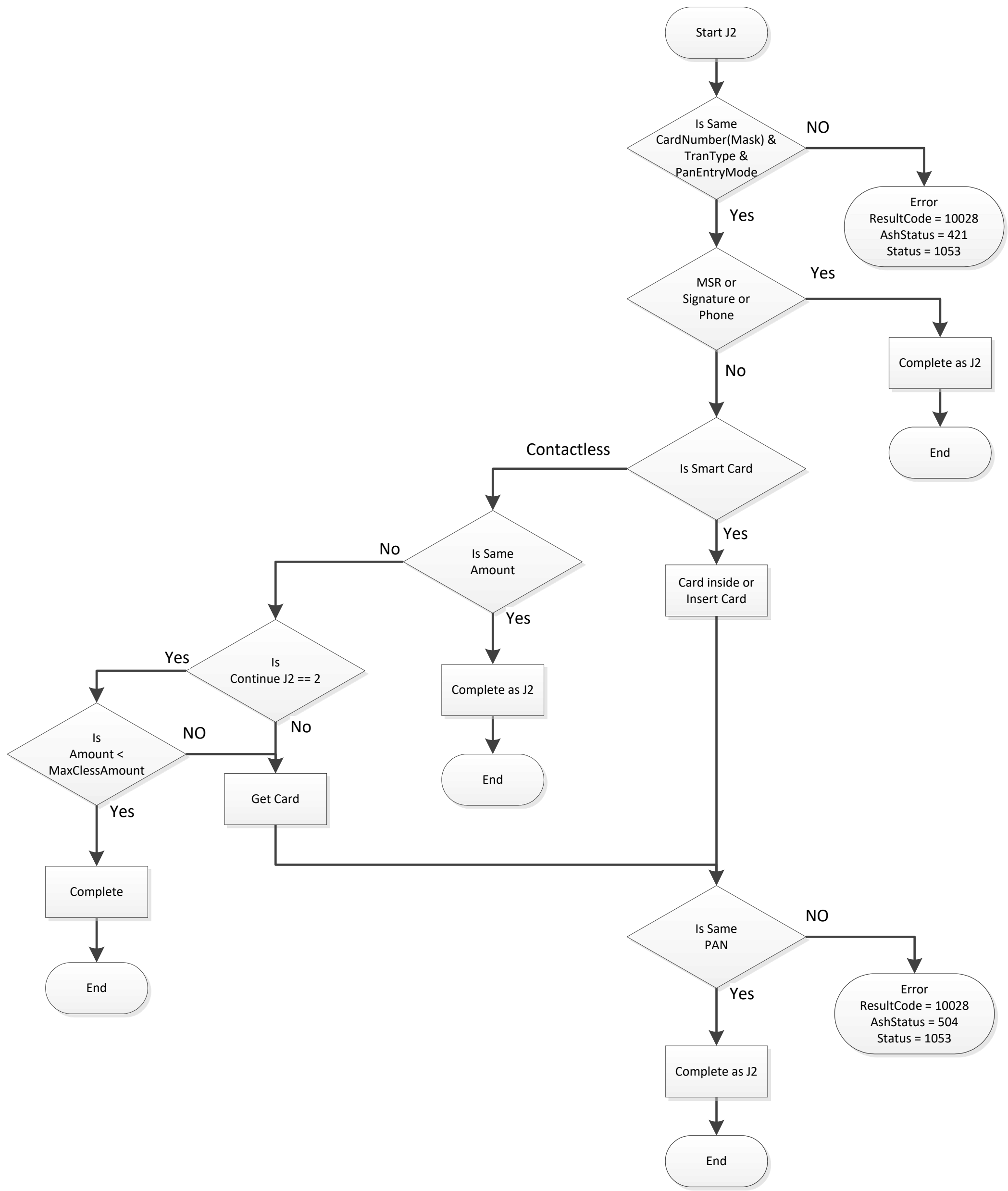
The request of the actual transaction will be like that:

<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>1</TranType>
<Amount>10000</Amount> **(Amount is in agorot)**
<Currency>376</Currency> **(Currency is in ISO currency code)**
<PanEntryMode>05</PanEntryMode>
<Pan>XXXXXXXXX1234</Pan>
<ContinueJ2>1</ContinueJ2>

```
<ParameterJ>4</ParameterJ>
<Xfield>20398049823</Xfield>
</Request>
```

The <PanEntryMode> should be the same <PanEntryMode> that was in the J2 transaction response XML (it will not always be 05. There are 4 different pan entry modes for contactless. See the possible values in the table in chapter 9.3).
<Pan> must be the same value as it came in the J2 transaction response XML.
<ContinueJ2> must also exist.
If the <Amount> is different, the cardholder will be asked to present his card again.

Continue J2 logic diagram



12. J5 – Pre authorization (בקשה לאישור ללא עסקה)

Use <ParameterJ>5</ParameterJ> to perform a pre authorization transaction.

```
<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>1</TranType>
<Amount>10000</Amount>  (Amount is in agorot)
<Currency>376</Currency> (Currency is in ISO currency code)
<PanEntryMode>PinPad</PanEntryMode>
<Xfield>20398049823</Xfield>
<ParameterJ>5</ParameterJ>
</Request>
```

13. Topup Magnetic Transaction. The Track2 comes from an external MSR.

The request XML (INT_IN):

```
<Request>
<Command>001</Command>

<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>

<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>55</TranType>
<Amount>10000</Amount>  (Amount is in agorot)
<Currency>376</Currency> (Currency is in ISO currency code)
<PanEntryMode>PinPad</PanEntryMode>

<Xfield>20398049823</Xfield>

<Track2>4580458045804580=1218203934827394928937</Track2>

</Request>
```

14. Discharge (פריקה)

Discharge transaction is for giftcards only.
The amount of the discharge must be equal to the total balance of the giftcard. Before doing a discharge, the merchant must do a balance transaction to get the balance of the giftcard. Then he will use that balance for the amount of the discharge transaction.
Tran Type is 2.


```
<Request>
    <Command>001</Command>
    <TerminalId>0880265</TerminalId>
    <TimeoutInSeconds>60</TimeoutInSeconds>
    <Mti>100</Mti>
    <TranType>2</TranType>
    <Amount>5000</Amount>
    <TermNo>1</TermNo>
    <PanEntryMode>PinPad</PanEntryMode>
    <AuthorizationCodeManpik>1</AuthorizationCodeManpik>
    <Currency>376</Currency>
    <XField>1234567890</XField>
    <CreditTerms>1</CreditTerms>
    <ParameterJ>4</ParameterJ>
    <RequestId>20170201141951</RequestId>
</Request>
```


15. A topup transaction with benefits (הטבות)

The benefits feature of Ashrait EMV is a complex issue. The usage of benefits may be different between different merchants because it depends on the financial agreement between the merchant and his credit company. I will show some examples of using benefits. There are 4 XML tags that are relevant to benefits, but not all of them are used all the time. The tags are: <IpayNumber>, <IpayCode>, <IpayAmount>, <IpayPercent>.

So, here is a topup transaction with points (the benefit in this case are the points).


```
<Request>
    <Command>001</Command>
    <TerminalId>0880265</TerminalId>
    <TimeoutInSeconds>60</TimeoutInSeconds>
    <Mti>100</Mti>
    <TranType>55</TranType>
    <Amount>5000</Amount>
    <TermNo>1</TermNo>
    <PanEntryMode>PinPad</PanEntryMode>
    <Currency>376</Currency>
    <XField>1234567890</XField>
    <CreditTerms>1</CreditTerms>
    <ParameterJ>4</ParameterJ>
    <IpayNumber>3</IpayNumber>
    <IpayCode>1</IpayCode>
    <RequestId>20170201141951</RequestId>
</Request>
```


The response receipt text will include the text:


```
<Line name="Benefit">    :מספר יחידות הטבה</Line>
<Line name="Benefit">              3</Line>
```

If you sent the wrong <IpayCode>, like that example:

```
<Request>
        <Command>001</Command>
        <TerminalId>0880265</TerminalId>
        <TimeoutInSeconds>60</TimeoutInSeconds>
        <Mti>100</Mti>
        <TranType>55</TranType>
        <Amount>5000</Amount>
        <TermNo>1</TermNo>
        <PanEntryMode>PinPad</PanEntryMode>
        <Currency>376</Currency>
        <XField>1234567890</XField>
        <CreditTerms>1</CreditTerms>
        <ParameterJ>4</ParameterJ>
        <IpayNumber>3</IpayNumber>
        <IpayCode>3</IpayCode>   (sending IpayCode 3 instead of 1)
        <RequestId>20170201141951</RequestId>
</Request>
```

You will get the following response:

```
<EMV_Output>
<Command>1</Command>
<TerminalId>0880265</TerminalId>
<ResultCode>10048</ResultCode>
<Status>1038</Status>
<AshStatus>313</AshStatus>
<XField>1234567890</XField>
<RequestId>20170201141951</RequestId>
</EMV_Output>
```

16. A regular transaction with points benefit

```
<Request>
        <Command>001</Command>
        <TerminalId>0880265</TerminalId>
        <TimeoutInSeconds>60</TimeoutInSeconds>
        <Mti>100</Mti>
        <TranType>1</TranType>
        <Amount>3000</Amount>
        <TermNo>1</TermNo>
        <PanEntryMode>PinPad</PanEntryMode>
       <AuthorizationCodeManpik>1</AuthorizationCodeManpik>
        <Currency>376</Currency>
        <XField>1234567890</XField>
        <CreditTerms>1</CreditTerms>
        <ParameterJ>4</ParameterJ>
   <IpayAmount>2000</IpayAmount>
   <IpayNumber>10</IpayNumber>
   <IpayCode>1</IpayCode>
        <RequestId>20170201141951</RequestId>
</Request>
```

The response receipt text will include the following text:

```
<Line name="Benefit">        סכום נטו:</Line>
<Line name="Benefit">          10.00</Line>
<Line name="Benefit">    סכום הנחה בקניה:</Line>
<Line name="Benefit">          20.00</Line>
<Line name="Benefit">    מספר יחידות הטבה:</Line>
<Line name="Benefit">            10</Line>
```

17. A regular transaction with a discount benefit (discount by amount)

```
<Request>
        <Command>001</Command>
        <TerminalId>0880265</TerminalId>
        <TimeoutInSeconds>60</TimeoutInSeconds>
        <Mti>100</Mti>
        <TranType>1</TranType>
        <Amount>4000</Amount>
        <TermNo>1</TermNo>
        <PanEntryMode>PinPad</PanEntryMode>
        <AuthorizationCodeManpik>1</AuthorizationCodeManpik>
        <Currency>376</Currency>
        <XField>1234567890</XField>
        <CreditTerms>1</CreditTerms>
        <ParameterJ>4</ParameterJ>
   <IpayAmount>2000</IpayAmount>
   <IpayNumber>0</IpayNumber>
   <IpayCode>0</IpayCode>
        <RequestId>20170201141951</RequestId>
</Request>
```

The response XML receipt text will include the following lines:

```
<Line name="Benefit">        סכום נטו:</Line>
<Line name="Benefit">          20.00</Line>
<Line name="Benefit">    סכום הנחה בקניה:</Line>
```

<Line name="Benefit"> 20.00</Line>
<Line name="Benefit"> אחוז הנחה בהטבה:</Line>
<Line name="Benefit"> 50.00</Line>

18. A regular transaction with a discount benefit (discount by percent)

<Request>
<Command>001</Command>
<TerminalId>0880265</TerminalId>
<TimeoutInSeconds>60</TimeoutInSeconds>
<Mti>100</Mti>
<TranType>1</TranType>
<Amount>5000</Amount>
<TermNo>1</TermNo>
<PanEntryMode>PinPad</PanEntryMode>
<AuthorizationCodeManpik>1</AuthorizationCodeManpik>
<Currency>376</Currency>
<XField>1234567890</XField>
<CreditTerms>1</CreditTerms>
<ParameterJ>4</ParameterJ>
<IpayPercent>30</IpayPercent>
<IpayNumber>0</IpayNumber>
<IpayCode>0</IpayCode>
<RequestId>20170201141951</RequestId>
</Request>

The response XML receipt text will include the following text:
<Line name="Benefit"> סכום נטו:</Line>
<Line name="Benefit"> 49.85</Line>
<Line name="Benefit"> סכום הנחה בקניה:</Line>
<Line name="Benefit"> 0.15</Line>
<Line name="Benefit"> אחוז הנחה בהטבה:</Line>
<Line name="Benefit"> 0.30</Line>

19. Cancel J5 – Cancel Pre authorization (ביטול בקשה לאישור ללא עסקה)

If you want to cancel a J5, you need to:
Send Mti 420
Send the original amount in tag <OriginalAmount>
Send the original Uid in tag <OriginalUid>
Send value 7 in tag <OriginalAuthorizationCodeManpik>

<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>420</Mti>
<OriginalAmount>10000</OriginalAmount>
<OriginalUid>23458934759823579387</OriginalUid>
<OriginalAuthorizationCodeManpik>7</OriginalAuthorizationCodeManpik>
<Xfield>20398049823</Xfield>
<ParameterJ>5</ParameterJ>
</Request>

20. Complete J5 – Complete Pre authorization (עסקת השלמה לבקשת אישור ללא עסקה)

If you want to complete the transactions that was preauthorized, you need to:
Send Mti 100
Send the original amount in tag <OriginalAmount>
Send the original Uid in tag <OriginalUid>
Send value 7 in tag <OriginalAuthorizationCodeManpik>
Send <ParameterJ>49</ParameterJ>

Example:


```
<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<OriginalAmount>10000</OriginalAmount>
<OriginalUid>23458934759823579387</OriginalUid>
<OriginalAuthorizationCodeManpik>7</OriginalAuthorizationCodeManpik>
<Xfield>20398049823</Xfield>
<ParameterJ>49</ParameterJ>
</Request>
```

9.3 The Response XML (INT_OT)

The following XML shows all the possible XML tags that may appear in the response XML. However, in most transactions, only part of the XML tags will appear in the response. Only the relevant XML tags.

On a successful transaction, the response XML will include also the full receipt text. The two parts of the receipt will be in the XML – the part that belong to the merchant and the second part that is supposed to be handed to the customer. The ECR may use that receipt text to print the receipt, or the ECR may choose to ignore the receipt text and to design and print its own style of receipt.

Although the receipt of the merchant and the receipt of the customer are very similar, they still may have small differences and so we decided to give the full text of them both in tag groups <ReceiptMerchant> and <ReceiptCustomer>.

See section 9.4 for the structure of the receipt XML parts.


```
<?xml version="1.0"?>
<EMV_Output>
<Command></Command>
<RequestId>230492039</RequestId>
<TerminalId></TerminalId>
<ResultCode>0</ResultCode>
<Status>0</Status>
<AshStatus>0</AshStatus>
<Mti>100</Mti>
<Pan>1234567890123456</Pan>
<PanEntryMode>40</PanEntryMode>
<CardName>מסטרקרד זהב</CardName>
<Manpik>01</Manpik>
<Brand>01</Brand>
<Solek>01</Solek>
<CardType>111</CardType>
<SpType>DEFAULT</SpType>
<Amount>000000001234</Amount>
<TranType>01</TranType>
<CreditTerms>1</CreditTerms>
<Currency>376</Currency>
<CurrencyName/>
<FileNo>17</FileNo>
<TermNo>001</TermNo>
<TermSeq>001</TermSeq>
<TerminalName>test emv 700</TerminalName>
<Retailer>0880700014</Retailer>
<ComRetailerNum>0071506</ComRetailerNum>
<HostVersion>ABS10</HostVersion>
<DateTime>0420102003</DateTime>
<ResponseId/>
<ResponseCvv2>0</ResponseCvv2>
<ResponseAvs>0</ResponseAvs>
<AuthManpikNo>0000541</AuthManpikNo>
<AuthCodeManpik>1</AuthCodeManpik>
<AuthSolekNo/>
<AuthCodeSolek>0</AuthCodeSolek>
<FirstPayment/>
<NotFirstPayment/>
<NoPayment><NoPayment/>
```

```
<Uid>15042010200308807003502</Uid>
<Rrn>511010000541</Rrn>
<TelAuthAbility/>
<TelNoCom/>
<Tip/>
<IpayAmount/>
<IpayNumber/>
<IpayPercent/>
<Cashback/>
<TVR>0080008000</TVR>
<AddendumSettl/>
<IndxPayment>0</IndxPayment>
<Authorizedamount/>
<Addendum1/>
<Addendum2/>
<ParameterJ>0</ParameterJ>
<AddDspBalance/>
<DspBalance/>
<ZData/>
<Track2/>
<Atc/>
<CardSeqNumber/>
<AID>A0000000031010</AID>
<TSI/>
<ARC/>
<VerifiedByPIN>0</VerifiedByPIN>
<AppVersion>ABS000089w</AppVersion>
<Xfield>39439849</Xfield>
<CaspitInternalError>0</CaspitInternalError>
<PinpadStep></PinpadStep>
<Token></Token>
<SwitchStatus></SwitchStatus>
<CardInside></CardInside>
<Field55></Field55>
<ExtraInfo></ExtraInfo>
<CardHash></CardHash>
<PossibleCreditTerms>
<OfflineSupport>1</OfflineSupport>
<OnlineReason>decode</OnlineReason>
<OfflineAmount>300</OfflineAmount>
<J5Support>1</J5Support>
<SlipLang>
<ReceiptMerchant>
<Line> Text </Line>
…
…
</ReceiptMerchant>

<ReceiptCustomer>
<Line> Text </Line>
…
…
</ReceiptCustomer>
</EMV_Output>
```

Understanding the status of the transaction

There are 3 "status" tags that might come back in the response XML. They are:

- <ResultCode>
- <Status>
- <AshStatus>

<Status> and <AshStatus> will have values as specified in Shva mefizim-EMV document (see Appendix 1 of mefizim-EMV document). <Status> and <AshStatus> will always "go" together. They will both be zero or both be non-zero.
As explained in Shva document, the <Status> value is a simplification of the <AshStatus> value.
In the SmartRetail solution we decided to give both <Status> and <AshStatus>

- The <Status> codes can be seen also in Appendix H in this document.
- The <AshStatus> codes can be seen also in Appendix I in this document.
- <ResultCode> will have status codes that belong to Caspit. See Appendix A. We tried to stick to Shva statuses, but because SmartRetail has much more features than Shva Ashrait PC, we had to add the <ResultCode> tag that will hold error codes that are special for Caspit.

A successful (approved) transaction is identified by a response XML in which all 3 "status" tags are zeros. Like that:

<ResultCode>0</ResultCode>
<Status>0</Status>
<AshStatus>0</AshStatus>

In case of error, you should first look at <AshStatus>. If <AshStatus> is not zero then you can look at Appendix I and get the description of the error.
If <AshStatus> and <Status> are zero then you need to look at <ResultCode> and consider that error code.

Actually, there is a fourth tag, named <CaspitInternalError> that will hold the internal error code of Caspit. This value can be used by Caspit support to identify problems in the field.

This complexity of status tags exists only in the Transaction response. All other commands of SmartRetail (like STATIS request, communication test, Pinpad configuration, etc) have only the <ResultCode> tag.

Example

For example, if the credit company declined the transaction then the following tags will be in the response:

<Status>1038</Status>
<AshStatus>4</AshStatus>
<ResultCode>10048</ResultCode> (RETVAL_CHECK_OTHER_STATUSES)
<CaspitInternalError>1403</CaspitInternalError>

The ECR must check first the <AshStatus> tag, the value is 4 and according to Shva mefizim EMV document the meaning of AshStatus 4 is that the transaction has been declined.

When the ECR keeps the transaction response, I recommend that it keeps all these "status" tags in its database, it can be used for future inquires.

Table of tags in transaction response XML (INT_OT)

Tag name	Description	Matching TLV tag (Caspit Internal)
<EMV_Output>		
Command	פקודה Command Should be the same as in the request	001
RequestId	מזהה בקשה. אמור לחזור עם אותו ערך כמו בבקשה. Should be the same as in the request	051
TerminalId	מספר מסוף Terminal Id Should be the same as in the request	006

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
ResultCode	Relevant result codes: 0 – Success 10003 – RETVAL_INCORRECT_TERMINAL_ID 10022 – RETVAL_INCORRECT_ECR_NUMBER 10036 – RETVAL_XML_MISSING_TERMINAL_ID_TAG 10041 – RETVAL_ERROR_CONNECTING_TO_PINPAD 10042 – RETVAL_XML_MISSING_REQUEST_ID_TAG 10043 – RETVAL_CASPIT_WINDOWS_SERVICE_NOT_RESPONDING 10044 – RETVAL_USER_ABORTED_TRANSACTION 10048 – RETVAL_CHECK_OTHER_STATUSES 10050 – RETVAL_DOUBLE_TRANSACTION 10053 – RETVAL_PINPAD_RETURNED_EMPTY_RESPONSE 10054 – RETVAL_ASHRAIT_INVALID_SETUP 10100 – RETVAL_INVALID_PAN_ENTRY_MODE 10101 – RETVAL_INVALID_MTI 10102 – RETVAL_INVALID_X_FIELD Other result codes may arrive; see Appendix A for a full list of result codes	
Status	סטטוס. ראה נספח H במסמך זה לתיאור הסטטוסים האפשריים Status of transaction See appendix H for all possible values	159
AshStatus	קודי פירוט שגיאה. קוד פירוט על הודעות השגיאה בשדה Status. ראה נספח I במסמך זה לתיאור הסטטוסים האפשריים Another status for transaction See appendix I for all possible values	160

Tag name	Description	Matching TLV tag (Caspit Internal)
Mti	קוד המסר: 100 – עסקה רגילה 400 – עסקת ביטול 420 – ביטול בקשה לאישור ללא עסקה (ביטול J5). 100 – Regular transation 400 – Void transaction 420 – Void Pre authorized transaction (void J5)	087
Pan	מספר כרטיס אשראי. אפסים מובילים. ממוסך. Card number. With leading zeros. Masked.	162
PanEntryMode	מקור נתוני הכרטיס. Pan Entry mode 00 – מגנטי (Magnetic) 04 – Contactless MSR 05 – Contactless EMV 06 – Mobile Contactless MSR 07 – Mobile Contactless EMV 10 – מספר טלפון נייד (Cellphone number) 40 – EMV contact 50 – טלפונית (Card not present) 51 – חתימה בלבד (Signature) 52 – אינטרנט (Internet) 80 – fallback 81 – Empty candidate list	138
CardName	שם כרטיס Card name	163

Tag name	Description	Matching TLV tag (Caspit Internal)
Manpik	מנפיק ערכים אפשריים: 00 – תייר 01 – ישראכרט 02 – כ.א.ל 06 – מקס Issuer 00 – Tourist 01 – Isracard 02 – CAL 06 – MAX	165

Tag name	Description	Matching TLV tag (Caspit Internal)
Brand	מותג. ערכים אפשריים: 0 – כרטיס פרטי של חברה מנפיקה (PL) 1 - מסטרכרד 2 – ויזה 3 – דיינרס 4 – אמקס 5 – מותג ישראכרט 6 – JCB 7 –Discover 8 – מאסטרו Brand 0 – Private label of issuer 1 - Mastercard 2 – Visa 3 – Diners 4 – Amex 5 – Isracard 6 – JCB 7 – Discover 8 - Maestro	166
Solek	סולק מספר החברה הסולקת את העסקה 1 - ישראכרט 2 – כ.א.ל 6 – מקס Acquirer 1 - Isracard 2 – CAL 3 – MAX	167

Tag name	Description	Matching TLV tag (Caspit Internal)
CardType	סוג כרטיס Card Type	168
SpType	סוג כרטיס מיוחד 00 – ברירת מחדל 01 – כרטיס חיוב מיידי 70 – מועדון 03 – כרטיס דלק 04 – כרטיס דואלי 74 – דואלי מועדון 73 – דלק מועדון 76 – מועדון נטען 06 – נטען 08 – דלקן 99 –כרטיס תייר Special card type 00 – Default 01 – Immediate 70 – Club 03 – Petrol card 04 – Dual card 74 – Dual club 75 – Petrol club 76 – Clube chrageable 06 – Chrageable 08 – Petrol 99 – Tourist card	169
Amount	סכום באגורות Amount in cents (agorot)	146

Tag name	Description	Matching TLV tag (Caspit Internal)
TranType	סוג עסקה 01 – עסקת חיוב 02 – עסקת פריקה 03 – עסקת חיוב מאולצת 06 – עסקת cashback 07 – עסקת מזומן 11 – עסקה עם הוראות קבע 30 – בירור יתרה 53 – עסקת זיכוי 55 – עסקת טעינה Transaction type 01 – Debit 02 – Discharge 03 – Forced 06 – Cashback 07 – Cash 11 – Recurring 30 – Get balance 53 – Refund 55 - Topup	140

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
CreditTerms	סוג אשראי 1 – אשראי רגיל 2 – ישראקרדיט/אמקסקרדיט/עדיף/30+ 3 – חיוב מיידי 6 – קרדיט 8 – תשלומים Credit Terms 1 - Regular 2 – Isracredit/AmexCredit/Adif/30+ 3 – Immediate 6 – Credit 8 – Installments	139
Currency	קוד מטבע קוד נומרי בהתאם לטבלת קודי מטבע ISO8583. קודים שכיחים: 376 – שקל 840 – דולר 978 – יורו Currency code according to ISO8583 376 – NIS 840 – USD 978 - EUR	148
CurrencyName	שם קוד מטבע ISO	170
FileNo	מספר קובץ העסקאות המנוהל ע"י הפינפד. מספר הקובץ גדל באחד בכל פעם שמשדרים עסקאות לשב"א. מספר הקובץ נע מ1 עד 99. לאחר מכן הוא חוזר ל1. במסוף חדש מספר הקובץ מתחיל ב1. The file number of the transactions batch. The file number increases at the end of each day The valid file numbers are from 1 to 99 in cyclic method. A new terminal starts with file number 1.	016

Tag name	Description	Matching TLV tag (Caspit Internal)
TermNo	מספר קופה\תחנה\עמדה. ברירת המחדל היא 001. ניתן לשנות את מספר הקופה בתפריט הפינפד. Should be the same as in the request	040
TermSeq	מספר סודר של העסקה בתוך קובץ העסקאות. Transaction sequence inside the transactions batch	171
TerminalName	שם המסוף כפי שהתקבל בפרמטרים משב"א The terminal name as received from the main switch	172
Retailer	מספר מסוף ארוך. משמש להזדהות המסוף מול המחשב המרכזי מספר המסוף הארוך הוא באורך של 10 ספרות. The long terminal Id.	173
ComRetailerNum	מספר ספק בחברה הסולקת Business number	174
HostVersion	מספר גירסה של המחשב בשב"א	175
DateTime	תאריך וזמן זמן העיסקה בפורמט MMDDHHMMSS Date/time of the transaction in MMDDHHMMSS format	176
ResponseId	תשובה לבדיקה ת"ז 0 – לא הוכנס 1 – הוכנס ותקין 2 – לא תקין 3 – לא נבדק The response for ID check 0 – Not entered 1 - Valid 2 – Invalid 3 – Not Checked	177

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
ResponseCvv2	תשובה לבדיקת CVV2 0 – לא הוכנס 1 – הוכנס ותקין 2 – לא תקין 3 – לא נבדק The response for CVV2 check 0 – Not entered 1 - Valid 2 – Invalid 3 – Not checked	178
ResponseAvs	תשובה לבדיקת כתובת 0 – לא הוכנס 1 – הוכנס ותקין 2 – לא תקין 3 – לא נבדק This field is irrelevant currently.	180
AuthManpikNo	מספר אישור מנפיק Issuer authorization number	181

Tag name	Description	Matching TLV tag (Caspit Internal)
AuthCodeManpik	הגורם שסיפק תשובה כמנפיק 0 – עסקה ללא מספר אישור מנפיק 1 – אושר ע"י המנפיק 2 – דחייה ע"י המנפיק 3 – אושר ע"י שב"א בשירות STIP 4 – דחייה ע"י שב"א בשירות STIP 5 – אושר ע"י המענה הקולי 6 – דחייה ע"י המענה הקולי 7 – אישור offline ע"י כרטיס חכם 8 – אושר ע"י כרטיס חכם – לא ניתן לבצע התקשרות The source for the issuer authorization 0 – A transaction without authorization number 1 – Authorized by issuer 2 – Declined by issuer 3 – Authorized by Shva 4 – Declined by Shva 5 – Authorized by voice 6 – Declined by voice 7 – Authorized Offline by chip 8 – Authorized by chip – not communication possible	143
AuthSolekNo	מספר אישור סולק השדה יועבר כשהוא מיושר לימין Authorization number from acquirer	182

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
AuthCodeSolek	הגורם שסיפק תשובה כסולק 0 – עסקה ללא מספר אישור סולק 1 – אישור ע"י הסולק 2 – דחייה ע"י הסולק 3 – אישור ע"י שב"א בשירות STIP לסולק 4 – דחייה ע"י שב"א בשירות STIP לסולק 5 – אושר ע"י המענה הקולי 6 – דחייה ע"י מענה קולי 7 – אושר ע"י החברה הסולקת באמצעות בקשה לאישור ללא עסקה The source for the acquirer authorization 0 – The transaction has no authorization number from acquirer 1 – Authorized by acquirer 2 – Declined by acquirer 3 – Authorized by Shva 4 – Declined by Shva 5 – Authorized by voice 6 – Declined by voice 7 – Atuhorized by acquirer for pre authorization	142
FirstPayment	סכום תשלום ראשון במאיות, אפסים מובילים, ללא נקודה עשרונית First installment amount. With leading zeros. Without decimal point.	123
NotFirstPayment	סכום תשלום קבוע בעסקת תשלומים, במאיות, אפסים מובילים, ללא נקודה עשרונית Not first installment. With leading zeros. Without decimal point.	124
NoPayment	מספר תשלומים בעסקת תשלומים, ללא תשלום ראשון. ערכים: 999- 0 Number of installments. Excluding the first installment. Value can be 0 - 999	126

Tag name	Description	Matching TLV tag (Caspit Internal)
Uid	מזהה עסקה. המספר ילווה את העסקה בכל מחזור חייה. הUid הוא מספר בן 23 ספרות. המורכב מהשדות הבאים: YYMMDDHHMMSSTTTTTTTRRRC חותמת זמן: YYMMDDHHMMSS מספר מסוף: TTTTTTT מספר קופה: RRR ספרת ביקורת: C Unique Id. The Uid is a 23 digits number. YYMMDDHHMMSSTTTTTTTRRRC Time stamp: YYMMDDHHMMSS Terminal Id: TTTTTTT ECR number: RRR Checksum: C	183
Rrn	מזהה עסקה מהחברה המאשרת. מזהה עסקה ייחודי שמוקצה ע"י החברה המאשרת את הבקשה. Transaction Id from the body that authorized the transaction	155
TelNoCom	מספר טלפון של החברה הסולקת. Phone number of the acquirer	184
TelAuthAbility	אינדיקציה האם ניתן להקליד מספר אישור מראש לעסקה מסוג זה 0 – לא ניתן להקליד מספר אישור מראש. 1 – ניתן להקליד מספר אישור מראש. The field is irrelevant currently.	
Tip	תשר במאיות, אפסים מובילים, ללא נקודה עשרונית The field is irrelevant currently.	093
IpayAmount	סכום הנחה בהטבות במאיות, ללא נקודה עשרונית Benefit amount	095
IpayNumber	מספר יחידות בהטבה Benefit units	096

Tag name	Description	Matching TLV tag (Caspit Internal)
IpayPercent	אחוז הנחה בהטבה. השדה יועבר במאיות, ללא נקוד עשרונית, עם אפסים מובילים. לדוגמא: 1.54% יועבר כ0154. Benefit percent	097
CashBack	סכום המזומן בעסקת cashback במאיות, אפסים מובילים, ללא נקודה עשרונית The amount for Cashback transaction. With leading zeros. Without decimal point.	092
TVR	תוצאת בדיקות המסוף EMV field – Terminal Verification results	186
AddendumSettl	שדה להוספת נתונים ברשומת העסקה Additional information that may return in the response. The content may be specific for each customer.	
IndxPayment	הצמדה לדולר/מדד. שימוש בעסקאות תשלומים או קרדיט	125
AuthorizedAmount	סכום עסקה מאושר בתשובה לבקשה לאישור The field is irrelevant currently.	
Addendum1	שדה להוספת טקסט חופשי בין המסוף לחברה המקבלת ובתשובה למסוף Free text field	
Addendum2	שדה להוספת טקסט חופשיי בין המסוף לחברה המקבלת ובתשובה למסוף Free text field	
ParameterJ	Same as in the request	150

Tag name	Description	Matching TLV tag (Caspit Internal)
AddDspBalance	הוראות תצוגה לכרטיס נטען. הנחיות לטיפול בהודעה הנשלחת מהחברה שמחזירה את היתרה בתשובה לבקשה לאישור בשדה DspBalance. יכיל אחד מהערכים הבאים: 1 – יש להקרין את המסר המופיע בשדה DspBalance ולהדפיסו בפתקית 2 – הקרנה בלבד של המסר המופיע בשדה DspBalance 3 – הדפסה על הפתקית של המסר המופיע בשדה DspBalance Display instructions for gift card 1 – Display the DspBalance message and print it on the receipt 2 – Only display the DspBalance message 3 – Only print the DspBalance message	187
DspBalance	שדה להוספת טקסט חופשי המכיל יתרה בתשובה מחברת האשראי למסוף (אפשרי לכל סוגי העסקאות). שימו לב שהשדה יכיל טקסט בסגנון של: *"יתרת הכרטיס היא: 230.40"* ככה הטקסט מגיע בתשובה משב"א. הטקסט לא קבוע. אם הקופה צריכה רק את הסכום, אז הקופה תצטרך לפרסר את הטקסט ולהוציא מתוכו את הסכום. A message from credit company about the balance of the giftcard	188
ZData	כמו ערך שדה Z באשראית 96 Field Z	
Track2	מידע הנקרא מהכרטיס בעסקאות בהן הכרטיס נוכח. המידע יהיה ממוסך. Masked Track2	065
Atc	EMV tag - Application Transaction Counter	189
CardSeqNumber	EMV tag - Card Sequence Number (CSN)	121
AID	זיהוי יישום בשבב בעסקה חכמה יודפס ב-HEX EMV tag – Application Id	199
TSI	תג 9B בשדה 55. בעסקה חכמה בלבד. EMV tag – Transaction Status Information	200

Tag name	Description	Matching TLV tag (Caspit Internal)
ARC	תג 8A בשדה 55. בעסקה חכמה בלבד. EMV tag – Application Response Code	201
VerifiedByPIN	ערך 1 או 0. ניתן לשימוש עבור הדפסת פתקית 1 – עסקה מאושרת חכמה בה נבדק קוד סודי – על פי מפרטי PP סעיף 3.7.6 יש להדפיס משפט "Verified by PIN" בתחתית של קבלה 0 – אין להדפיס "Verified by PIN" Instruction about printing "Verified by PIN" on the receipt 1 – The PIN was verified. You need to print "Verified by PIN" on the receipt. 0 – The PIN was not verified. Don't print "Verified by PIN"	202
AppVersion	גרסת התוכנה כפי שיש להדפיס על הקבלה Application version of the Ashrait EMV	210
Xfield	מספר השובר (או מספר החשבונית) של הקופה. זהה לשדה X מאשראית 96. This is the ECR voucher number (or ECR invoice number). This value should be the same as in the request	212 The X value will eventually be put in ISO field 126 B2
CaspitInternalError	קוד שגיאה פנימי של כספיט. Internal error code of Caspit	022
PinpadStep	The last step (state) of the Pinpad. See Appendix J for a list of steps. The ECR can use this information to know better where the transaction failed (if it failed).	217
Token	**<u>Switch Support</u>** – Relevant only when using a Switch The Switch may return a token that the ECR will keep. Tokens will not always be supported, so this tag may be empty or will not exist at all.	218

Tag name	Description	Matching TLV tag (Caspit Internal)
SwitchStatus	**Switch Support** – Relevant only when using a **Switch** This tag gives extra information about the transaction record with regard to the Switch. This tag **DOES NOT** say anything about the success or failure of the transaction. Possible values: **0** – Everything is OK. Transaction **record** was sent to the Switch **1** – Transaction record was **NOT** sent to the Switch. Transaction may have been successful, but the record was not sent to the Switch. If the request was done with <SwitchSend>0</SwitchSend> then you will always get this <SwitchStatus>1</SwitchStatus>. If the request was done with <SwitchSend>1</SwitchSend> and you got <SwitchStatus>1</SwitchStatus>, it may mean that **a)** transaction failed so there was not need to send the record to the Switch or **b)** transaction was successful but, for some reason, the transaction record was not sent to the switch. If you use <SwitchSend>1</SwitchSend> in every request then the unsent records will be sent on the following transaction (or you may use Generic command with sub command 012 to send the records). **Do not** use the <SwitchStatus> tag as an indicator to the status (success or failure) of the transaction. The status of the transaction must be derived as explained in the beginning of this chapter. The <SwitchStatus> tag just give some extra information with regard to the Switch.	219
CardInside	0 – No card inside 1 – Card inside contact reader This is an important issue. With EMV, it is very common for the cardholder to forget his card inside the Pinpad. The ECR should always check if the card is still inside and alert the merchant about it.	220
Field55	EMV field 55. A field with many EMV related information about the transaction	

Tag name	Description	Matching TLV tag (Caspit Internal)
ExtraInfo	Extra information about the transaction. For example, for LifeStyle support, this field will contain the following string: LifeStyleCheck=X Where X can be: 0 – Undetermined 1 – LifeStyle Single 2 – LifeStyle Multi For example, the tag will look like that: <ExtraInfo>LifeStyleCheck=1</ExtraInfo>	251
CardHash	MD5 or Sha256 Hash result on the card number (the PAN) and PosHash.	252

Tag name	Description	Matching TLV tag (Caspit Internal)
PossibleCreditTerms	This tag is relevant to J2 response only. The J2 response will include all the possible credit terms that are available to the credit card that was used. The possible credit terms are selected by Ashrait EMV logic that takes into account the amount, credit card issuer, acquirer, transaction types and more factors. The ECR can then display all the possible credit terms and the cashier can select one of the credit terms to be used in the J4 completion of the transaction. The format of the tag's value is like that: <PossibleCreditTerms> <CreditTerm Code="1" Avail="True" MinPayments="0" MaxPayments="0"/> <CreditTerm Code="2" Avail="False" MinPayments="0" MaxPayments="0"/> <CreditTerm Code="3" Avail="False" MinPayments="0" MaxPayments="0"/> <CreditTerm Code="6" Avail="True" MinPayments="4" MaxPayments="20"/> <CreditTerm Code="8" Avail="True" MinPayments="2" MaxPayments="30"/> </PossibleCreditTerms> The attributes are: "Avail" – if "True" then this credit term is available. If "False" then this credit term is no available. "MinPayments" – if "0" then payments are not allowed. If bigger than zero then it is the minimum number of payments (installments) allowed. "MaxPayments" – if "0" then payments are not allowed. If bigger than zero then it is the maximum number of payments (installments) allowed. The credit term codes are the valid credit terms of Ashrait EMV. סוג אשראי 1 – אשראי רגיל 2 – ישראקרדיט/אמקסקרדיט/עדיף/30+ 3 – חיוב מיידי 6 – קרדיט 8 – תשלומים Credit terms 1 - Regular 2 – Isracredit/AmexCredit/Adif/30+ 3 – Immediate 6 – Credit 8 - Installments	253

Tag name	**Description**	**Matching TLV tag (Caspit Internal)**
PossibleCurrency	This tag is relevant to J2 response only. The J2 response will include all the possible currency for each available credit term. The ECR can then display all the possible currency for selected credit term and the cashier can select one of the currency to be used in the J4 completion of the transaction. The format of the tag's value is like that: <PossibleCurrencyCreditTerm> Code="5" CurrencyCode1="376" Currency1="NIS" Index1="0" CurrencyCode2="840" Currency1="USD" Index1="1" </PossibleCurrencyCreditTerm> <PossibleCurrencyCreditTerm> Code="2" Code1="376" Currency1="NIS" Index1="0" Currency2="USD" </PossibleCurrencyCreditTerm>	259
SlipLang	The language of the slip "iw_IL" – Hebrew "en_US" - English	255
ReceiptMerchant	קבוצת XML זאת תכיל את כל שורות הקבלה. קבלה אשר מיועדת לבית העסק. ראו סעיף הבא לפרטים. This XML group will contain all the lines of the merchant receipt	N/A
ReceiptCustomer	קבוצת XML זאת תכיל את כל שורות הקבלה. קבלה אשר מיועדת ללקוח. ראו סעיף הבא לפרטים This XML group will contain all the lines of the customer receipt	N/A
DCC	Is DCC transaction done	
ConversionAmount	The DCC offered/foreign amount. Foreign Amount from Acquirer/DCC provider	101
ConversionRate	The DCC offer exchange rate.	102
ConversionCurrency	CHC offered by Acquirer/DCC provider	103
OfferValidity	The DCC offer validity in hours	262
MarginRate	The DCC margin rate	263
CurrencyAlphaCode	The DCC currency alpha code	264
ExchangeRateTime	The DCC exchange rate time stamp	265
ConversionProvider	The DCC conversion provider	266

Tag name	Description	Matching TLV tag (Caspit Internal)
Commission	The DCC commission	267
DeferMonths	Delayed payment in months	153
DueDate	The date YYMMDD of customer charge.	154
ReceiptPrint	תג באורך 2 ספרות XY , המציין האם נדרש להדפיס פיתקית . X– האם נדרש להדפיס פיתקית בית העסק Y- האם נדרש להדפיס פיתקית לקוח אם הערך 1 אזי להדפיס . אם הערך 0 אזי לא להדפיס	
OfflineSupport	Dose shva parameters enable offline transaction for the card	
OnlineReason	Why offline transaction not supported for the card	
OfflineAmount	shva parameters offline amount for the card	
J5Support	Dose shva parameters enable J5/J59 transaction for the card	

9.4 Receipt XML structure

1. The structure

The structure of the XML is very simple. We have only one tag <Line>, which will be used for each line of the receipt. In some lines, we will add the "name" attribute with a value that will make the line more meaningful. The same attribute value can exist in more than one line.

XML template with the "name" attribute:

<ReceiptMerchant>

<Line name="value"> Text of line </Line>

<Line name="value"> Text of line </Line>

</ReceiptMerchant>

<ReceiptCustomer>

Same structure as ReceiptMerchant

</ReceiptCustomer>

The Hebrew characters will be encoded with ISO8859-8.

2. Possible values for the "name" attribute

Value	Hebrew Meaning	English Meaning
Merchant	שם בית העסק	The name of the merchant

Value	**Hebrew Meaning**	**English Meaning**
TerminalId	מספר מסוף	Terminal Id
SWVersion	מספר גרסת תוכנה בשב"א	Software version
BusinessNumber	מספר ספק בחברת אשראי	Business number
TranDateTime	זמן ביצוע העסקה	Transaction date and time
PrintDateTime	זמן הדפסת הקבלה	Print date and time
CardName	שם כרטיס	Card name
CardNumber	מספר כרטיס	Card number (PAN)
Voucher	מספר שובר	Transaction voucher
TranType	סוג עסקה	Tran type
AuthNumber	מספר אישור	Authorization number (if exists)
PanEntryMode	אופן ביצוע	Pan entry mode
CreditTerms	תנאי אשראי	Credit terms
Amount	סכום	Amount
Currency	מטבע	Currency
Uid	מזהה עסקה	Uid
EMV	נתוני EMV	Some EMV details
Aid	מזהה אפליקציה על כרטיס חכם	Chip Application Id
RRN	מספר עסקה אצל הסולק	Acquirer transaction number
Verified	ההודעה שהעסקה אומתה ע"י קוד סודי	"Verified by PIN" message
Signature	מקום לחתימת הלקוח	Signature placeholder
Phone	מקום לרישום הטלפון	Phone number placeholder
Payments	נתוני עסקת תשלומים	Details about installments
Balance	נתוני יתרת הכרטיס (לכרטיס גיפטקארד)	Details about card balance
Copy	ההודעה על העתק ללקוח	A message that this is a another copy of the receipt
Goodbye	הודעת תודה ללקוח	A thank you message to the customer
Empty	שורה ריקה	An empty line
PreAuth	הודעה של אישור מוקדם	A message that this transaction is a pre authorization transasction
Benefit	נתוני הטבה	Details about benefits

3. Example with the "name" attribute

The merchant receipt: the receipt that must be kept by the merchant. It may contain the full PAN (card number) or just the last 4 digits of the PAN (it depends on a parameter).

<ReceiptMerchant>

<Line name="Merchant">המכולת של שמוליק</Line>

<Line name="Empty"></Line>

<Line name="TerminalId">מס. כספיט: 0880023</Line>

<Line name="SWVersion">תוכנה: CST000078T</Line>

<Line name="BusinessNumber">מספר עסק: 0300012</Line>

<Line name="Empty"></Line>

<Line name="DateTime">11/12/16 10:16</Line>

<Line name="Empty"></Line>

<Line name="CardName">ויזה רגיל</Line>

<Line name="CardNumber"> Card number </Line> (Full PAN or partial PAN. Depends on a parameter).

<Line name="Voucher">מס' שובר: 31001001</Line>

<Line name="TranType">עסקה: חובה</Line>

<Line name="AuthNumber">0778164 חברה</Line>

<Line name="PanEntryMode">ביצוע: חכם</Line>

<Line name="CreditTerms">אשראי: רגיל</Line>

<Line name="Amount">סכום עיסקה:</Line>

<Line name="Amount"> 1.00 ש"ח</Line>

<Line name="Currency">מטבע: ש"ח</Line>

<Line name="Uid">UID:16121110162408800236240</Line>

<Line name="Aid">AID:A0000000031010</Line>

<Line name="RRN">RRN:502211772</Line>

<Line name="EMV">ATC:0033, CSN:01, TSI:F800, ARC:3030</Line>

<Line name="EMV">TVR:0080001000</Line>

<Line name="Verified">Verified by PIN</Line>

<Line name="Empty"></Line>

<Line name="Signature">חתימה____________________</Line>

<Line name="Empty"></Line>

<Line name="Phone">טלפון____________________</Line>

</ReceiptMerchant>

The customer receipt: the receipt that is given to the customer. It will contain a masked PAN (card number). Only the last 4 digits of the PAN (the card number) will be visible.

<ReceiptCustomer>

<Line name="Merchant">המכולת של שמוליק</Line>

<Line name="Empty"></Line>

<Line name="TerminalId">מס. כספיט: 0880023</Line>

<Line name="SWVersion">תוכנה: CST000078T</Line>

<Line name="BusinessNumber">מספר עסק: 0300012</Line>

<Line name="Empty"></Line>

<Line name="DateTime">11/12/16 10:16</Line>

<Line name="Empty"></Line>

<Line name="CardName">ויזה רגיל</Line>

<Line name="CardNumber"> Last of 4 digits of card number </Line> (Partial PAN. Last 4 digits).

<Line name="Voucher">מס' שובר: 31001001</Line>

<Line name="TranType">עסקה: חובה</Line>

<Line name="AuthNumber">0778164 חברה</Line>

<Line name="PanEntryMode">ביצוע: חכם</Line>

<Line name="CreditTerms">אשראי: רגיל</Line>

<Line name="Amount">סכום עיסקה:</Line>

<Line name="Amount"> 1.00 ש"ח</Line>

<Line name="Currency">מטבע: ש"ח</Line>

<Line name="Uid">UID:16121110162408800236240</Line>

<Line name="EMV">RRN:502211772,ATC:0033, CSN:01, TSI:F800, ARC:3030</Line>

<Line name="EMV">, TVR:0080001000,AID:A0000000031010</Line>

<Line name="Verified">Verified by PIN</Line>

<Line name="Copy">*** עותק ללקוח ***</Line>

<Line name="Goodbye">תודה ולהתראות</Line>

</ReceiptCustomer>

9.5 Smart Swipe

There are three "Swipe" features in the Pinpad, some may find it a bit confusing, and so I will summarize them here:

- The **Swipe Command** is described in chapter 21. It is a command that is initiated by the PC.
- The **Smart Swipe** is a feature that changes a transaction request into a Swipe response. It is described here.
- The **Idle Swipe** is a feature that allows swiping cards while the Pinpad is in idle mode. See Appendix G.

Smart Swipe can be enabled/disabled using the Pinpad configuration command (using the <EnableSmartSwipe> tag)

<Request>

<Command>013</Command> (Pinpad Configuration)

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<EnableSmartSwipe>1</EnableSmartSwipe>

<RequestId>20160216155540</RequestId>

</Request>

The purpose of Smart Swipe is that if the ECR tells the Pinpad to start a transaction, and the cardholder swiped a card that is **NOT** a credit card, the Pinpad will **not** return an error saying that card is unknown. Instead of returning an error, the Pinpad will change the response to that of Swipe command response.

Some retailers need this option. They want to start a transaction, but then, if the cardholder swipes his loyalty card (club card) then the details of the loyalty card will return to the ECR. The ECR will do what it needs to do with the loyalty card details, sometimes giving a discount to the cardholder, and then the ECR will start again the transaction, possibly with a lower price due to the discount.

The **Smart Swipe** feature helps the merchant. Instead of supporting two commands to handle different cards, the Swipe Command and the Payment Command, the merchant will have to support only the Payment Command. The merchant will not have to select the appropriate command according to the card in the hand of the cardholder. Smart Swipe handles all types of cards, whether credit cards or non-credit cards, thus making the life of the merchant easier.

Example

The ECR sends the following XML, to perform a regular transaction. The ECR doesn't care if the cardholder is going to swipe a credit card or a non-credit card.

```
<Request>
<Command>001</Command>
<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>
<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>1</TranType>
<Amount>10000</Amount>
<Currency>376</Currency>
<PanEntryMode>PinPad</PanEntryMode>
<Xfield>20398049823</Xfield>
</Request>
```

If the cardholder swiped a valid credit card then the response will be the regular command 001 response.

If The cardholder swipes a loyalty card (a non credit card), the response depends on the <EnableSmartSwipe> parameter.

If Smart Swipe is disabled the response XML will be something like that (the AshStatus may be different):

<EMV_Output>

<AshStatus>12</AshStatus> (כרטיס לא מורשה במסוף)

…

More tags may exist in the response

…

</EMV_Output>

But if Smart Swipe **is** enabled then the Pinpad will return the track2 of the card, like this:

<Response>

<Command>023</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<ResultCode>0</ResultCode>

<RequestId>23049820984</RequestId>

<Track2>4580458045804580230298304294809З</Track2>

</Response>

The ECR needs to identify the response, according to the Command (023) and parse it accordingly.

10. Retrieve TOTAL File

The TOTAL file contains data about (and only) the **last deposit** to Shva. It is in XML format, as described in Shva Mefizim-EMV document. It is the same XML format as in STATIS (you can see the XML format also here in chapter 10.2).

The ECR can parse the TOTAL response and then print the summary of the last deposit. Pay attention that the TOTAL file (as well as the STATIS file) contain only a summary of the transactions. There is **no** detailed information about each transaction.

Instead of using the TOTAL file, you can use the Retrieve Deposit Report to get the deposit report that was created by Shva. See the chapter about Deposit Report.

Switch Support

When using a Switch, the TOTAL file is irrelevant because transactions are not kept in the Pinpad and the Pinpad is not responsible for making the deposit, therefore it is not possible to ask the Pinpad for the last deposit when working with as Switch.

10.1 TOTAL Request

```
<Request>
<Command>005</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<RequestId>20160216134455</RequestId>
</Request>
```

Caspit Internal (This part is intended for Caspit developers)

Command flow:

- When SmartRetail gets this request XML, it will:
  - Delete the TOTAL.ASH file from the HOST folder (if it exists)
  - Convert the XML to TLV
  - Then it will send the TLV to Ashrait EMV (using IAC).
- Ashrait EMV will prepare the TOTAL.ASH file in the HOST folder and will return the control to SmartRetail. The TOTAL.ASH file is a XML file (see Ashrait EMV spec for more details).
- SmartRetail will:
  - Read the TOTAL.ASH file from the HOST folder
  - Parse the first line of TOTAL.ASH and make sure that the RequestId from the first line is the same RequestId that was sent to Ashrait EMV in TLV 051.
  - Build the response XML. SmartRetail just needs to read the TOTAL.ASH file and insert the file text into the response XML (discarding the first line of the file that contains the RequestId). SmartRetail doesn't need to parse the TOTAL.ASH file. It will just concatenate it to the response XML.
- SmartRetail will return the response XML back to the PC ECR.

Request XML tags:

Tag name	Description	Matching TLV tag (Caspit internal)

Tag name	Description	Matching TLV tag (Caspit internal)
Request	The opening tag of the request XML	N/A
Command	פקודה. לבקשת קובץ TOTAL, מספר הפקודה היא 005 The command number for TOTAL request is 005	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה Station number (ECR number)	040
RequestId	מזהה בקשה Request Id	051
Request	Closing tag of the request XML	N/A

10.2 /HOST/TOTAL.ASH file format (Caspit Internal)

```
20160215164322                     (this is the first line that contains the RequestId)
<StaticObj>
    <Date>2015-02-25 13:55:06</Date>
    <Session-Number>4449948</Session-Number> (אסמכתא של שב"א)
    <File-Number>19</File-Number> (Batch number, מספר קובץ)
    <File-Open-Date>2015-02-25 00:00:00</File-Open-Date>
    <All-Debit-Num-Recs>2</All-Debit-Num-Recs> (כמות עסקאות חיוב)
    <All-Debit-Sums>2000</All-Debit-Sums> (סכום עסקאות חיוב)
    <All-Credit-Num-Recs>0</All-Credit-Num-Recs>
    <All-Credit-Sums>0</All-Credit-Sums>
    <Solek-List>
        <Solek id="1">   (The <Solek> element will repeat for each solek)
            <Currency-Sums-List>
                <Currency Code="376"> (The <Currency> element will repeat for each currency)
                    <Debit-Num-Recs>2</Debit-Num-Recs>
                    <Debit-Sums>2000</Debit-Sums>
                    <Credit-Num-Recs>0</Credit-Num-Recs>
                    <Credit-Sums>0</Credit-Sums>
                </Currency>
            </Currency-Sums-List>
        </Solek>
    </Solek-List>
</StaticObj>
```

10.3 Total Response

```
<Response>
<Command>005</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<RequestId>20160216134455</RequestId>
<ResultCode>0</ResultCode>
(All the <StaticObj> element, until </StaticObj> is read from the TOTAL.ASH file)
<StaticObj>
    <Date>2015-02-25 13:55:06</Date>
    <Session-Number>4449948</Session-Number> (אסמכתא של שב"א)
    <File-Number>19</File-Number> (מספר קובץ)
    <File-Open-Date>2015-02-25 00:00:00</File-Open-Date>
    <All-Debit-Num-Recs>2</All-Debit-Num-Recs> (כמות כללית עסקאות חיוב)
    <All-Debit-Sums>2000</All-Debit-Sums> (סכום כללי עסקאות חיוב)
    <All-Credit-Num-Recs>0</All-Credit-Num-Recs> (כמות כללית עסקאות זיכוי)
    <All-Credit-Sums>0</All-Credit-Sums> (סכום כללי עסקאות זיכוי)
    <Solek-List>
        <Solek id="1"> (The <Solek> element will repeat for each solek)
            <Currency-Sums-List>
                <Currency Code="376"> (The <Currency> element will repeat for each currency)
                    <Debit-Num-Recs>2</Debit-Num-Recs>
                    <Debit-Sums>2000</Debit-Sums>
                    <Credit-Num-Recs>0</Credit-Num-Recs>
                    <Credit-Sums>0</Credit-Sums>
                </Currency>
            </Currency-Sums-List>
        </Solek>
    </Solek-List>
</StaticObj>
</Response>
```

Response XML tags (the tags that are above the <StatisObj> element)

Tag name	Description	Matching TLV tag (Caspit internal)
Response	The opening tag of the response XML	N/A
Command	פקודה. לבקשת קובץ TOTAL, מספר הפקודה היא 005 Should be the same as in the request	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה Should be the same as in the request	040
RequestId	מזהה בקשה Should be the same as in the request	051
ResultCode	0 – Success 10106 – No TOTAL file See Appendix A for other result codes	010
Response	Closing tag of the response XML	N/A

11. Retrieve STATIS File

The STATIS file is a XML file that contains data about all the deposits that the Pinpad did. The Pinpad will keep up to 99 deposits in history and so the Pinpad may have up to 99 STATIS records.

Switch Support

When using a Switch, the STATIS file is irrelevant because transactions are not kept in the Pinpad.

11.1 STATIS Request

```
<Request>
<Command>014</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<CurrentRecord>000</CurrentRecord>
<RequestId>20160215173042</RequestId>
</Request>
```

Caspit Internal – This part is intended for Caspit Developers.

Command flow:

- SmartRetail will check if <CurrentRecord> is 0 (zero). If it is zero then SmartRetail will:
  - Delete STATIS.ASH file if it exists in the HOST folder
  - Convert the XML to TLV
  - Send the TLV to Ashrait EMV.
- Ashrait EMV will prepare the STATIS.ASH file in the HOST folder and will return to SmartRetail. The STATIS.ASH file is a XML file. See Ashrait EMV spec for more details. But you can see it also in the STATIS response down below. It is the same XML from TOTAL request but it repeats for each deposit that the Pinpad has data about. If the Pinpad has, for example, 73 deposits in history, then the element <StatisObj> will repeat for 73 times inside the STATIS.ASH file.
- The first line of the STATIS.ASH will contain the RequestId.
- If the <CurrentRecord> is not 0 (zero) then there is no need to speak with Ashrait EMV, because the STATIS.ASH should already be in the HOST folder (of course we must check that it exists in the HOST folder, but I don't get into this details).
- SmartRetail will read the STATIS.ASH file from the HOST folder and will build the response XML.
- SmartRetail must compare the RequestId from the request XML with the RequestId from the first line of the STATIS.ASH file. They must be identical.
- The response XML will contain **only one** <StatisObj> each time. It means that the PC ECR will need to continue requesting for the rest of the STATIS records.
- SmartRetail will return the response XML back to the PC ECR.

Request XML tags:

Tag name	**Description**	**Matching TLV tag** **(Caspit internal)**

Tag name	Description	Matching TLV tag (Caspit internal)
Request	The opening tag of the request XML	N/A
Command	פקודה. לבקשת קובץ STATIS, מספר הפקודה היא 014 The command for SSTATIS request is 014	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה Station number (ECR number)	040
CurrentRecord	רשומת STATIS מבוקשת. הערכים האפשריים הם 01 – 99. אבל בבקשה הראשונה צריך לשלוח ערך 00 כדי להודיע לפינפד שרוצים להתחיל לקבל קובץ STATIS The request STATUS record. The possible value are 01-99 But on the first request, the ECR must send 00 to tell the Pinpad that we want to start getting the STATIS file.	190
RequestId	The request Id must be the same during all the STATIS retrieval session.	051
Request	Closing tag of the request XML	N/A

11.2 STATIS Response

<Response>

<Command>014</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<CurrentRecord>01</CurrentRecord>

<NumberOfRecords>73</NumberOfRecords>

<RequestId>20160215173042</RequestId>

<ResultCode>0</ResultCode>

(All the <StaticObj> element, until </StaticObj) is read from the TOTAL.ASH file)

<StaticObj>

<Date>2015-02-25 13:55:06</Date>

<Session-Number>4449948</Session-Number>

<File-Number>19</File-Number>

<File-Open-Date>2015-02-25 00:00:00</File-Open-Date>

<All-Debit-Num-Recs>2</All-Debit-Num-Recs>

<All-Debit-Sums>2000</All-Debit-Sums>

<All-Credit-Num-Recs>0</All-Credit-Num-Recs>

<All-Credit-Sums>0</All-Credit-Sums>

<Solek-List>

<Solek id="1"> **(The <Solek> element will repeat for each solek)**

<Currency-Sums-List>

<Currency Code="376"> **(The <Currency> element will repeat for each currency)**

<Debit-Num-Recs>2</Debit-Num-Recs>

<Debit-Sums>2000</Debit-Sums>

<Credit-Num-Recs>0</Credit-Num-Recs>

<Credit-Sums>0</Credit-Sums>

<Currency/>

<Currency-Sums-List/>

<Solek/>

<Solek-List/>

<StaticObj/>

</Response>

Response XML tags, above the <StatisObj> element:

Tag name	Description	Matching TLV tag (Caspit internal)
Response	The opening tag of the response XML	N/A
Command	פקודה. לבקשת קובץ STATIS, מספר הפקודה היא 014 Should be the same as in the request	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה Should be the same as in the request	040

Tag name	Description	Matching TLV tag (Caspit internal)
CurrentRecord	מספר הרשומה הנוכחית. שימו לב שמספר הרשומה לא חייב להיות חופף למספר הקובץ של שב"א כפי שהוא מופיע בXML. יתכן בהחלט שברשומה מספר 1, יהיה קובץ מספר 54. The current record number. Pay attention that the record number doesn't match the Shva file number. It is possible that in record number 1, there will be the data of file number 54.	190
NumberOfRecords	מספר הרשומות הקיימות בקובץ הSTATIS. The number of records exist in the STATIS file. The STATIS file can hold up to 99 records, which are deleted and replaced in cyclic method.	191
RequestId	מזהה בקשה Should be the same as in the request	051
ResultCode	See Appendix A	010
Response	Closing tag of the response XML	N/A

12. Retrieve TRAN File

- The TRAN file is a binary file that contains data about all the transactions that exist in a specific deposit.
- The TRAN file is kept in the secure memory of the Pinpad.
- The Pinpad keeps the TRAN files of the last 99 deposits. Older TRANS are deleted in a cyclic manner.
- The **current** TRAN can also be retrieved. The **current** is the TRAN that holds the transactions before they are transmitted to Shva.
- Some retailers like to get the **current TRAN** before transmitting it to Shva. They go over all the transactions and compare them to the transactions saved in the ECR database. If there is a need, they may decide to cancel one or more transactions. It is possible to cancel a transaction using its Uid, so if you find one or more transactions in the TRAN file that you would like to cancel, you can parse the Uid of these transactions and send cancellation requests for each transaction you want to cancel.
- Some retailers also like to retrieve all the transactions from the TRAN file, after transmission to Shva, because they need to compare between the records of the ECR and the actual records in the Pinpad.
- Each record in the TRAN file is formatted according to appendix 1 in Shva document (ashEMV_appx). In a nutshell, each record has a header and then the rest of the record is in ISO8583. The PC ECR will need to be able to parse ISO8583.
- Because we use XML and the records are in binary format, each record will be encoded in Base64 encoding.
- Caspit will provide an ISO8580 parser and a Base64 decoder in the DLL.
- When you retrieve a TRAN, the original stays in the Pinpad. It is not deleted.

<u>Switch Support</u>

When using a Switch, the TRAN file is irrelevant because transactions are not kept in the Pinpad

12.1 TRAN Request

<Request>

<Command>007</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<FileNo>26</FileNo>

<CurrentRecord>000</CurrentRecord>

<RecordsPerRequest>5</RecordsPerRequest>

<RequestId>20160215174522</RequestId> **(this same RequestId must repeat for the whole session)**

</Request>

Caspit Internal – This part is intended for Caspit developers.

Command flow:

- If the <CurrentRecord> is 0 (zero) then SmartRetail will:
  - Convert the XML to TLV and then it will send the TLV to Ashrait EMV.
  - Ashrait EMV will prepare the TRAN.ASH file in the HOST folder and will return to SmartRetail.
  - The TRAN.ASH file is a text file with a line for each transaction that exists in the deposit (each deposit is identified by a batch number/file number). See Ashrait EMV spec for more details.

  - Ashrait EMV will put the RequestId as the first line of the TRAN.ASH file
- If the <CurrentRecord> is not 0 (zero) then there is no need to speak with Ashrait EMV, because the TRAN.ASH should already be in the HOST folder.
- SmartRetail will read the TRAN.ASH file from the HOST folder and will build the response XML. The response XML will contain one ore more records in each response. It means that the PC ECR will need to continue requesting for the rest of the TRAN records.
- SmartRetail will return the response XML back to the PC ECR.

Request XML tags:

Tag name	**Description**	**Matching TLV tag** **(Caspit internal)**
Request	The opening tag of the request XML	N/A
Command	Command number. To get TRAN, use command number 007	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה\תחנה\עמדה Station number (ECR number)	040
FileNo	מספר קובץ TRAN. ערכים אפשריים 00-99 00 – מציין את העסקאות הקיימות במכשיר שעדיין לא שודרו לשב"א The file number. Possible values are 00 – 99 Use 00 to get the transaction in the current file.	016
CurrentRecord	רשומה מבוקשת. הערכים האפשריים הם 001 – 999. **אבל בבקשה הראשונה חובה לשלוח ערך 000 כדי להודיע לפינפד שרוצים להתחיל לקבל קובץ TRAN** The requested record number. The possible values are 001 – 999 But at first, you must send 000 to tell the Pinpad that we want to start getting the TRAN file.	191

Tag name	Description	Matching TLV tag (Caspit internal)
RecordsPerRequest	• Default is 1 • Max value: 5 • If value is more than 5 then the Pinpad will make it 5. • If value is less that 1, the Pinpad will make it 1. • This tag set the number of records that will return in the response for each request. • For example, if the TRAN has 50 transactions (i.e. records) then the ECR can ask to get one record for each request, so a total number of 50 requests (and responses) must be done to retrieve the complete TRAN. • But the ECR may choose to get 5 records for each request and thus the total number of requests will be reduced to 10. • Another example, if the <CurrentRecord> is 6 and the <RecordsPerRequest> is 5 then the ECR is asking to get 5 records starting from record 6. • If the ECR requests 5 records but only 3 records are left in the TRAN file then the Pinpad will return only 3 records. • The maximum number of 5 records is imposed because of the Pinpad buffer constraints.	
RequestId	**The same identical request Id must be used for all the session of TRAN retrieval. i.e. if the TRAN retrieval requires more than one request (and this is mostly the case) then all subsequent request will use the same request Id.**	051
Request	Closing tag of the request XML	N/A

12.2 /HOST/TRAN.ASH (Caspit Internal)

TODO: Add description of TRAN file

12.3 TRAN Response

<Response>

<Command>007</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<FileNo>26</FileNo>

<RecordFrom>001</RecordFrom>

<RecordTo>005</RecordTo>

<TotalNumberOfRecords>73</TotalNumberOfRecords>

<Record> record text in Base64 </Record>

<Record> another record text in Base64</Record>

<Record> another record text in Base64</Record>

<Record> another record text in Base64</Record>
<Record> another record text in Base64</Record>
<Base64>Yes</Base64>
<ResultCode>0</ResultCode>
<RequestId>20160215174522</RequestId>
</Response>

Response XML tags

Tag name	Description	Matching TLV tag (Caspit internal)
Response	The opening tag of the response XML	N/A
Command	פקודה. לבקשת קובץ TRAN, מספר הפקודה היא 007 Should be the same as in the request	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה Should be the same as in the request	040
RecordFrom	The records in the response are starting from this record number (including) The first record is number 1.	190
RecordTo	The records in the response are ending in this record number (including)	
TotalNumberOfRecords	מספר הרשומות הקיימות בקובץ הTRAN. The total number of records that exist in the TRAN file	191
Record	The transaction details in ISO8583 format. The record will be encoded in Base64 (depending on tag <Base64>)	N/A
Record	This tag can repeat 5 times	
Base64	Yes – Records are encoded in Base64 No – Records are NOT encoded in Base64	
ResultCode	0 – Success 10006 - RETVAL_CANT_FIND_REQUESTED_TRAN_FILE 10039 - RETVAL_INVALID_TRAN_RECORD_NUMBER Else, See Appendix A	010
RequestId	Must be the same as in the request	051

Tag name	Description	Matching TLV tag (Caspit internal)
Response	Closing tag of the response XML	N/A

12.4 Example of getting <u>current</u> TRAN

PC sends the following request:

<Request>

<Command>007</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<FileNo>00</FileNo> **(FileNo is 00 for current TRAN)**

<CurrentRecord>000</CurrentRecord> **(Current record must be 000 for the first request)**

<RecordsPerRequest>5</RecordsPerRequest>

<RequestId>20160215174522</RequestId> **(Same request id will be used for all the session of TRAN retrieval).**

</Request>

Lets' assume the Pinpad has a current TRAN with 8 transactions that have not been transmitted to Shva.

Pinpad responds with first 5 records from the current TRAN:

<Response>

<Command>007</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<FileNo>00</FileNo>

<RecordFrom>001</RecordFrom>

<RecordTo>005</RecordTo>

<TotalNumberOfRecords>8</TotalNumberOfRecords>

<Record> A5FFAE234534645798735333AE23459586BBC9D0A5FFAE23459586BBC9D0 </Record>

<Record> A5FFAE23459586BBC9D0A5FFAE2345958242424242424234586BBC9D0 </Record>

<Record> A5F23423BBC9D0A5FFAE23459586BBC9D0A5FFAE23459586BBC9D0 </Record>

<Record> A5FFAE23459586BB234242423423423959586BBC9D0A5FFAE23459586BBC9D0 </Record>

<Record> A5FFAE23459586BBC9D0A5FFAE234567567567567567567575759586BB3454D0 </Record>

<Base64>Yes</Base64>

<ResultCode>0</ResultCode>

<RequestId>20160215174522</RequestId>

</Response>

In the response we can see that we got the first 5 records and that there are a total of 8 records (<TotalNumberOfRecords>8</TotalNumberOfRecords>)

We have 5 times the <Record> tag.

The ECR will do the following for **<u>each</u>** <Record> tag in the response:

1. The <Record> comes encoded with a Base64 encoding. So the ECR must decode the <Record>. Caspit DLL includes a method to decode Base64.
2. The result of the Base64 decoding is a binary byte array. This array is in ISO8583 format.
3. The Caspit DLL have a method to parse ISO8583. The ECR may use this method to parse the record.
4. The Caspit DLL method will return a map object with all the ISO8583 fields.
5. The ECR may save this map.
6. The ECR now moves to the next <Record> and handle it also, and so on until all <Record> tags are parsed

After all 5 <Record> tags are parsed, the ECR can send a request to the Pinpad asking to get the remaining records.

Here is the request:

<Request>

<Command>007</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<FileNo>00</FileNo> **(FileNo is 00 for current TRAN)**

<CurrentRecord>006</CurrentRecord> **(start from record 6)**

<RecordsPerRequest>5</RecordsPerRequest>

<RequestId>20160215174522</RequestId> **(<u>same request Id</u>)**

</Request>

The response of the Pinpad will be like that:

<Response>

<Command>007</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<FileNo>00</FileNo>

<RecordFrom>006</RecordFrom>

<RecordTo>008</RecordTo>

<TotalNumberOfRecords>8</TotalNumberOfRecords>

<Record> A5FFAE23453464576945634538BBC9D0A5FFAE23459586BBC9D0 </Record>

<Record> A5FFAE23459586BBC9D0A5FFAE2345645632234234242345 86BBC9D0 </Record>

<Record> A5F23423BBC9D0A34536576853C9D0A5FFAE23459586BBC9D0 </Record>

<Base64>Yes</Base64>

<ResultCode>0</ResultCode>

<RequestId>20160215174522</RequestId>

</Response>

Again, the ECR will do the following for each <Record> tag in the response:

1. The <Record> comes encoded with a Base64 encoding. So the ECR must decode the <Record>. Caspit DLL includes a method to decode Base64.
2. The result of the Base64 decoding is a binary byte array. This array is in ISO8583 format.
3. The Caspit DLL have a method to parse ISO8583. The ECR may use this method to parse the record.
4. The Caspit DLL method will return a map object with all the ISO8583 fields.
5. The ECR may save this map.
6. The ECR now moves to the next <Record> and handle it also, and so on until all <Record> tags are parsed

This concludes the retrieval of 8 record from the current TRAN.

Of course, if the TRAN had more records, then more requests and responses would be needed to get the entire TRAN.

12.5 Example of getting a specific TRAN

To get a specific TRAN, the flow is almost the same, you just need to put the specific TRAN number in the <FileNo> tag. In the following example, I ask to get TRAN number 43. I show here just the request.

<Request>

<Command>007</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<FileNo>**43**</FileNo>

<CurrentRecord>000</CurrentRecord> **(Current record must be 000 for the first request)**

<RecordsPerRequest>5</RecordsPerRequest>

<RequestId>20160315174522</RequestId> **(Same request id will be used for all the session of TRAN retrieval).**

</Request>

If you don't know the TRAN number, you can use the STATIS request to get the STATIS file. The STATIS file contains the list of all the TRANs that exist in the Pinpad. If you find what you are looking for in the STATIS file, you can get the TRAN number from there and then use it with the Retrieve TRAN command.

13. Retrieve JENR File

The JENR file is a text file that contains mostly the last communication details and fields that specify the time and generation of the data retrieved from Shva.

SmartRetail will support the same JENR file format that is described in Shva Mefizim-EMV document (see that document). All of the JENR file data will arrive in XML tag <JENR>. Actually the JENR will be one long record.

JENR file format:

שם השדה	מס' תווים	מקום יחסי	הערות
מספר מסוף	10	1	
דור חסומים בטנדם	4	11	
תאריך פרמטרים כלליים	8	15	YYYYMMDD
תאריך פרמטרים לכרטיסים ישראלים	8	23	YYYYMMDD
תאריך פרמטרים לתיירים	8	31	YYYYMMDD
תאריך תקפים לישראכרט	8	39	YYYYMMDD
תאריך תקפים לכ.א.ל	8	47	YYYYMMDD
תאריך תקפים למנפיקים עתידיים	10 * 8	55	YYYYMMDD 80 תווים
תאריך מועדון ישראכרט	8	135	YYYYMMDD
תאריך מועדון לאומי קארד	8	143	YYYYMMDD
תאריך מועדון כ.א.ל	8	151	YYYYMMDD
תאריך פרמטרים של מסופי דלק	8	159	לא בשימוש
תאריך תקרות לכרטיסי PL	8	167	YYYYMMDD
תאריך תקרות לשיטה 0 לכרטיסים ישראלים שאינם PL	8	175	YYYYMMDD
תאריך תקרות לשיטה 1 לכרטיסים ישראלים שאינם PL	8	183	YYYYMMDD
תאריך תקרות לכרטיסי תייר	8	191	YYYYMMDD
דור ווקטורי ישראכרט	4	199	
דור ווקטורי כ.א.ל	4	203	
דור ווקטורי לאומי קארד	4	207	
דור ווקטורים כלליים	4	211	
דור ווקטור 90	4	215	
דור ווקטורים למנפיקים עתידיים	4	219	
קוד המשך שידור קובץ חסומים	1	223	0 – שידור מהתחלה 1 – המשך שידור קובץ מלא

2 – המשך שידור קובץ ביטולים 3 – המשך שידור קובץ תוספות			

13.1 JENR File Request

<Request>

<Command>002</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<RequestId>20160216153432</RequestId>

</Request>

Caspit Internal – This part is intended for Caspit developers.

Command flow:

- If SmartRetail gets this request XML, it will convert the XML to TLV and then it will send the TLV to Ashrait EMV.
- Ashrait EMV will prepare the JENR.ASH file in the HOST folder and will return to SmartRetail. The JENR.ASH file is a plain text file. On the first line of JENR.ASH file, Ashrait EMV will put the RequestId.
- SmartRetail will read the JENR.ASH file from the HOST folder and will build the response XML.
- SmartRetail will validate that the RequestId on the first line of the file is the same RequestId from the request.
- SmartRetail just needs to read the JENR.ASH file and insert the file text into the response XML (without the first line). SmartRetail doesn't need to parse the JENR.ASH file.
- SmartRetail will return the response XML back to the PC ECR.

Request XML tags:

Tag name	**Description**	**Matching TLV tag** **(Caspit internal)**
Request	The opening tag of the request XML	N/A
Command	פקודה. לבקשת קובץ JENR, מספר הפקודה היא 002 The command for JENR file is 002	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה Station number (ECR number)	040
RequestId	מזהה בקשה Request Id	051

Tag name	Description	Matching TLV tag (Caspit internal)
Request	Closing tag of the request XML	N/A

13.2 JENR File Format (Caspit Internal)

20160216153432 **(first line is the RequestId)**

(The JENR line. 224 characters long)

13.3 JENR File Response

<Response>

<Command>002</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<ResultCode>0</ResultCode>

<RequestId>20160216153432</RequestId>

<JENR> here will be the text of the JENR file. It is 224 characters long </JENR>

</Response>

Response XML tags:

Tag name	Description	Matching TLV tag (Caspit internal)
Response	The opening tag of the response XML	N/A
Command	פקודה. לבקשת קובץ JENR, מספר הפקודה היא 002 Should be the same as in the request	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה Should be the same as in the request	040
ResultCode	See Appendix A	010
RequestId	מזהה בקשה Should be the same as in the request	051
JENR	The text of the JENR file 224 characters long	N/A
Response	Closing tag of the response XML	N/A

14. Retrieve DATA File

The DATA file is a text file that keeps information about some important details.

SmartRetails supports the same DATA file format that is described in Shva Mefizim-EMV document (see that document).

All of the DATA file data will arrive in XML tag <DATA>. So actually, the DATA file is a one long string.

DATA file format:

שם השדה	מס' תווים	מקום יחסי	הערות
תאריך תנועה ראשונה	8	01	
תאריך שידור תנועות אחרון	8	09	
תאריך עדכון חסומים	8	17	
מספר קובץ שידור תנועות	2	25	משתנה מ- 01-99 בכל שידור מוצלח של תנועות.
FILLER	1	27	רווחים
הפרש בדורות חסומים	4	28	במקרה של ניתוק בקבלת חסומים, מראה את ההפרש בין דורות של קובץ חסומים במסוף לבין המחשב המרכזי של שב"א
FILLER	1	32	
קוד סיום שידור	1	33	0 – תקין 1 – לא התחבר 2 – ניתוק בפרמטרים 3 – ניתוק בתנועות 4 – ניתוק בחסומים 5– ניתוק בעדכון תוכנה
FILLER	13	34	רווחים

15. Transmit Transactions to Shva

The PC ECR can request the Pinpad to transmit all transaction to Shva.

By default, the PinPad keeps all transactions in its secure memory. Transactions are kept until the ECR decides to transmit the transactions to Shva. If the ECR does not tell the Pinpad to transmit the transactions, the transactions will accumulate in the Pinpad indefinitely. In most cases, the ECR tells the Pinpad to transmit the transactions during the ECR end-of-day procedure.

However, the Pinpad can be configured to send the transactions automatically every day. That way, you can be sure that the transactions are safely transmitted to Shva. However, using the automatic feature, you lose the synchronization between the ECR end-of-day process and the Pinpad end-of-day.

During the call to Shva, not only the transactions are sent to Shva, but the Pinpad also gets updated parameters and an updated black list.

After transmitting the transactions, the ECR may use one or more of the following features:

- Get Total file to see the summary of transactions
- Get Deposit Report to see (and print) the summary report that was created by Shva
- Get TRAN file (record after record) to get the full details of all the transactions
- Using or not using these features, it all depends on the requirements of the ECR.

<u>Switch Support</u>

If the Pinpad is working with a Switch then this command, Transmit Transactions to Shva, will have a different meaning. It will mostly be useless, unless some transactions are stuck in the Pinpad and there is a need to send them manually to the Switch.

In a regular flow with the Switch, transactions will be kept in the Switch, but it is possible, that because of a communication problem between the Pinpad and the Switch, transactions will be authorized **offline** by the Pinpad. Usually, when communication is restored, the Pinpad will automatically send the transactions, that were authorized offline, to the Switch. But to make sure that there are no "stuck" transactions in the Pinpad, the ECR may call this command.

15.1 Transmit Transactions Request

```
<Request>
<Command>006</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<RequestId>20160216155540</RequestId>
</Request>
```

Caspit Internal – This part is intended for Caspit developers.

Command flow:

- SmartRetail will convert the XML to TLV and then it will send the TLV to Ashrait EMV.
- Ashrait EMV will call Shva and will transmit all transaction to Shva. During the call, Ashrait will update also the parameters and black list.
- Ashrait will prepare the DATA.ASH file in the HOST folder and will return to SmartRetail. The DATA.ASH file is a plain text file. It is the same DATA.ASH file that is created during DATA file request command.

- SmartRetail will read the DATA.ASH file from the HOST folder and will build the response XML. SmartRetail just needs to read the DATA.ASH file and insert the file text into the response XML. SmartRetail doesn't need to parse the DATA.ASH file.
- SmartRetail will return the response XML back to the PC ECR.

Request XML tags:

Tag name	**Description**	**Matching TLV tag** **(Caspit internal)**
Request	The opening tag of the request XML	N/A
Command	פקודה. לביצוע שידור לשב"א, מספר הפקודה היא 006 The command to Transmit transaction is 006	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה Station number (ECR number)	040
RequestId	מזהה בקשה Request Id	051
Request	Closing tag of the request XML	N/A

15.2 Transmit Transactions Response

```
<Response>
<Command>006</Command>
<TerminalId>0880264</TerminalId>
<ResultCode>0</ResultCode>
<Session-Number>29238392</Session-Number>
<File-Number>17</File-Number>
<Date>YYYY-MM-DD</Date>
<RequestId>20160216155540</RequestId>
<DATA> here will be the text of the DATA file. It is 47 characters long </DATA>
</Response>
```

The ECR should keep the <Session-Number>, <File-Number> and <Date> tags for future use.

These tags (<Session-Number>, <File-Number> and <Date>) will exist in the response XML only if transactions were sent to Shva.

Switch Support

If using a Switch, the response will **not include** the following tags:

- <Session-Number>
- <File-Number>
- <Date>
- <DATA>

Response XML tags:

Tag name	Description	Matching TLV tag (Caspit internal)
Response	The opening tag of the response XML	N/A
Command	פקודה. לביצוע שידור לשב"א, מספר הפקודה היא 006 Should be the same as in the request	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה Should be the same as in the request	040

Tag name	Description	Matching TLV tag (Caspit internal)
ResultCode	Relevant result codes: 0 – Success 10003 – RETVAL_INCORRECT_TERMINAL_ID 10022 – RETVAL_INCORRECT_ECR_NUMBER (<TermNo> tag) 10036 – RETVAL_XML_MISSING_TERMINAL_ID_TAG Other result codes may arrive. Refer to Appendix A.	010
Session-Number	If transactions were sent successfully, the <Session-Number> will hold the Asmachta from Shva (it is the reference of the deposit). This tag will exist in the response XML only if there were transactions sent to Shva.	015
File-Number	If transactions were sent successfully, then <File-Number> will hold the batch/file number of the transactions batch. This tag will exist in the response XML only if there were transactions sent to Shva.	016
Date	If transactions were sent successfully, then <Date> will hold the current date. Format: YYYY-MM-DD This tag will exist in the response XML only if there were transactions sent to Shva.	213
RequestId	Must be the same as in the request.	
DATA	The text of the DATA file 47 characters long	The value of this tag will be taken from file DATA.ASH
Response	Closing tag of the response XML	N/A

15.3 DATA File Request

```
<Request>
<Command>008</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<RequestId>20160216154920</RequestId>
</Request>
```

Caspit Internal – This part is intended for Caspit developers.

Command flow:

- If SmartRetail gets this request XML, it will convert the XML to TLV and then it will send the TLV to Ashrait EMV.
- Ashrait EMV will prepare the DATA.ASH file in the HOST folder and will return to SmartRetail.
- The DATA.ASH file is a plain text file. Ashrait EMV will add the RequestId on the first line of the file.
- SmartRetail will read the DATA.ASH file from the HOST folder and will build the response XML. SmartRetail just needs to read the DATA.ASH file and insert the file text into the response XML (without the first line, which is the RequestId). SmartRetail doesn't need to parse the DATA.ASH file.
- SmartRetail will return the response XML back to the PC ECR.

Request XML tags:

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Request	The opening tag of the request XML	N/A
Command	פקודה. לבקשת קובץ DATA, מספר הפקודה היא 008 The command file DATA file request is 008	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה Station number (ECR number)	040
RequestId	מזהה בקשה Request Id	051
Request	Closing tag of the request XML	N/A

15.4 DATA File Format (Caspit Internal)

20160216154920 **(first line is the RequestId)**

(The DATA line. 47 characters long)

15.5 DATA File Response

<Response>

<Command>008</Command>

<TerminalId>0880264</TerminalId>

<ResultCode>0</ResultCode>

<RequestId>20160216154920</RequestId>

<DATA> here will be the text of the DATA file. It is 47 characters long </DATA>

</Response>

Response XML tags:

Tag name	Description	Matching TLV tag (Caspit internal)
Response	The opening tag of the response XML	N/A
Command	פקודה. לבקשת קובץ DATA, מספר הפקודה היא 008 Should be the same as in the request	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה Should be the same as in the request	040
ResultCode	See Appendix A	010
RequestId	מזהה בקשה Should be the same as in the request	051
DATA	The text of the DATA file 47 characters long	N/A
Response	Closing tag of the response XML	N/A

16. Communication Test

The PC ECR can make communication tests to test the connection to the Pinpad itself and also to test the various hosts that the Pinpad uses.

Use this command to check if the ECR is configured with the correct Terminal Id and TermNo (ECR number/station number).

16.1 Communication Test Request

<Request>

<Command>003</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<CheckShva>1</CheckShva>

<CheckSwitch>0</CheckSwitch>

<CheckCaspitHost>0</CheckCaspitHost>

<CheckTMS>0</CheckTMS>

<CheckCertServer>0</CheckCertServer>

<CheckReportServer>0</CheckReportServer>

<CheckFTP>0</CheckFTP>

<TimeoutInSeconds>10</TimeoutInSeconds>

<RequestId>20160216155540</RequestId>

</Request>

If you just want to test the connection to the Pinpad (and check if there is card inside the reader) then you can send the command like this:

<Request>

<Command>003</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<RequestId>20160216155540</RequestId>

</Request>

Caspit Internal – This part is intended for Caspit developers.

Command flow:

- SmartRetail will convert the XML to TLV and then it will send the TLV to Ashrait EMV.
- Ashrait EMV will:

- If <CheckShva> is 1 then Ashrait EMV will perform a communication test all the way to Shva. Ashrait EMV will just open a TCP socket to see if the connection is OK. There is no need to send/receive anything from Shva. If the connection to Shva requires SSL, then the check will include a complete connection (including the SSL).
- Check the status of the EPA application (Contact EMV application)
- Check the status of the CPA application (Contactless application)
- Ashrait EMV must consider the Timeout and return the control to SmartRetail if timeout expired.
- If <CheckSwitch> is 1 then AshraitEMV will also perform a communication test all the way to the Switch. This will be possible only if AshraitEMV is working in **Switch** mode. Only the connection stage will be tested. There is no need to do any parameters session with the Switch. **This feature is not ready yet**
- If <CheckCaspitHost> is 1 then AshraitEMV will also perform a communication test all the way to Caspit Host (the technicians' center). Only the connection stage will be tested. There is no need to do any parameters session.
- If <CheckTMS> is 1 then AshraitEMV will also perform a communication test all the way to the TMS. Only the connection stage will be tested. Note that AshraitEMV will **not** start the regular function that performs a full TMS communication. Ashrait EMV just need to open a TCP socket to the TMS server.
- If <CheckCertServer> is 1 then AshraitEMV will also perform a communication test all the way to the cert server. Only the connection stage will be tested. **No** certificate update will be done. **This feature is not ready yet.**
- If <CheckReportServer> is 1 (and if we have a report server) then AshraitEMV will also perform a communication test all the way to the report server. Only the connection stage will be tested. **No** report request will be done. **This feature is not ready yet.**

- SmartRetail will return the response XML back to the PC ECR.

Request XML tags:

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Request	The opening tag of the request XML	N/A
Command	פקודה. לבדיקת תקשורת, מספר הפקודה היא 006 The command for communication test is 006	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה (מספר עמדה\מספר תחנה). Station number/ECR number	040
CheckShva	0 – Don't check connection to Shva. 1 – Check connection to Shva. Only the connection step is checked (TCP connection and SSL handshake). **No** parameters or transactions are sent. So this check **doesn't** tell us if the Terminal Id is configured correctly in Shva. If you want to make a full communication to Shva (including transactions and parameters) then you need to call the Transmit Transaction Command. If you want to make a partial communication to Shva (only parameters without transactions) then you need to use the Generic Command.	194

Tag name	Description	Matching TLV tag (Caspit internal)
CheckSwitch	0 –Don't check connection to the Switch. 1 – Check connection to the Switch. Only the connection step is checked (TCP connection and SSL handshake). This check **doesn't** tell use if the Terminal Id is configured in the Switch. **Relevant only when using a Switch. <u>This option is not ready yet.</u>** If you want to make a full parameters call to the Switch, then you need to use the Generic Command.	225
CheckCaspitHost	0 – Don't check connection to Caspit 1 – Check connection to Caspit technician server. Only the connection step is checked (TCP connection and SSL handshake). If you want to make a full parameters call to Caspit tech center, then you need to use the Generic Command	226
CheckTMS	0 – Don't check connection to the TMS (Terminal Management System) 1 – Check connection to the TMS. Only the connection step is checked (TCP connection and SSL handshake). **<u>No</u>** downloads will be done. If you want to make a full call to the TMS (including getting upgrades, if available), then you need to use the Generic Command	227
CheckCertServer	0 – Don't check the certificates server. 1 – Check the certificates server. Only the connection step is checked (TCP connection and SSL handshake). **<u>No</u>** certificate will be downloaded. **Relevant only when using a certificates server. <u>This option is not ready yet</u>** If you want to make a full call to the certificates server, including getting new certificates, then you need to use the Generic Command	232
CheckReportServer	0 – Don't check the reports server 1 – Check the reports server. Only the connection step is checked (TCP connection and SSL handshake). **Relevant only when using a reports server. <u>This option is not ready yet.</u>** If you want to make a full call to the reports server, including getting a report, then you need to use the Generic Command	233
CheckFTP	0 – Don't check the FTP server 1 – Check the FTP server. Only the connection is checked (TCP connection). **NO** files will be sent. If you want to make a full call to the FTP server, including sending the log files, then you need to use the Generic Command	228
TimeoutInSeconds	Timeout value for the entire communication test. The SmartRetail must return a response within the requested timeout period.	005

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
RequestId	מזהה בקשה Request Id	051
Request	Closing tag of the request XML	N/A

16.2 Communication Test Response

<Response>
<Command>003</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<SerialNumber>30303PP3039393</SerialNumber>
<SmartRetailVersion>4872830001</SoftwareVersion>
<AshraitEMVVersion>5959590040</AshraitEMVVersion>
<EPAVersion>33939392122</EPAVersion>
<CPAVersion>20202022020</CPAVersion>
<RouterVersion>30303030303</RouterVersion>
<SDKVersion>9.22.0</SDKVersion>
<ResultCode>0</ResultCode>
<ShvaCommStatus>1</ShvaCommStatus>
<SwitchCommStatus>9</SwitchCommStatus>
<CaspitCommStatus>9</CaspitCommStatus>
<TMSCommStatus>9</TMSCommStatus>
<CertServerStatus>9</CertServerStatus>
<ReportServerStatus>9</ReportServerStatus>
<FTPServerStatus>9</CheckFTPStatus>
<ContactStatus>1</ContactStatus>
<ContactlessStatus>1</ContactlessStatus>
<CardInside>0</CardInside>
<RequestId>20160216155540</RequestId>
</Response>

Response XML tags:

Tag name	**Description**	**Matching TLV tag (Caspit intenral)**
Response	The opening tag of the response XML	N/A
Command	פקודה. לבדיקת תקשורת, מספר הפקודה היא 003 Should be the same as in the request	001

Tag name	**Description**	**Matching TLV tag (Caspit intenral)**
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה Should be the same as in the request	040
SerialNumber	Serial number of the Pinpad	N/A There is no TLV tag because it will be done locally by SmartRetail
SmartRetailVersion	Software version of SmartRetail app	N/A There is no TLV tag because it will be done locally by SmartRetail
AshraitEMVVersion	Software version of Ashrait EMV	013
EPAVersion	Software version of EPA (Pinpad EMV Contact application) This is another application on the Pinpad that takes part in the EMV transaction.	192
CPAVersion	Software version of CPA (Pinpad Contactless application) This is another application on the Pinpad that takes part in the EMV transaction.	193
RouterVersion	Software version of the Router application This is an application on the Pinpad that is responsible to route packets between applications and the ECR.	TODO
SDKVersion	SDK version of the Pinpad	N/A There is no TLV tag because it will be done locally by SmartRetail
ResultCode	See Appendix A	
ShvaCommStatus	0 – Connection error 1 – Connection OK 9 – Not checked	195

Tag name	Description	Matching TLV tag (Caspit intenral)
SwitchCommStatus	0 – Connection error 1 – Connection OK 9 – Not checked **<u>This feature is not ready yet</u>**	229
CaspitCommStatus	0 – Connection error 1 – Connection OK 9 – Not checked	230
TMSCommStatus	0 – Connection error 1 – Connection OK 9 – Not checked	231
CertServerStatus	0 – Connection error 1 – Connection OK 9 – Not checked **<u>This feature is not ready yet</u>**	234
ReportServerStatus	0 – Connection error 1 – Connection OK 9 – Not checked **<u>This feature is not ready yet</u>**	235
FTPServerStatus	0 – Connection error 1 – Connection OK 9 – Not checked	236
ContactStatus	0 – Smart card reader is not OK 1 – Smart card reader is Ok 2 – No Smart card reader device	196
ContactlessStatus	0 – Contactless reader is not OK 1 – Contactless reader is OK 2 – No Contactless reader	197
CardInside	0 – No card inside 1 – Card inside contact reader This is an important issue. With EMV, it is very common for the cardholder to forget his card inside the Pinpad. The ECR should always check if the card is still inside and alert the merchant about it.	220

Tag name	Description	Matching TLV tag (Caspit intenral)
RequestId	מזהה עסקה Should be the same as in the request	051
Response	Closing tag of the response XML	N/A

17. Pinpad Configuration

Using the Pinpad configuration command the ECR can set various parameters of the Pinpad. The response XML will include also many interesting information about the Pinpad hardware and software.

17.1 Pinpad Configuration Request

```
<Request>
<Command>013</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<ShvaTime>HHMM</ShvaTime>
<ShvaTimerType>0</ShvaTimerType>
<EnableIdleSwipe>1</EnableIdleSwipe>
<EnableSmartSwipe>1</EnableSmartSwipe>
<BlockDoubleTrans>1</BlockDoubleTrans>
<AccessControl>0</AccessControl>
<EnableFullPAN>0</EnableFullPAN>
<IdleSwipeAddress>192.168.0.1/1200</IdleSwipeAddress>
<EnableTranSteps>0</EnableTranSteps>
<TranStepsAddress>192.168.0.1/1200</TranStepsAddress>
<MustTranSteps>0</MustTranSteps>
<BlockAutoSwitchBatchDelete>0</BlockAutoSwitchBatchDelete >
<PinpadLock>0</PinpadLock>
<RequestId>20160216155540</RequestId>
</Request>
```

If you just want to get the configuration and not to set anything then you can use a shorter XML request:

```
<Request>
<Command>013</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<RequestId>20160216155540</RequestId>
</Request>
```

If the ECR doesn't know the terminal id of the Pinpad, it can send an even shorter XML request:

```
<Request>
```

<Command>013</Command>
<RequestId>20160216155540</RequestId>
</Request>

Without a terminal id, the ECR cannot change the configuration of the Pinpad but the response will be the same response of the Pinpad Configuration.

The request tags

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Request	The opening tag of the request XML	N/A
Command	פקודה. להגדרת פינפד, מספר הפקודה היא 013 The command for Pinpad Configuration is 013	001
TerminalId	מספר מסוף To change a Pinpad configuration you must supply the correct terminal id. If you don't supply the correct terminal id then the parameters you set in the configuration request will not be accepted by the Pinpad.	006
TermNo	מספר קופה To change a Pinpad configuration you must supply the correct TermNo (aka station number/ECR number). If you don't supply the correct TermNo then the parameters you set in the configuration request will not be accepted by the Pinpad.	040
ShvaTime	שעת התקשרות לשב"א. רלוונטי רק אם הפינפד מתקשר אוטומטית לשב"א (ולא על פי פקודה מהקופה) ורק אם סוג טיימר שב"א מוגדר כ2. פורמט: HHMM The time to call Shva.	035

Tag name	Description	Matching TLV tag (Caspit internal)
ShvaTimerType	סוג טיימר לשב"א. 0 – אין טיימר. התקשורת לשב"א תתבצע רק לפי פקודה מהקופה. אם הקופה לא תפעיל את פקודת התקשורת לשב"א (פקודה מספר 006, ראה פרק 15), אז העסקאות יישארו בפינפד. יש גם אפשרות להפעיל את התקשורת לשב"א מתוך תפריט הפינפד, אבל זה עלול לגרום לחוסר סנכרון בין סגירת היום החשבונאית של הקופה לבין שידור עסקאות האשראי, מה שחלק מהקופות לא אוהבות שקורה. 1 – טיימר פעיל. התקשורת לשב"א תהיה אוטומטית ויזומה על ידי הפינפד. והזמן לתקשורת לשב"א יהיה לפי הזמן שהתקבל בפרמטרים משב"א. ניתן לפנות לשב"א ולקבוע זמן אחר. כמובן שגם עם טיימר פעיל, הקופה עדיין יכולה לשלוח פקודה לביצוע תקשורת. 2 – טיימר פעיל. התקשורת לשב"א תהיה אוטומטית ויזומה על ידי הפינפד. הזמן לתקשורת לשב"א יהיה לפי הזמן שהוגדר בתגית ShvaTime. הפינפד יתעלם מהזמן שנקבע בפרמטרים שהגיעו משב"א. ברירת המחדל בפינפד חדש היא 0. כלומר בפינפד שלא קיבל עדיין אף פקודת Configuration, סוג הטיימר יהיה 0 (אין טיימר). השימוש הנפוץ הוא ברירת המחדל (0, אין טיימר). **0** – Not timer. Communication to Shva will be done only using a command from the ECR. **1** – Timer is on. Automatic call to Shva will be made according to the schedule that was received in the parameters from Shva **2** – Timer is on. Automatic call to Shva will be made according to the schedule that was received in tag ShvaTime.	198
EnableIdleSwipe	**0** – Idle swipe is disabled **1** – Idle swipe is enabled You need to set also the <IdleSwipeAddress> See Appendix G for a full description of the Idle Swipe.	N/A It is done locally in SmartRetail. No need to send to Ashrait EMV
EnableSmartSwipe	**0** – Smart swipe is disabled **1** – Smart swipe is enabled. If you start a transaction and the pinpad is waiting for a card. If a card is swiped and the card is **not** recognized as a credit card then the pinpad will not perform a transaction but it will return a response suitable for Swipe Command (command number 023) The default in a new Pinpad is 0. Meaning, in a Pinpad that didn't get any Pinpad configuration command, the Smart Swipe is disabled. See chapter 9.5 for a full description of Smart Swipe.	083

Tag name	Description	Matching TLV tag (Caspit internal)
BlockDoubleTrans	**0** – Block double transaction is disabled **1** – Block double transaction is enabled. The pinpad will refuse the transaction if the same card number, same amount and same voucher number are used as in the last transaction. The default in a new Pinpad is 1. Meaning, in a Pinpad that didn't get any Pinpad configuration command, the Block Double Transaction feature is enabled. The ECR can temporarily override this block, for a specific transaction, by sending the tag <SkipDoubleTranCheck>1</SkipDoubleTranCheck> in the transaction request.	084
AccessControl	**0** – Access control is disabled. **1** – Access control is enabled. Entering to SmartRetail application will require the user to enter a password or to swipe a technician card. The default in a new Pinpad is 0. Meaning, in a Pinpad that didn't get any Pinpad configuration command, the Access Control will be disabled.	N/A
EnableFullPAN	**0** – Full PAN is disabled. The Pinpad will never send out the full PAN of a credit card. The PAN will always be masked. The PAN will be masked in the transaction response XML, in the receipt XML, in the TRAN record. Everywhere. **1** – Full PAN is enabled. The transaction response will be sent with the full PAN visible. The full PAN will be in the <Pan> tag and also in the <ReceiptMerchant> receipt text. The PAN will still be masked in the TRAN record. The default in a new Pinpad is 0. Meaning, in a Pinpad that didn't get any Pinpad configuration command, The full PAN will be disabled.	215
IdleSwipeAddress	The IP address and the port number of the listener for the Idle Swipe data. Format: XXX.XXX.XXX.XXX/YYYYY YYYYY is the port number. For example: 192.168.0.1/1200 See appendix G for a full description of Idle Swipe	

Tag name	Description	Matching TLV tag (Caspit internal)
EnableTranSteps	This tag enable/disable the transaction steps **0** – Transaction steps are disabled **1** – Transaction steps are enabled. The Pinpad will update the PC about the current step of the transaction. The ECR is responsible to set up a TCP listener See Appendix J for a full description of Transaction Steps	222
TranStepsAddress	The IP address and the port number of the listener for the transaction steps. Format: XXX.XXX.XXX.XXX/YYYYY YYYYY is the port number For example: 192.168.0.1/24321 See Appendix J for a full description of Transaction Steps	223
MustTranSteps	This flag can make the transaction steps mandatory. **0** – If transaction steps are enabled, but the listener is not working, the Pinpad will continue with the transaction. **1** – If transaction steps are enabled, but the listener is not working, the Pinpad will stop the transaction with result code 10026.	242
BlockAutoSwitchBatchDelete	<u>Switch Support</u> 0 – Default value. If the Switch batch is empty, the Switch batch will be deleted automatically on midnight. 1 – Switch batch will not be deleted automatically. The ECR will be responsible to delete the Switch batch using Generic Command sub command 013.	256
PinpadLock	0 – Pinpad is not locked. 1 – Pinpad is locked. It will not be possible to enter the system menu or the application menu. The Pinpad lock can be configured also from the Pinpad menu.	257
RequestId	מזהה בקשה	051
DCCAuthorization	Dynamic Curency Change (DCC) version, starts with 1, supported by CR	268
BListTime	Date/Time of black list file	
Request	Closing tag of the request XML	N/A

17.2 Pinpad Configuration Response

<Response>
<Command>013</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<SerialNumber>30303PP3039393</SerialNumber>
<PartNumber>WERWERE</PartNumber>
<SmartRetailVersion>4872830001</SoftwareVersion>
<AshraitEMVVersion>5959590040</AshraitEMVVersion>
<EPAVersion>33939392122</EPAVersion>
<CPAVersion>20202022020</CPAVersion>
<SDKVersion>9.22.0</SDKVersion>
<ResultCode>0</ResultCode>
<ShvaTime>HHMM</ShvaTime>
<ShvaTimerType>0</ShvaTimerType>
<CommType>1</CommType>
<ShvaAddress>255.255.255.255/9000</ShvaAddress>
<SwitchAddress1>255.255.255.255/9000</SwitchAddress1>
<SwitchAddress2>255.255.255.255/9000</SwitchAddress2>
<CaspitAddress>255.255.255.255/9000</CaspitAddress>
<TMSAddress>255.255.255.255/9000</TMSAddress>
<ReportServerAddress>255.255.255.255/9000</ReportServerAddress>
<CertServerAddress>255.255.255.255/9000</CertServerAddress>
<LogServerAddress>255.255.255.255/9000</LogServerAddress>
<DateTime>YYYYMMDDHHMMSS</DateTime>
<EnableIdleSwipe>1</EnableIdleSwipe>
<ShvaBusinessName>Caspit Test</ShvaBusinessName>
<ShvaVersion>CST00000033T</ShvaVersion>
<EnableSmartSwipe>1</EnableSmartSwipe>
<BlockDoubleTrans>1</BlockDoubleTrans>
<SetupDateTime>YYYYMMDDHHMMSS</SetupDateTime>
<LastTranDateTime>YYYYMMDDHHMMSS</LastTranDateTime>
<LastShvaDateTime>YYYYMMDDHHMMSS</LastShvaDateTime>
<LastSwitchDateTime>YYYYMMDDHHMMSS</LastSwitchDateTime>

```
<LastCaspitDateTime>YYYYMMDDHHMMSS</LastCaspitDateTime>
<LastTMSDateTime>YYYYMMDDHHMMSS</LastTMSDateTime>
<AccessControl>0</AccessControl>
<RequestId>20160216155540</RequestId>
<AshraitSSL>1</AshraitSSL>
<UnsentTransactions>0</UnsentTransactions>
<CardInside>0</CardInside>
<BlacklistStatus>0</BlacklistStatus>
<ShvaParamStatus>0</ShvaParamStatus>
<IdleSwipeAddress>192.168.0.1/1200</IdleSwipeAddress>
<EnableFullPan>0</EnableFullPan>
< EnableTransSteps >0</EnableTransSteps >
<MustTransSteps>0</MustTransSteps>
<TransStepsAddress>192.168.0.1/24321</TransStepsAddress>
<SwitchEnabled>0</SwitchEnabled>
<CardEntryTimeout>60</CardEntryTimeout>
<PinEntryTimeout>90</PinEntryTimeout>
<SwitchName>CreditGuard</SwitchName>
<SwitchId>1</SwitchId>
<BlockAutoSwitchBatchDelete>0</BlockAutoSwitchBatchDelete>
<PinpadLock>0</PinpadLock>
</Response>
```


Response XML tags:

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Response	The opening tag of the response XML	N/A
Command	פקודה. להגדרת פינפד, מספר הפקודה היא 013 Should be the same as in the request	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Should be the same as in the request	040

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
SerialNumber	Serial number (S/N) of the Pinpad The serial number can also be seen on the back of the Pinpad. This is a read-only tag.	N/A
PartNumber	Part Number (P/N) of the Pinpad This is a read-only tag.	N/A
SmartRetailVersion	Software version of SmartRetail app This is a read-only tag.	N/A
AshraitEMVVersion	Software version of Ashrait EMV app This is a read-only tag.	
EPAVersion	Software version of EPA (Pinpad EMV Contact application) This is a read-only tag.	
CPAVersion	Software version of CPA (Pinpad Contactless application) This is a read-only tag.	
SDKVersion	SDK version of the Pinpad This is a read-only tag.	N/A
ResultCode	0 – Success See Appendix A	010
ShvaTime	שעת התקשרות לשב"א. רלוונטי רק אם הפינפד מתקשר אוטומטית לשב"א (ולא על פי פקודה מהקופה) ורק אם סוג טיימר שב"א מוגדר כ2. The time to call Shva.	035

Tag name	Description	Matching TLV tag (Caspit internal)
ShvaTimerType	סוג טיימר לשב"א. 0 – אין טיימר. התקשורת לשב"א תתבצע רק לפי פקודה מהקופה. אם הקופה לא תפעיל את פקודת התקשורת לשב"א (פקודה מספר 006, ראה פרק 15), אז העסקאות יישארו בפינפד. יש גם אפשרות להפעיל את התקשורת לשב"א מתוך תפריט הפינפד, אבל זה עלול לגרום לחוסר סנכרון בין סגירת היום החשבונאית של הקופה לבין שידור עסקאות האשראי, מה שחלק מהקופות לא אוהבות שקורה. 1 – טיימר פעיל. התקשורת לשב"א תהיה אוטומטית ויזומה על ידי הפינפד. והזמן לתקשורת לשב"א יהיה לפי הזמן שהתקבל בפרמטרים משב"א. ניתן לפנות לשב"א ולקבוע זמן אחר. כמובן שגם עם טיימר פעיל, הקופה עדיין יכולה לשלוח פקודה לביצוע תקשורת. 2 – טיימר פעיל. התקשורת לשב"א תהיה אוטומטית ויזומה על ידי הפינפד. הזמן לתקשורת לשב"א יהיה לפי הזמן שהוגדר בתגית ShvaTime. הפינפד יתעלם מהזמן שנקבע בפרמטרים שהגיעו משב"א. ברירת המחדל בפינפד חדש היא 0. כלומר בפינפד שלא קיבל עדיין אף פקודת Configuration, סוג הטיימר יהיה 0 (אין טיימר). השימוש הנפוץ הוא ברירת המחדל (0, אין טיימר). **0** – Not timer. Communication to Shva will be done only using a command from the ECR. **1** – Timer is on. Automatic call to Shva will be made according to the schedule that was received in the parameters from Shva **2** – Timer is on. Automatic call to Shva will be made according to the schedule that was received in tag ShvaTime.	198
CommType		
ShvaAddresss	The IP address of Shva server	
SwitchAddress1	For Switch support Only The IP address of the Switch server (can be URL)	N/A
SwitchAddress2	For Switch support only The backup IP address of the Switch server (can be URL)	N/A
CaspitAddress	The IP address of the Technician server	N/A
TMSAddress	The IP address of the TMS server	N/A
ReportServerAddress	The IP address of the report server (can be URL). Only if a report server is being used.	N/A
CertServerAddress	The IP address of the certificates server (can be URL) Only if a certificates server is being used.	N/A

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
LogServerAddress	The IP address of the log server (can be URL)	N/A
DateTime	The current date time stamp of the Pinpad	N/A
EnableIdleSwipe	0 – Idle swipe is disabled 1 – Idle swipe is enabled See request for explanation.	N/A
ShvaBusinessName	The name of the business as received from Shva (in the parameters from Shva)	
ShvaVersion	The version of the Ashrait EMV application as it is sent to Shva.	
EnableSmartSwipe	0 – Smart swipe is disabled 1 – Smart swipe is enabled. If you start a transaction and the pinpad is waiting for a card. If a card is swiped and the card is not recognized as a credit card then the pinpad will not perform a transaction but it will return a response suitable for "swipe only" command.	083
BlockDoubleTrans	0 – Block double transaction is disabled 1 – Block double transaction is enabled. The pinpad will refuse the transaction if the same card number, same amount and same voucher number are used as in the last transaction.	084
SetupDateTime	The time stamp of when the Ashrait EMV setup was done (YYYYMMDDHHNNSS) Ashrait EMV must keep this information in the params file.	N/A
LastTranDateTime	The time stamp of the last transaction made	N/A
LastShvaDateTime	The time stamp of the last Shva communication	
LastSwitchDateTime	The time stamp of the last Switch communication (FFU)	N/A
LastCaspitDateTime	The time stamp of the last Caspit communication (to the Technician Server)	N/A
LastTMSDateTime	The time stamp of the last TMS communication	N/A
AccessControl	0 – Access control is disabled. 1 – Access control is enabled. Entering to SmartRetail application will require the user to enter a password or to swipe a technician card.	N/A
RequestId	Must be the same value as in the request	
AshraitSSL	0 – Ashrait EMV is not using SSL 1 – Ashrait EMV is using SSL	

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
UnsentTransactions	If there are unsent transactions in the memory of the Pinpad then this tag will return the number of unsent transactions. If there are no transactions in the memory, then this tag will return zero.	214
EnableFullP**an**	0 – Full PAN is disabled. The Pinpad will never send out the full PAN of a credit card. The PAN will always be masked. The PAN will be masked in the transaction response XML, in the receipt XML, in the TRAN record. Everywhere. 1 – Full PAN is enabled. The transaction response will be sent with the full PAN visible. The full PAN will be in the <Pan> tag and also in the <ReceiptMerchant> receipt text. The PAN will still be masked in the TRAN record. The default in a new Pinpad is 0. Meaning, in a Pinpad that didn't get any Pinpad configuration command, The full PAN will be disabled.	215
CardInside	0 – No card inside 1 – Card inside contact reader This is an important issue. With EMV, it is very common for the cardholder to forget his card inside the Pinpad. The ECR should always check if the card is still inside and alert the merchant about it.	220
BlacklistStatus	0 – Everything is OK with the black list 1 – There is a problem with the black list. The ECR may start a communication to Shva to get again the black list. If the ECR doesn't want to transmit the transactions, it can use the Generic Command with sub command 006, to call Shva and get blacklist, but without sending the transactions	246
ShvaParamStatus	0 – Everything is OK with the Shva parameters 1 – There is a problem with Shva parameters. The may start a communication to Shva to get again he parameters. If the ECR doesn't want to transmit the transactions, which may exist in the Pinpad, it can use the Generic Command with sub command 006, to call Shva and get the Shva parameters.	247
EnableTran**s**Steps	0 – Transaction steps are disabled 1 – Transaction steps are enabled. The Pinpad will update the PC about the current step of the transaction. The ECR is responsible to set up a TCP listener.	222

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
TransStepsAddress	The IP address and the port number of the listener for the transaction steps. Format: XXX.XXX.XXX.XXX/YYYYY YYYYY is the port number For example: 192.168.0.1/24321	223
MustTransSteps	This flag can make the transaction steps mandatory. 0 – If transaction steps are enabled, but the listener is not working, the Pinpad will continue with the transaction. 1 – If transaction steps are enabled, but the listener is not working, the Pinpad will stop the transaction with result code 10026.	242
SwitchEnabled	0 – Switch is disabled 1 – Switch is enabled. The Pinpad is using a Switch to authorize the transactions. This is a read-only tag. The Switch flag is set during the Pinpad setup wizard.	243
CardEntryTimeout	The timeout in seconds to wait for the cardholder to swipe/tap/insert his card. This value is set by Shva and the Pinpad get this value during the daily call to Shva. This is a read-only tag. The ECR should consider this timeout value when it waits for the transaction to end.	244
PinEntryTimeout	The timeout in seconds to wait for the cardholder to enter his PIN. This value is set by Shva and the Pinpad get this value during the daily call to Shva. This is a read-only tag The ECR should consider this timeout value when it waits for the transaction to end. Together with the previous timeout (the CardEntryTimeout) the total time of the transaction can be CardEntryTimeout + PinEntryTimeout.	245
SwitchName	Switch Support The name of the Switch	249

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
SwitchId	Switch Support The ID of Switch.	250
BlockAutoSwitchBatchDelete	Switch Support 0 – Default value. If the Switch batch is empty, the Switch batch will be deleted automatically on midnight. 1 – Switch batch will not be deleted automatically. The ECR will be responsible to delete the Switch batch using Generic Command sub command 013.	256
PinpadLock	0 – Pinpad is not locked. 1 – Pinpad is locked. It will not be possible to enter the system menu or the application menu. The Pinpad lock can be configured also from the Pinpad menu.	257
Response	Closing tag of the response XML	N/A

18. Generic Command

As the name suggests, this command is used to perform various tasks (sub commands).

18.1 Generic Command Request

<Request>

<Command>010</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<SubCommand>001</SubCommand>

<Address>192.168.1.1</Address>

<RequestId>20160216155540</RequestId>

</Request>

Caspit Internal (This part is intended for Caspit developers)

Command flow:

- When Terminal Id is 0000000 (RAV SAPAK), it will:
  - verify that the sub command is 12 or 13, otherwise error
  - verify that the terminal is RAV SAPAK
  - transmit unsent transactions to switch (12) or delete tran file (13), for all terminal Ids
  - return success if all succeeded (0,10019,10021), otherwise error 10025 for 12 or 10024 for 13
  - return all failed terminals

Ashrait EMV will prepare the TOTAL.ASH file in the HOST folder and will return the control to SmartRetail. The TOTAL.ASH

Tag name	Description	Matching TLV tag (Caspit internal)
Request	The opening tag of the request XML	N/A
Command	מספר פקודה. לפקודת ג'נרית, מספר הפקודה היא 010 The command number for Generic Command is 010	001
TerminalId	מספר מסוף או 0000000 במקרה של תת פקודות 12/13 במקרה של רב ספק Terminal id or 000000 for sub commands 12/13 in case of RAV-SAPAK	006
TermNo	מספר קופה Station number (ECR number)	40
SubCommand	**001** – Call Switch parameters. **If Switch is supported**, SmartRetail will call the Switch to get parameters. Our current Switch doesn't support parameters so **<u>This feature is not ready yet</u>** **002** – Call Caspit Technician server. Ashrait EMV will call the Caspit technician server to get parameters. Note that Ashrait EMV has an automatic timer that calls Caspit Technician server every night. This	238

Tag name	Description	Matching TLV tag (Caspit internal)
	command can be used if another call is necessary. **003** – Send logs to FTP. Ashrait EMV will connect to an FTP server and will upload the logs. Note that Ashrait EMV can be configured (using the Ashrait EMV menu) to send the logs automatically every night. **004** – Call certificates server. If a certificates server is supported, SmartRetail will call the certificates server to obtain a new certificate. **<u>This feature is not ready yet.</u>** **005** – Call reports server. If a reports server is supported, Ashrait EMV will call the reports server to get a report. **<u>This feature is not ready yet</u>.** **006** – Call Shva. **<u>WITHOUT TRANSMITTING TRANSACTIONS</u>**. But get updated parameters and updated black list. If you want to call Shva to send the transactions then you should use the Transmit Transactions Command (chapter 15). If you just want to test the connection to Shva, you should use the Communication Test Command. **007** – Call TMS (Terminal Management System). This call may take time. The Pinpad will return a response immediately saying that it got the request and then the Pinpad will start the TMS call. If a new software version exists for the Pinpad then the Pinpad will download the new version and will install it. The ECR can wait a few minutes and then start to poll the Pinpad in order to check if the Pinpad is back online. The Pinpad may a reboot itself during the call to the TMS. This is normal behavior. The ECR can use the Communication Test Command to poll the Pinpad **008** – Restart Pinpad (reboot). The Pinpad will return a response immediately saying that it got the request and then the Pinpad will restart itself. The ECR can wait a few seconds and then start to poll the Pinpad in order to check if the Pinpad is back online. The ECR can use the Communication Test Command to poll the Pinpad. **009** – Ping address. The Pinpad will try to ping the address from the tag <Address> **010** – Delete Shva parameters. After deletion of Shva parameters, the ECR must tell the Pinpad to call Shva to get new Shva parameters. It is not possible to do transactions without Shva parameters. **011** – Delete Shva black list. After deletion of Shva black list, the ECR must tell the Pinpad to call Shva to get new Shva black list. **<u>Switch Support</u>** **012** – **If Switch is supported**. **Call the Switch** and notify the Switch about all the transactions records that were not notified. The Switch is accumulating these transaction records in the Switch database (transactions are still NOT sent to Shva). The Pinpad TRAN file will NOT be deleted after this command. It will remain as it is and new transactions will continue to accumulate in the same	

Tag name	Description	Matching TLV tag (Caspit internal)
	Pinpad TRAN file. The ECR may call this command after every transaction or after a couple of transaction. It is up to the ECR to decide. **013** – **If Switch is supported**. **Delete the current Pinpad TRAN file**. The Pinpad TRAN file can be deleted only if the Switch was notified about all transactions. If not all transactions are notified, the ECR will need to use Generic Command 012 to notify the Switch about the unnotified transactions and then try again Generic Command 013. The Pinpad will not do all of this automatically. It is up to the ECR to handle this situation of unnotified transaction. Note that there is also an automatic timer event that deletes the TRAN on a predefined hour. This generic command 013 is used in case there is a need to delete the TRAN now and not to wait for the automatic timer event. **014** – **If Switch is supported**. The Pinpad will call the Switch and will tell the Switch to transmit all accumulated transactions to Shva. Pay attention that this is a Switch operation. The Switch is the one that transmits the transactions, the Pinpad just tells the Switch to do it now. You should spend some time for a better understanding of the difference between Generic Command 012 and Generic Command 014. It is very important to understand the difference. **015** – Abort J2 transaction. This sub command tells the Pinpad that the transaction that started with J2 (see explanation of J2 in chapter 9.2) will not be completed. If a J2 transaction was performed with the smart card (with the chip) then the Pinpad will display "Do not remove card" while waiting for J4 to complete the transaction. But if the ECR decided to NOT complete the transaction then we want to tell the Pinpad that it should not wait anymore for J4 and that the card can be removed.	
Address	IP Address. Relevant to sub command 009 - Ping	
RequestId	מזהה בקשה. הקופה תייצר את המזהה הזה (למשל תאריך ושעה). אותו מזהה אמור לחזור בתשובה, וכך ניתן לוודא שהתשובה מתאימה לבקשה. Request Id	N/A
Request	Closing tag of the request XML	N/A

18.2 Generic Command Response

<Response>

<Command>010</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<SubCommand>001</SubCommand>

<ResultCode>000</ResultCode>

<RequestId>20160216155540</RequestId>

</Response>

RAV SAPAK (error response)

<Response>

<Command>010</Command>

<TerminalId>0000000</TerminalId>

<TermNo>001</TermNo>

<SubCommand>001</SubCommand>

<ResultCode>10025</ResultCode> for subcommand 12 or 10024 for subcommand 13

<RequestId>20160216155540</RequestId>

<FailedTerminal>0880264,0880269</FailedTerminal>

</Response>

Response XML tags:

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Response	The opening tag of the response XML	N/A
Command	לפקודה ג'נרית, מספר הפקודה היא 010 Should be the same as in the request	N/A
TerminalId	מספר מסוף Should be the same as in the request	N/A
TermNo	מספר קופה Should be the same as in the request	N/A
SubCommand	Should be the same as in the request	N/A
ResultCode	Relevant result codes: 0 – Success If the sub command is invalid: 10056 - RETVAL_UNSUPPORTED_GENERIC_COMMAND_CODE	N/A

Tag name	Description	Matching TLV tag (Caspit internal)
	For sub command 001: 10038 - RETVAL_SWITCH_CALL_FAILED For sub command 002: 10060 - RETVAL_CASPIT_TECH_CENTER_FAILED For sub command 003: 10057 - RETVAL_TRANSMIT_LOGS_FAILED For sub command 004: 10037 - RETVAL_CERTIFICATES_SERVER_CALL_FAILED For sub command 005: 10034 - RETVAL_REPORTS_SERVER_CALL_FAILED For sub command 006: 10033 - RETVAL_SHVA_CALL_FAILED For sub command 007: 10062 - RETVAL_STARTING_TMS_CALL_FAILED For sub command 008: 10061 - RETVAL_TERMINAL_REBOOT_FAILED For sub command 009: 10055 - RETVAL_NO_PING For sub command 010: 10030 - RETVAL_SHVA_PARAMETERS_DELETION_FAILED For sub command 011: 10031 - RETVAL_SHVA_BLACKLIST_DELETION_FAILED For sub command 012: 10025 - RETVAL_SWITCH_TRANSACTIONS_SENT_FAILED 10021 - RETVAL_SWITCH_TRAN_FILE_IS_EMPTY 10019 - RETVAL_SWITCH_TRAN_FILE_NOTHING_TO_SEND For sub command 013: 10024 - RETVAL_SWITCH_TRAN_HAS_UNSENT_TRANSACTIONS For other result codes, see Appendix A	
RequestId	Should be the same as in the request	N/A
Response	Closing tag of the response XML	N/A

19. Transaction Query

The PC ECR can query the Pinpad about the transactions made on the **same day**. Transaction query will be possible as long as the transactions batch is not transmitted to Shva.

Transaction Query is useful for times when the status of a transaction is unknown. For example, if the ECR didn't get the response of the transaction and the ECR doesn't know if the transaction was approved or not.

Transaction Query is based on the XML tag <Xfield>. For those of you that are familiar with Ashrait 96, the <Xfield> has the same meaning as the X tag in the Ashrait 96 INT_IN string. Every transaction the ECR starts must include the XML tag <Xfield>. The <Xfield> value is made by the ECR. Usually, the <Xfield> will be the ECR voucher number (or transaction id, invoice number, or whatever you want to call it). The <Xfield> must be unique for each transaction in the ECR. However, the Pinpad doesn't enforce this uniqueness, the Pinpad will allow transactions to have the same <Xfield>. But if the <Xfield> will NOT be unique, the Transaction Query will not function properly, because the Transaction Query will return the first result that it finds, the first transaction record that matches this <Xfield>.

19.1 Transaction Query Request

To query a transaction using a specific <Xfield>

<Request>

<Command>012</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<RequestId>20160216155540</RequestId>

<Xfield>ECR303093</Xfield>

</Request>

To query the **last** transaction in the TRAN

<Request>

<Command>012</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<RequestId>20160216155540</RequestId>

<Xfield>LAST</Xfield>

</Request>

Tag name	Description	Matching TLV tag (Caspit internal)
Request	The opening tag of the request XML	N/A

Tag name	Description	Matching TLV tag (Caspit internal)
Command	מספר פקודה. לשאילתת עסקה, מספר הפקודה היא 012 The command number for Transaction Query is 012	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה Station number (ECR number)	040
RequestId	מזהה בקשה Request Id	051
Xfield	מספר השובר של הקופה (בחלק מהקופות זה נקרא מספר עסקה\מספר חשבונית) באחריות הקופה לשלוח מספר שובר ייחודי עבור כל עסקה. מספר השובר נשמר ביחד עם פרטי עסקת האשראי וכך יכולה הקופה לבדוק את סטטוס העסקה ולשחזר את פרטיה. ניתן גם לשלוח את המילה "LAST" והפינפד יחזיר את העסקה האחרונה שרשומה לו בקובץ TRAN. <Xfield>LAST</Xfield>	144
Uid	מספר מזהה עסקת מקור השדה הזא הוא אופציונלי ומופיע רק כאשר Xfield לא נשלח. The Original Uid from Shva is used to query transaction. This field is optional and exist only if Xfield is missing.	127
Request	Closing tag of the request XML	N/A

19.2 Transaction Query Response

If the transaction is found, the response XML will include an inner XML that will be the full XML of the transaction response. If the found transaction was a successful transaction then the response XML will include also the receipt text.

For example:

<Response>

<Command>012</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<ResultCode>0</ResultCode>

<RequestId>20160216155540</RequestId>
<Xfield>ECR303093</Xfield>
<EMV_Output>
…
… **(Here will come the complete response XML of the transaction)**
…
…
<ReceiptMerchant>
Here will come the text lines of the merchant receipt
</ReceiptMerchant>
<ReceiptCustomer>
Here will come the text lines of the customer receipt
</ReceiptCustomer>
</EMV_Output>
</Response>

If the transaction cannot be found, then the response will be like that:

<Response>
<Command>012</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<ResultCode>10049</ResultCode>
<RequestId>20160216155540</RequestId>
<Xfield>ECR303093</Xfield>
</Response>

Tag name	Description	Matching TLV tag (Caspit internal)
Response	The opening tag of the response XML	N/A
Command	מספר פקודה. לשאילתת עסקה, מספר הפקודה היא 012 Should be the same as in the request	001

Tag name	Description	Matching TLV tag (Caspit internal)
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה Should be the same as in the request	040
EMV_Output		
ResultCode	0 – Success 10049 - RETVAL_TRANSACTION_NOT_FOUND (the Pinpad searched for a transaction with this <Xfield> but didn't find any matching transaction. It is possible that the transaction has already been sent to Shva and so it cannot be queried. Also, it is possible that the transaction has failed and was not written in the TRAN file). 10102 - RETVAL_INVALID_X_FIELD (<Xfield> tag is missing) See Appendix A for more result codes.	N/A
RequestId	Should be the same as in the request	N/A
Response	Closing tag of the response XML	N/A

20. Retrieve File

<u>This feature is not ready yet</u>

- The ECR can retrieve files from the Pinpad. These files will mostly be log files (text files) but some files may be binary files.
- The Pinpad will read a block of data from the file (according to the BlockSize tag), the Pinpad will encode the block with Base64 and return the encoded result to the ECR in tag BlockData.
- The ECR needs to decode BlockData using Base64 and keep the decoded block in a file.
- Then the ECR will ask for the next block of data.
- There will be a white list of files that are allowed to be retrieved. The requested filename will be checked against that white list and only if it is on the white list then the ECR will get that file. For security reasons, the white list cannot be changed. It is hardcoded.
- The white list can be seen below.
- Caspit will provide a feature, in the DLL, to decode Base64.
- The file retrieval session will most likely consists of many Retrieve File Requests and Retrieve File Responses until the full file is retrieved.

20.1 Retrieve File Request

```
<Request>
<Command>016</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<Filename>EPA.LOG</Filename>
<BlockNumber>1</BlockNumber>
<BlockSize>1024</BlockSize>
<RequestId>20160216155540</RequestId>
</Request>
```

Tag name	Description	Matching TLV tag (Caspit internal)
Request	The opening tag of the request XML	N/A
Command	Command number. For Retrieve File command, the command number is 016	001
TerminalId	מספר מסוף	006
TermNo	מספר קופה	040
RequestId	מזהה בקשה	051
Filename	The filename that the ECR wants to retrieve. The filename will be checked against a white list of files.	
BlockNumber	Block numbers starts with 1.	

Tag name	Description	Matching TLV tag (Caspit internal)
BlockSize	The BlockSize is the size of data **before encoding it to Base64.** Obviously, after encoding the size will be larger. Default: 1024 Maximum: 4096	
Request	Closing tag of the request XML	N/A

20.2 Retrieve File Response

```
<Response>
<Command>016</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<ResultCode>0</ResultCode>
<RequestId>20160216155540</RequestId>
<BlockNumber>1</BlockNumber>
<BlockSize>1024</BlockSize>
<MoreBlocks>Yes</MoreBlocks>
<BlockData>In Base64 endcoding </BlockData>
<Base64>Yes</Base64>
</Response>
```

Tag name	Description	Matching TLV tag (Caspit internal)
Response	The opening tag of the Response XML	N/A
Command	Command number. For Retrieve File command, the command number is 016	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה Should be the same as in the request	040
RequestId	מזהה בקשה Should be the same as in the request	051

Tag name	Description	Matching TLV tag (Caspit internal)
ResultCode	0 – Success 10040 - RETVAL_FILE_CANNOT_BE_RETRIEVED (this error will happen if the file cannot be found in the Pinpad or if it is not in the white list of files) 10052 - RETVAL_INVALID_BLOCK_NUMBER_TO_RESUME Other result codes may arrive. See Appendix A	010
BlockNumber	Should be the same as in the request	064
BlockSize	The size of data before base64 encoding. If it is the last block then it might be less than the block size of the request. After decoding the <BlockData>, the ECR should check that the decoded size is equal to the <BlockSize>	065
MoreBlocks	Yes – there are more blocks read in the file No – no more blocks in the file If <MoreBlocks> is Yes then the ECR will increment the <BlockNumber> and will call again the Retrieve File command.	066
BlockData	The block data of the file (could be In Base64 encoding)	
Base64	Some files may not need to be encoded by Base64, so this tag will tell the ECR if the BlockData is encoded by Base64 or not. The decision if to encode in Base64 (or not encode) is made by the Pinpad. Yes – BlockData is encoded by Base64 No – BlockData is not encoded.	
Response	Closing tag of the response XML	N/A

20.3 Retrieve File White List

The next table lists the currently supported files that can be retrieved using this command.

Filename	Description
ASH_EMV.LOG	Main log file of Ashrait EMV application
ASH_EMV.LG0	Previous log file of Ashrait EMV
EPA.LOG	Main log file of the EPA application (EMV Contact application)
EPA.LG0	Previous log file of the EPA application (EMV Contact application)
CPA.LOG	Main log file of the CPA application (EMV Contactless application)
CPA.LG0	Previous log file of the CPA application (EMV Contactless application)

Filename	**Description**
SMART_RET.LOG	Main log file of the SmartRetail application
SMART_RET.LG0	Previous log file of the SmartRetail application

21. Swipe Command

The PC ECR can use the Pinpad as a simple Magnetic Stripe Reader (MSR). This can be useful if there is a need to read a **non** credit card. If a **non** credit card is swiped, the Pinpad will response with the magnetic data. However, if the card is identified as a credit card then the Pinpad will return with an error.

Examples for non-credit-cards: Loyalty club card, timeclock card, employee card, technician card, etc.

The command also enable the user to enter number by keypad.

There are three "Swipe" features in the Pinpad, some may find it a bit confusing, and so I will summarize them here:

- The **Swipe Command** is described here, in this chapter. It is a command that is initiated by the PC.
- The **Smart Swipe** is a feature that changes a transaction request into a Swipe response. See chapter 9.5.
- The **Idle Swipe** is a feature that allows swiping cards while the Pinpad is in idle mode. See Appendix G.

In default, a message "please swipe a card or enter number" is displayed.

Incace ECR sends Tag Line1 and/or Line2 it will replace the default message.

Caspit Internal

The Swipe command will be done by SmartRetail. Ashrait EMV doesn't need to take part in this command.

Algorithm for detecting credit cards

A magnetic card is identified as a credit card, if the track2 data includes the "=" character and if there are 11 to 19 digits before the "=".

For special cards, we will add special handling (like Praxell cards).

To support Praxell card, if the card is identified as a credit card, SmartRetail will make another validation and it will check if the Track2 data contains the following string "=1962". If this string exists in the Track2 data then this card is a Praxell card and the Track2 can be sent out.

21.1 Swipe Request


```
<Request>
  <Command>023</Command>
  <TimeoutInSeconds>30</TimeoutInSeconds>
  <TerminalId>0880264</TerminalId>
  <TermNo>001</TermNo>
  <Line1>8OAg5Pfs4y/pIDQg8fT45fo=</Line1>
  <Line2>4Of45fDl+iD57CD6LuY=</Line2>
  <RequestId>20230912173747</RequestId>
</Request>
```


Tag name	Description	Matching TLV tag (Caspit internal)
Request	The opening tag of the request XML	N/A

Tag name	Description	Matching TLV tag (Caspit internal)
Command	Command number. For Swipe command, the command number is 023	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number	040
TimeoutInSeconds	Timeout in seconds. If tag doesn't exist then a default timeout of 30 seconds will be used.	005
Line1	20 characters Base 64 (Hebrew iso8859-8 encoding) - optional	
Line2	20 characters Base 64 (Hebrew iso8859-8 encoding) - optional	
RequestId	מזהה בקשה	051
Request	Closing tag of the request XML	N/A

21.2 Swipe Response

Successful swipe of a non-credit card.

<Response>

<Command>023</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<ResultCode>0</ResultCode>

<RequestId>20160216155540</RequestId>

<Track1>23093809348503</Track1>

<Track2>45804580458045802302983042948093</Track2>

<Track3>3095830958309530459384O</Track3>

<PanEntryMode>0</PanEntryMode>

</Response>

(Not all the <trackX> tags will be filled. It depends on the specific card.)

If the swiped card is identified as a credit card, the response will be like this:

<Response>

<Command>023</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<ResultCode>11000</ResultCode>

<RequestId>20160216155540</RequestId>

<CardHash>2F9A8B7C2F9A8B7C2F9A8B7C2F9A8B7C</CardHash>

<PanEntryMode>0</PanEntryMode>

</Response>

If keypad is used to enter number, the response will be like this:

<Response>

<Command>23</Command>

<TerminalId>0883073</TerminalId>

<TermNo>001</TermNo>

<ResultCode>0</ResultCode>

<RequestId>20231210162435</RequestId>

<Track2>1234</Track2>

<PanEntryMode>51</PanEntryMode>

</Response>

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Response	The opening tag of the Response XML	N/A
Command	Command number. For Swipe command, the command number is 023	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number	040
RequestId	מזהה בקשה Should be the same as in the request	051

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
ResultCode	Relevant result codes: 0 - Success 10003 – RETVAL_INCORRECT_TERMINAL_ID 10022 – RETVAL_INCORRECT_ECR_NUMBER 10036 – RETVAL_XML_MISSING_TERMINAL_ID_TAG 11000 - RETVAL_CANNOT_SEND_OUT_CREDIT_CARDS For other result codes, see Appendix A	010
Track1	Track1 data, if exists	064
Track2	Track2 data, if exists, or number from keypad entry	065
Track3	Track3 data, if exists	066
CardHash	MD5 of the card number. The hash is relevant to valid credit cards. There will be no hash for non-credit cards.	
PanEntryMode	0 for card swipe, 51 for keypad entry	
Response	Closing tag of the response XML	N/A

22. Retrieve Deposit Report

- The deposit report is received from Shva. The report is **not** made by the Pinpad.
- If the ECR wants, it can request the deposit report from the Pinpad and print it in the ECR printer.
- The input for the deposit report request is the Session-Number (Asmachta number) and the date of the deposit.
- The Pinpad may need to perform a communication to Shva to get the deposit report, because only the last deposit report is kept in the Pinpad.
- The deposit report is **NOT** a detailed report. You will **not** find there the details of every single transaction. The report is a summary of all the transactions. The summary is grouped by card types and transaction types. If you need a detailed report, have a look at Retrieve Tran File command.

Switch Support

When using a Switch, the Deposit Repot is irrelevant because transactions are not kept in the Pinpad.

Caspit Internal (This part is intended for Caspit developers)

Command flow:

- If XML Session-Number valu is LAST:
  - The valus of Session-Number and Date to be sent to SHVA shell be taken from last Transmit transaction to SHVA respons

22.1 Deposit Report Request

```
<Request>
<Command>015</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<RequestId>20160216155540</RequestId>
<Session-Number>29238392</Session-Number>
<Date>2016-11-27</Date>
</Request>
```

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Request	The opening tag of the request XML	N/A
Command	Command number. For Depsit Report, the command number is 015	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number.	040

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Session-Number	מספר אסמכתא של שב"א כפי שהתקבל בתשובת הTOTAL או הSTATIS. או ערך LAST The session number from Shva or LAST	015
Date	תאריך אסמכתא. כפי שהתקבל בתשובת הTOTAL או הSTATIS. YYYY-MM-DD The date of the deposit	213
RequestId	מזהה בקשה Request Id	051
Request	Closing tag of the request XML	N/A

22.2 Deposit Report Response

```
<Response>
<Command>015</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<ResultCode>0</ResultCode>
<RequestId>20160216155540</RequestId>
<Session-Number>29238392</Session-Number>
<Date>2016-11-27</Date>
<DepositReport>
    <Line> Text Text Text </Line>
    <Line> Text Text Text </Line>
        .....
        .....
        More lines will come here
        ....
        ....
</DepositReport>
</Response>
```

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Response	The opening tag of the Response XML	N/A
Command	Command number. For Deposit Report, the command number is 015	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה Should be the same as in the request	040
RequestId	מזהה בקשה Should be the same as in the request	051
ResultCode	**0** – Success **10003** – RETVAL_INCORRECT_TERMINAL_ID **10022** – RETVAL_INCORRECT_ECR_NUMBER **10036** – RETVAL_XML_MISSING_TERMINAL_ID_TAG **10103** – Error. Cannot find deposit report (RETVAL_CANT_FIND_DEPOSIT_REPORT) **10104** – Error. <Session-Number> tag is missing or invalid (RETVAL_INVALID_SESSION_NUMBER_XML_TAG) **10105** – Error. <Date> tag is missing or invalid (RETVAL_INVALID_DATE_XML_TAG) **Other** – See Appendix A	010
Session-Number	The response value must be identical to the request	015
Date	The response value must be identical to the request	213
DepositReport	XML Group that will contain all the lines of the deposit report. <DepositReport> <Line> Text Text Text </Line> <Line> Text Text Text </Line> ……. <DepositReport> The Hebrew characters will be encoded with ISO8859-8. The Pinpad should be configured in Shva as having a 80mm printer.	
Response	Closing tag of the response XML	N/A

Example of deposit report:


```
<DepositReport>
<Line> המכולת של שמוליק</Line>
<Line>מספר מסוף: 0880023</Line>
<Line>תאריך: 11/12/2016 </Line>
<Line>שעה: 10:16</Line>
<Line></Line>
<Line>מספר עסק ישראכרט</Line>
<Line>0071506</Line>
<Line>מספר עסק כאל</Line>
<Line>7007245</Line>
<Line>מספר עסק לאומיקארד</Line>
<Line>0300012</Line>
<Line> מספר קובץ: 30</Line>
<Line>אסמכתה: 05017766 </Line>
<Line>===================</Line>
<Line>סה"כ שקל חובה</Line>
<Line>0004     10.00</Line>
<Line>סה"כ שקל זכות</Line>
<Line>0002     3.00</Line>
<Line> ............................</Line>
<Line>===============</Line>
<Line>סה"כ סולק לאומי קארד</Line>
<Line>===============</Line>
<Line>לאומי קארד שקל חובה</Line>
<Line>0004    10.00</Line>
<Line>===============</Line>
<Line>ויזה      שקל חובה</Line>
<Line>0004       10.00<Line>
</DepositReport>
```

23. UI Command

<u>The UI Command is not implemented yet.</u>

- The ECR can use the UI command to display some messages on the Pinpad screen.
- There is a predefined list of messages that can be displayed. You can see this list in paragraph 23.3, below.
- Some of the messages will be displayed on the Pinpad screen until timeout or until key press.
- But some messages, the ones with "Remove Card", will also wait until timeout or until card removal.
- It is possible to add the waiting for card removal for all messages by using the tag <WaitCardRemoval>

23.1 UI Command Request


```
<Request>
<Command>019</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<RequestId>20160216155540</RequestId>
<MessageCode>001</MessageCode>
<WaitCardRemoval>1</WaitCardRemoval>
<WaitForKey>1</WaitForKey>
<TimeoutInSeconds>5</TimeoutInSeconds>
<Language>iw_IL</Language>
</Request>
```


Tag name	Description	Matching TLV tag (Caspit internal)
Request	The opening tag of the request XML	N/A
Command	Command number. For UI command, the command number is 019	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number.	040
MessageCode	The code of the message to display. The code must be taken from the list of predefined messages. You can see the list below.	239

Tag name	Description	Matching TLV tag (Caspit internal)
WaitCardRemoval	**0** – The Pinpad will **NOT** wait for card removal, unless the message code has a "build in" wait for card, like message code 001. **1** – The Pinpad will add the line "Please Remove Card" in the lower line of the screen and will wait for card removal (If the message code has a "build in" wait for card, like message code 001, then this tag will have no effect, because waiting for card removal is already implied by message code 001). Pay attention that if you ask to wait for card removal, but there is no card inside, then the message that you want to display (like "Tran Success") will not be displayed at all, because the Pinpad will sense that there is no card inside and will decide not to wait and not to show the message at all.	241
WaitForKey	**0** – The Pinpad will not accept any key press during the UI command. So timeout or card removal will be the only way to exit this command. **1** – The Pinpad will accept the RED key during the UI command. Pressing the RED key will abort the UI command. Of course, timeout or card removal will also exit the UI command.	248
TimeoutInSeconds	How many seconds to display the message or how many seconds to wait for card removal	005
Language	"iw_IL" – Hebrew "en_US" - English	240
RequestId	מזהה בקשה	051
Request	Closing tag of the request XML	N/A

23.2 UI Command Response

<Response>

<Command>019</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<ResultCode>0</ResultCode>

<MessageCode></MessageCode>

<RequestId>20160216155540</RequestId>

<CardInside>0</CardInside>

</Response>

Tag name	Description	Matching TLV tag (Caspit internal)
Response	The opening tag of the Response XML	N/A

Tag name	Description	Matching TLV tag (Caspit internal)
Command	Command number. For UI command, the command number is 019	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number.	040
RequestId	מזהה בקשה Should be the same as in the request	051
ResultCode	**0** – Success **10003** – RETVAL_INCORRECT_TERMINAL_ID **10022** – RETVAL_INCORRECT_ECR_NUMBER **10036** – RETVAL_XML_MISSING_TERMINAL_ID_TAG **10027** – RETVAL_INVALID_MESSAGE_CODE **10017** - RETVAL_USER_PRESSED_ON_RED_KEY **Other** – See Appendix A	010
MessageCode	Should be the same as in the request	239
CardInside	0 – No card inside 1 – Card inside contact reader The ECR should examine this tag and see if the card is still inside. If it is inside, the ECR may alert the cashier or ask again to remove the card.	220
Response	Closing tag of the response XML	N/A

23.3 List of messages

Message Code	Explanation	Hebrew Text	English Text
001	The ECR will display the message and will wait until card remove or until timeout	העסקה אושרה נא הוצא כרטיס	Transaction Approved Remove Card
002	The ECR will display the message and will wait until card remove or until timeout	העסקה לא אושרה נא הוצא כרטיס	Transaction Declined Remove Card
009	The ECR will display the message and will wait until card remove or until timeout	נא הוצא כרטיס	Remove card
013	The ECR will display the message and will wait until timeout or key press	כרטיס לא תקין	Card error

Message Code	Explanation	Hebrew Text	English Text
014	The ECR will display the message and will wait until timeout or key press	כרטיס לא נתמך	Unrecognized card
016	The ECR will display the message and will wait until timeout or key press	העסקה אושרה	Transaction Approved
017	The ECR will display the message and will wait until timeout or key press	העסקה לא אושרה	Transaction Declined
022	The ECR will display the message and will wait until timeout or key press	מספר כרטיס שגוי	Invalid card number
024	The ECR will display the message and will wait until timeout or key press	כרטיס לא בתוקף	Card expired
032	The ECR will display the message and will wait until timeout or key press	כרטיס מועדון לא תקין	Invalid Loyalty card
033	The ECR will display the message and will wait until timeout or key press	כרטיס מתנה לא תקין	Invalid Giftcard

24. QR Code Display / Scan Command

Caspit Internal:

The Display QR Code Command will response immediately to ECR and will display the QR until getting any other command, or it will wait for a given time out or until Break key pressed or until Break command received with Step response.

The same command, without QR data, will be used to Scan QR by camera. In this case the response will wait till reading the QR or till a given time out or until Break key pressed or until Break command received with Step response.

24.1 QR Code Display / Scan Command Request

<Request>

<Command>020</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<RequestId>20160216155540</RequestId>

<QrCode>1236543957395790573294574329573945</QrCode>

<TimeoutInSeconds>30</TimeoutInSeconds>

</Request>

Tag name	Description	Matching TLV tag (Caspit internal)
Request	The opening tag of the request XML	N/A

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Command	Command number. For QR code display command, the command number is 020	001
TerminalId	מספר מסוף Terminal Id	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number.	040
QrCode	The QR code to display. This Tag should not be sent for Scanning QR by camera	239
TimeoutInSeconds	How many seconds to display the message or how many seconds to wait for card removal	005
RequestId	מזהה בקשה	051
Request	Closing tag of the request XML	N/A

24.2 QR Code Display Command Response

<Response>

<Command>020</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<ResultCode>0</ResultCode>

<QrCode>12365439573957905732945743295739445</QrCode>

<RequestId>20160216155540</RequestId>

</Response>

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Response	The opening tag of the Response XML	N/A
Command	Command number. For QR code display command, the command number is 020	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number.	040

Tag name	Description	Matching TLV tag (Caspit internal)
RequestId	מזהה בקשה Should be the same as in the request	051
ResultCode	**0** – Success **10003** – RETVAL_INCORRECT_TERMINAL_ID **10022** – RETVAL_INCORRECT_ECR_NUMBER **10036** – RETVAL_XML_MISSING_TERMINAL_ID_TAG **10027** – RETVAL_INVALID_QR_CODE **10017** - RETVAL_USER_PRESSED_ON_RED_KEY **Other** – See Appendix A	010
QrCode	This Tag will be recived only when Scanning QR by camera	239
Response	Closing tag of the response XML	N/A

25. Delek Commands extension

Caspit Internal:

The Payment session in Dlek is performed in two parts (phaseRequest2 = 1)

- **The first part is J59 request with amount that was agreed between the credit card company and the delek company**
- **The secomd part is J49 to update obligo and registering the transaction.**
- **Unlike J5, J59 can be approved offline upto card offline amount**
- **J5 with parameter Tidluk is equels to J59**

25.1 Delek J5 /J59 Command Request

Tag name	Description	Matching TLV tag (Caspit internal)
Request	The opening tag of the request XML	N/A
Command	Command number. For J5/J59 the command number is 001	001
RequestId	מזהה בקשה	051
TerminalId	מספר מסוף	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number.	040
TimeoutInSeconds	How many seconds to display the message or how many seconds to wait for card removal	005
Mti	100	

Tag name	Description	Matching TLV tag (Caspit internal)
CreditTerms	1	
TranType	1	
Amount	The fixed amount that was agreed between the credit card company and delek company	
Currency	376	
PanEntryMode	pinpad	
Xfield	ECR transaction id	
ParameterJ	The value is 5 to perform a pre authorization transaction	
Tidluk	Indication for fuelling transaction	
Request	Closing tag of the request XML	N/A

```
<Request>
        <Command>001</Command>
        <TerminalId>0880349</TerminalId>
        <TimeoutInSeconds>60</TimeoutInSeconds>
        <Mti>100</Mti>
        <TranType>1</TranType>
        <Amount>50000</Amount>
        <Currency>376</Currency>
        <TermNo>1</TermNo>
        <PanEntryMode>40</PanEntryMode>
        <XField>0880349</XField>
        <CreditTerms>1</CreditTerms>
        <AddendumSettl>{"IF REQUIRED"}</AddendumSettl>
        <RequestId>20211005105800</RequestId>
        <Pan>0004557449999990233</Pan>
        <AllowCardInsideBefore>1</AllowCardInsideBefore>
        <Tidluk>1</Tidluk>
        <ParameterJ>5</ParameterJ>
        <ContinueJ2>1</ContinueJ2> optional (if J2 was used before)
</Request>
```

25.2 Delek J5 Command Response

Tag name	Description	Matching TLV tag (Caspit internal)

Tag name	Description	Matching TLV tag (Caspit internal)
Response	The opening tag of the Response XML	N/A
Command	Command number. For J5 the command number is 001	001
TerminalId	מספר מסוף Should be the same as in the request	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number.	040
RequestId	מזהה בקשה Should be the same as in the request	051
ResultCode	**0** – Success. (See Appendix A)	010
AuthorizedAmount	The authorized amount for processing	
amount	The amount in the request	
AuthorizatinNo	The authorizarion number from issuer	
RRN	Unique number for transaction if approved	
TwoPhaseRequest	The value is 1 if online obligo update is supported	
Uid		
Response	Closing tag of the response XML	N/A


```
<EMV_Output>
<Command>1</Command>
<TerminalId>0880349</TerminalId>
<ResultCode>0</ResultCode>
<Status>0</Status>
<AshStatus>0</AshStatus>
<CardHash>B7C6A4A6F3524875C476FA0F5ACD46D7</CardHash>
<Mti>100</Mti>
<CardName>בהז הזיו       </CardName>
<CardType>041</CardType>
<Currency>376</Currency>
<DateTime>1005151842</DateTime>
<AuthManpikNo>4646913</AuthManpikNo>
<AuthCodeManpik>1</AuthCodeManpik>
<Rrn>127815646913</Rrn>
<TranType>1</TranType>
<CreditTerms>1</CreditTerms>
<Amount>50000</Amount>
<AuthorizedAmount>35000</AuthorizedAmount>
<TwoPhaseRequest >1</TwoPhaseRequest >
<Pan>0004557449999990233</Pan>
<PanEntryMode>40</PanEntryMode>
<Manpik>1</Manpik>
```

<Brand>2</Brand>
<Solek>1</Solek>
<Uid>21100515184208803490011</Uid>
<spType>0</spType>
<Xfield>08802650012110</Xfield>
<RequestId>20211005151911</RequestId>
<Tidluk>1</Tidluk>
<ParameterJ>5</ParameterJ>

25.3 Delek J49 Command Request

Caspit Internal:

The fuel filling done untill AuthorizedAmount is reached, or tank is full, the minimum of both.

The obligo is updated and the transaction is registered.

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Request	The opening tag of the request XML	N/A
Command	Command number. For J5 the command number is 001	001
RequestId	מזהה בקשה	051
TerminalId	מספר מסוף	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number.	040
Mti	100	
OriginalAmount	The fixed amount that was agreed between the credit card company and delek company from the first step	
Amount	The real amount	
OriginalUid	Uid from first step	
Xfield	ECR transaction id	
Tidluk	Indication for fuelling transaction	
SwitchSend	Value 1 for immediate send of transaction record to the GW	
ParameterJ	The value is 49 for final charge completion	

Tag name	Description	Matching TLV tag (Caspit internal)
GasolineType GasolineLiter OilLiter OilAmount Speedometer CarNumber ServiceAmount	Fuelling parametrs	
Request	Closing tag of the request XML	N/A


```
<Request>
<Command>001</Command>
<TerminalId>0880349</TerminalId>
<TimeoutInSeconds>60</TimeoutInSeconds>
<Mti>100</Mti>
<TermNo>1</TermNo>
<XField>0880349510</XField>
<OriginalUid>21100515184208803490011</OriginalUid>
<OriginalAmount>50000</OriginalAmount>
<Tidluk>1</Tidluk>
<Amount>23500</Amount>
<SwitchSend>1</SwitchSend>
<GasolineType>02</GasolineType>
<GasolineLiter>00533</GasolineLiter>
<OilLiter>051</OilLiter>
<OilAmount>0007345</OilAmount>
<Speedometer>095248</Speedometer>
<CarNumber>36912401</CarNumber>
<ServiceAmount>0001250</ServiceAmount>
<RequestId>20211005152000</RequestId>
<ParameterJ>49</ParameterJ>
</Request>
```

25.4 Delek J49 Command Response

Caspit Internal:

The transaction verified that it was approved and the AuthManpikNo from first step will be received.

If fuel was not filled at all, the reversal transaction will be sent as following:

- The **MTI** is 420
- The **Amount** is fixed amount fro first step or approved amount
- The **OriginalAuthorizationCodeManpik** is value from first response
- The RRN is value from first step
- The **OriginalAuthNum** is value from first step
- The **OriginalAuthorizationCodeManpik** is approval issuer

<EMV_Output>

<Command>1</Command>

<TerminalId>0880349</TerminalId>

<ResultCode>0</ResultCode>

<Status>0</Status>

<AshStatus>0</AshStatus>

<CardHash>B7C6A4A6F3524875C476FA0F5ACD46D7</CardHash>

<Mti>100</Mti>

<CardName>ויזה זהב   </CardName>

<Currency>376</Currency>

<DateTime>1005151842</DateTime>

<AuthManpikNo>4646913</AuthManpikNo>

<AuthCodeManpik>7</AuthCodeManpik>

<Rrn>127815646913</Rrn>

<TranType>1</TranType>

<CreditTerms>1</CreditTerms>

<Amount>35000</Amount>

<Pan>0004557449999990233</Pan>

<PanEntryMode>40</PanEntryMode>

<Manpik>1</Manpik>

<Brand>2</Brand>

<Solek>1</Solek>

<Uid>21100515184208803490011</Uid>

<spType>0</spType>

<GasolineType>02</ GasolineType>

<GasolineLiter>00533</ GasolineLiter>

<OilLiter>051</OilLiter>

<OilAmount>0007345</OilAmount>

<Speedometer>095248</Speedometer>

<CarNumber>36912401</CarNumber>

<ServiceAmount>0001250</ServiceAmount>

<Xfield>08802650012110</Xfield>

<RequestId>20211005151911</RequestId>

<ParameterJ>49</ParameterJ>

<EMV_Output>

26. Rav-Sapak Configuration

This command will provid the status of all terminals under RavSapak terminal

Caspit Internal:

The request XML will include only the command number.

Send this request to non-RAV SAPAK terminal will reply with an error

Terminals that are defined in device but not exsist in CTEM RavSapak new list, will be shown in Not Active list

Terminal will not be shown in Not Active list if Tran file is empty (all trans were sent and deleted by command ot time event)

Tag name	Description	Matching TLV tag (Caspit internal)
Request	The opening tag of the request XML	N/A
Command	Command number. For RAV SAPAK the command number is 030	001
RequestId	מזהה בקשה	051

<Request>

<Command>030</Command>

<RequestId>20230215173741</RequestId>

</Request>

The response XML will include information about the Pinpad terminals.

Tag name	Description	Matching TLV tag (Caspit internal)
RavSapak	Rav Sapak terminal Id	N/A
sp	Sapak - מספר סידורי רץ של ספק במקרה של ספק המקבץ כמה מסופים (ספליט) לכל המסופים יהיה אותו ערך	
tr	Unsent Transactions - כמות עסקאות שלא שודרו למתג או לשב"א	
as	Session Number - מספר אסמכת בשידור לשב"א (ללא מתג)	
xxxxxxx	terminal Id	
NotActive	List of terminals that exsist on device but not in CTEM	

<Response>
<Command>30</Command>
<ResultCode>0</ResultCode>
<RequestId>20230215173741</RequestId>
<RavSapak>0880264</RavSapak>
<Term sp="nn" tr="yyy" as="zzzzz">xxxxxxx </Term>
<Term sp="nn" tr="yyy" as="zzzzz">xxxxxxx </Term>
<Term sp="nn" tr="yyy" as="zzzzz">xxxxxxx </Term>
<NotActive>
<Term sp="nn" tr="yyy" as="zzzzz">xxxxxxx </Term>
</NotActive>
</Response>

27. VAS of Apple and Google

This command is used for reading information of loyalty card, from Apple and Google wallets

Caspit internal:

The reading will be performed according to the parameters sent in XML, if no parameters have been sent to a certain wallet, do not try to read this wallet

Since it is not known which wallet will be used, a Google wallet will be read first, and if no answer was received, try to read from an Apple wallet

The read will be made if the VAS service is configured in the CTEM profile for the terminal

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Request	The opening tag of the request XML	N/A
Command	Command number. For VAS the command number is 026	001
RequestId	מזהה בקשה	051
TerminalId	מספר מסוף	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number.	040
TimeoutInSeconds	How many seconds to display the message or how many seconds to wait for card removal	005
AppleMerchantId	Apple merchant Id	
AppleUrl	url for registration to apple wallet loyalty - optional	
GoogleMerchantId	Google Collector ID	
GoogleUrl	url for registration to google wallet loyalty - optional	


```xml
<Request>
<Command>026</Command>
<TerminalId>0880340</TerminalId>
<TermNo>001</TermNo>
<RequestId>20230507153012</RequestId>
<TimeoutInSeconds>30</TimeoutInSeconds>
<AppleMerchantId >pass.xxx.com </AppleMerchantId >
<AppleUrl>paz.co.il</AppleUrl>
<GoogleMerchantId >2147483647</GoogleMerchantId >
<GoogleUrl>xxx.co.il</GoogleUrl>
</Request>
```

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Response	The opening tag of the request XML	N/A
Command	Command number. For VAS the command number is 026	001
RequestId	מזהה בקשה	051
TerminalId	מספר מסוף	006
TermNo	מספר קופה (מספר תחנה\מספר עמדה) Station number/ECR number.	040
ResultCode	The following result codes will be returned: SUCCESS0 , where errors 10000- TIMEOUT 10044- USER_ABORT 12001- CARD_READ_ERROR 12003- CONTINUE_WITH_GOOGLE 12005- CONTINUE_WITH_PAYMENT 12008- NOT_ACTIVATED 12009- OTHER_WALLET	
Coupon	Raw Card information	
MerchantId	Apple or Google merchant Id	
Wallet	Type of wallet, Apple or Google	
CouponType	Type of card read from wallet: loyalty, gift etc.	

Example of Apple loyalty card

<Response>
<Command>25</Command>
<TerminalId>0880340</TerminalId>
<TermNo>001</TermNo>
<ResultCode>0</ResultCode>
<RequestId>20230507153012</RequestId>
<Coupon> 0003554082,7670005286067</Coupon>
<CouponType>Loyalty</CouponType>
<MerchantId >pass.xxx.com</MerchantId >
<Wallet>Apple</Wallet>
</Response>

Appendix A – Result Codes

This is the list of result codes that can come back in the tag <ResultCode>.

As I mentioned before, the Caspit SmartRetail solution has much more features than the standard "Shva" Ashrait PC solution, and so I needed many more error codes. Because I didn't want to change the Shva error codes (Status and AshStatus) I had to create another tag.

For more information about each result code, refer to Appendix M – Troubleshooting. There I will try to elaborate on each code and how to fix it.

Also, you should try and search the specific result code in this document. You may find additional information in the relevant command that caused this result code.

Code	Define	English	Hebrew
0	RETVAL_SUCCESS	Success	הצלחה
10000	RETVAL_TIMEOUT	Timeout	טיימאאוט
10001	RETVAL_AMOUNT_TOO_BIG	Amount too big	סכום גדול מדי
10002	RETVAL_AMOUNT_TOO_SMALL	Amount too small	סכום קטן מדי
10003	RETVAL_INCORRECT_TERMINAL_ID	Incorrect Terminal	מספר מסוף שגוי
10004	RETVAL_ERROR_GETTING_TOKEN	Error getting token	שגיאה בקבלת טוקן
10005	RETVAL_NO_LAST_TRAN_FILE	No last TRAN file	אין קובץ TRAN אחרון
10006	RETVAL_CANT_FIND_REQUESTED_TRAN_FILE	Can't find requested TRAN file	לא ניתן למצוא קובץ TRAN
10007			
10008			
10009			
10010	RETVAL_NOT_RAV_SAPAK_TERMINAL		
10011			
10012			
10013			
10014			
10015	RETVAL_SWITCH_BATCH_IS_TOO_OLD	Switch batch is too old. Make sure to transmit all transaction to the batch and to delete the batch	קובץ עסקאות מתג ישן מדי. יש לשדר עסקאות למתג ולמחוק את הקובץ
10016	RETVAL_INVALID_CURRENCY_CODE	Invalid currency code	קוד מטבע שגוי
10017	RETVAL_USER_PRESSED_ON_RED_KEY	User pressed on red key	משתמש לחץ על כפתור אדום

Code	Define	English	Hebrew
10018	RETVAL_INVALID_AUTHORIZATION_CODE_MANPIK	Invalid authorization code manpik	ערך לא תקין בשדה AuthorizationCodeManpik
10019	RETVAL_SWITCH_TRAN_FILE_NOTHING_TO_SEND	No records to send from Switch TRAN file	אין רשומות לשליחה בקובץ TRAN של מתג
10020	RETVAL_NO_PAPER_IN_PRINTER	No paper in printer	חסר נייר במדפסת
10021	RETVAL_SWITCH_TRAN_FILE_IS_EMPTY	Switch TRAN file is empty	קובץ TRAN של מתג – ריק
10022	RETVAL_INCORRECT_ECR_NUMBER	Incorrect ECR number	מספר קופה שגוי
10023	RETVAL_NO_STATIS_FILE_ON_POS	No STATIS file on Pinpad	אין קובץ STATIS
10024	RETVAL_SWITCH_TRAN_HAS_UNSENT_TRANSACTIONS	There are unsent transactions in the Switch TRAN	ישנן רשומות שלא נשלחו בקובץ TRAN של מתג
10025	RETVAL_SWITCH_TRANSACTIONS_SENT_FAILED	Failure in sending Switch transactions	כשלון בשליחה של רשומות של מתג
10026	RETVAL_TRAN_STEPS_LISTENER_NOT_WORKING	Transaction-steps-listener is not working	מאזין שלבים אינו פעיל
10027	RETVAL_INVALID_MESSAGE_CODE	Invalid message code	קוד הודעה שגוי
10028	RETVAL_CANT_CONTINUE_J2	Can't continue a J2 transaction	לא ניתן להמשיך J2
10029	RETVAL_INVALID_REQUEST_ID	Invalid request Id	מזהה בקשה שגוי
10030	RETVAL_SHVA_PARAMETERS_DELETION_FAILED	Failure when trying to delete Shva parameters	כשלון במחיקת פרמטרים של שב"א
10031	RETVAL_SHVA_BLACKLIST_DELETION_FAILED	Failure when trying to delete Shva blacklist	כשלון במחיקת רשימת חסומים של שב"א
10032	RETVAL_TIMEOUT_EVENT_FROM_CASPIT_DLL	Timeout event from Caspit DLL	הDLL החזיר טיימאאוט
10033	RETVAL_SHVA_CALL_FAILED	Shva communication failed	כשלון בתקשורת לשב"א
10034	RETVAL_REPORTS_SERVER_CALL_FAILED	Failure to call reports server	כשלון בתקשורת לשרת דוחות
10035	RETVAL_XML_INVALID_METHOD	Invalid method in XML	מתודה שגויה
10036	RETVAL_XML_MISSING_TERMINAL_ID_TAG	Missing terminal Id tag in XML	חסר תג Terminal Id
10037	RETVAL_CERTIFICATES_SERVER_CALL_FAILED	Failure calling certificates server	כשלון בתקשורת לשרת סרטיפיקטים
10038	RETVAL_SWITCH_CALL_FAILED	Failure calling the Switch	כשלון בתקשורת למתג

Code	Define	English	Hebrew
10039	RETVAL_INVALID_TRAN_RECORD_NUMBER	Invalid TRAN record number	מספר רשומה שגוי
10040	RETVAL_FILE_CANNOT_BE_RETRIEVED	File cannot be retrieved	לא ניתן לאחזר קובץ
10041	RETVAL_ERROR_CONNECTING_TO_PINPAD	Error connecting to Pinpad	שגיאה בהתחברות לפינפד
10042	RETVAL_XML_MISSING_REQUEST_ID_TAG	Missing request Id tag in XML	חסר תג Request Id
10043	RETVAL_CASPIT_WINDOWS_SERVICE_NOT_RESPONDING	Windows service not responding	Windows Service לא מגיב
10044	RETVAL_USER_ABORTED_TRANSACTION	User aborted transaction	משתמש ביטל את העסקה
10045	RETVAL_TIMEOUT_CHECK_TRAN	Timeout. You should check the status of the last transaction	טיימאאוט. יש לבדוק את את העסקה האחרונה כי הסטטוס שלה לא ברור
10046	RETVAL_XML_MISSING_TIMEOUT_TAG	Missing timeout tag in XML	חסר תג Timeout in seconds
10047	RETVAL_XML_MISSING_INT_IN_TAG	Missing INT_IN tag in XML	חסר תג INT_IN
10048	RETVAL_CHECK_OTHER_STATUSES	Unknown error. You should check other statuses	שגיאה לא ברורה. יש לבדוק שדות סטטוסים אחרים
10049	RETVAL_TRANSACTION_NOT_FOUND	Transaciton not found	עסקה לא נמצאה
10050	RETVAL_DOUBLE_TRANSACTION	Double transaction	זוהתה עסקה כפולה
10051	RETVAL_BATCH_NUMBER_DOESNT_EXIST	Batch number doesn't exist	מספר קובץ לא קיים
10052	RETVAL_INVALID_BLOCK_NUMBER_TO_RESUME	Invalid block number to resume	לא ניתן להמשיך מבלוק זה
10053	RETVAL_PINPAD_RETURNED_EMPTY_RESPONSE	Pinpad returned empty response	פינפד החזיר תשובה ריקה
10054	RETVAL_ASHRAIT_INVALID_SETUP	Invalid setup in Ashrait	הקמה לא תקינה של אשראית
10055	RETVAL_NO_PING	No Ping	אין פינג
10056	RETVAL_UNSUPPORTED_GENERIC_COMMAND_CODE	Unsupported generic command code	קוד פקודה כללית לא נתמך
10057	RETVAL_TRANSMIT_LOGS_FAILED	Failure transmitting logs	כשלון בשידור לוגים
10058	RETVAL_NO_CONNECTION_TO_SHVA	No connection to Shva	אין חיבור לשב"א
10059	RETVAL_DELETE_LOGS_FAILED	Failure deleting logs	כשלון במחיקת לוגים
10060	RETVAL_CASPIT_TECH_CENTER_FAILED	Failure calling technician	כשלון בתקשורת למרכז

Code	Define	English	Hebrew
		center	טכנאים של כספיט
10061	RETVAL_TERMINAL_REBOOT_FAILED	Terminal reboot failed	כשלון בביצוע reboot
10062	RETVAL_STARTING_TMS_CALL_FAILED	TMS call failed	כשלון בביצוע תקשורת למרכז טעינות תוכנה
10063	RETVAL_CLEAN_COMMUNICATION_ENGINE_FAILED	Failure cleaning communication engine	כשלון בניקוי מנוע תקשורת
10064	RETVAL_SETTING_DATE_TIME_FAILED	Failure setting date and time	כשלון בעדכון שעון
10065	RETVAL_TRANSACTIONS_CENTER_CALL_FAILED	Transaction center call failed	כשלון בתקשורת למרכז טרנזקציות של כספיט
10066	RETVAL_TRANSACTIONS_CENTER_TOGGLE_FAILED	Transaction center toggle failed	כשלון בהפעלת שירות מרכז טרנזקציות
10099	RETVAL_UNKNOWN_ERROR	Unknown error	שגיאה לא ברורה
10100	RETVAL_INVALID_PAN_ENTRY_MODE	Invalid PanEntryMode	ערך שגוי בPanEntryMode
10101	RETVAL_INVALID_MTI	Invalid MTI	ערך שגוי בMTI
10102	RETVAL_INVALID_X_FIELD	Invalid Xfield	ערך שגוי בXfield
10103	RETVAL_CANT_FIND_DEPOSIT_REPORT	Can't find deposit report	לא ניתן למצוא דוח הפקדה
10104	RETVAL_INVALID_SESSION_NUMBER_XML_TAG	Invalid session number	מספר session שגוי
10105	RETVAL_INVALID_DATE_XML_TAG	Invalid date XML tag	תאריך שגוי
10106	RETVAL_NO_TOTAL_FILE	No TOTAL file	קובץ TOTAL לא קיים
10501	RETVAL_TIMEOUT_WHILE_WAITING_FOR_SWIPE	Timeout while waiting for card	ארע טיימאאוט בזמן המתנה לכרטיס
10502	RETVAL_TIMEOUT_AFTER_FAILED_SWIPE_ATTEMPT	Timeout while waiting for card after a filed attempt	ארע טיימאאוט בזמן המתנה לכרטיס לאחר לפחות ניסיון כושל אחד להעביר כרטיס
10503	RETVAL_CANCEL_WITHOUT_SWIPING	User cancel the transaction before presenting card	המשתמש ביטל את העסקה לפני העברת כרטיס
10504	RETVAL_CANCEL_AFTER_FAILED_SWIPE_ATTEMPT	User cancel the transaction after failed attempt of presenting card	המשתמש ביטל את העסקה לאחר לפחות ניסיון כושל אחד להעביר כרטיס
11000	RETVAL_CANNOT_SEND_OUT_CREDIT_CARDS	Pinpad cannot send out	לא ניתן להוציא מספרי

Code	Define	English	Hebrew
		credit card details	כרטיס אשראי מהפינפד
11001	RETVAL_INVALID_CARD_SWIPE	Invalid card swipe	העברת כרטיס שגויה
11002	RETVAL_FILE_BAD_RECORD	Invalid record in file	רשומה לא תקינה בקובץ
12001	RETVAL_INVALID_VAS_CARD	Card read error	
12002			
12003	RETVAL_CONTINUE_WITH_GOOGLE	Continue with google	
12004			
12005	RETVAL_CONTINUE_WITH_PAYMENT	Continue with payment	
12006			
12007			
12008	RETVAL_NOT_ACTIVATED	Not activated	
12009	RETVAL_OTHER_WALLET	Other wallet	
>=900000	Switch error	Switch error	תקלת מתג

Appendix B - PTL Header

The XML string that is sent from the ECR to the Pinpad must be prefixed with a PTL header. If the ECR uses also the DLL from Caspit then the DLL will take care of adding the PTL header. However, if the ECR doesn't use the DLL from Caspit then the ECR must add the PTL header by itself.

Because the Pinpad can support multiple applications, the Pinpad needs the PTL header so it can know which application to route the XML to.

Pay attention, the PTL header exists only on the way from the ECR to the Pinpad. On the way back, from the Pinpad to the ECR, there is NO need for a PTL header.

1. Generic Packet format:

PTL Header (16 bytes)	Packet Body
^PTL!00#<LLLL><TT><FF>	Data in length LLLL hex. This data can include anything, even binary data. But in our case, the packet body will be the XML string.

2. Packet example: Sending 387 bytes from the ECR to the Ashrait EMV application. The total packet from the ECR to the Pinpad will be 387 bytes of data + 16 bytes of header.

Packet Header (16 bytes)	**Packet Body**
^PTL!00#01835102 Target = 51 (Ashrait EMV application) Source = 02 (The ECR)	387 bytes of XML data (0x0183h = 387d). <Request> …. </Request>

3. The PTL header format in details:

Constant string	**LLLL**	**TT**	**FF**
^PTL!00#	Length of packet body (4 bytes). In HEX.	Target host/app code (2 bytes)	From host/app code (2 bytes)

4. The length of the packet will be represented in ASCII HEX, here are a few examples:
   1. 0001 – Packet body length is 1 bytes (minimum packet length)
   2. FFF0 – Packet body length is 65520 bytes (which is the maximum packet length)
   3. 01BA – Packet body length is 442 bytes.

5. The following table show the current host/app codes used with PTL header. More codes will be added in the future as we develop more applications for the Pinpad.

Host/App Code	Destination	Telium Application type (Caspit Internal)
00	The Router application	51037
01	Not used	N/A
02	ECR	N/A
03	Caspit Payment Windows Service	
51	Ashrait EMV app	51033
52	SmartRetail app	51042
53	Caspit ATM app	51032
54	Customer Display app	51039
55	Contactless application (obsolete)	51025
56	Signature Capture app	51045

Appendix C – Commands Table

Command	Description
001	Payment transaction
002	Get JENR file
003	Communication test
004	FFU
005	Get Total file
006	Transmit transactions to Shva
007	Get TRAN file
008	Get DATA file
009	FFU
010	Generic command
011	FFU
012	Transaction query
013	Pinpad configuration
014	Get Statis
015	Get deposit report
016	Retrieve file
017	FFU
018	FFU
019	UI Command
020	FFU
021	FFU
022	FFU
023	Swipe request

Appendix D – Glossary

Term	Description
Ashrait EMV app	An application on the Pinpad. This is the core application that does all the EMV stuff.
CPA	Contactless Payment Application – The application on the Pinpad that handles the contactless transaction
ECR	Electronic Cash Register קופה ממוחשבת
EMV	The all ecosystem that deals with smart cards An Acronym for: **E**uropay **M**asterCard **V**isa
EPA	EMV Payment Application – The application on the Pinpad that handles the contact transaction
Ingenico	The world leading manufacturer of payment devices
MSR	Magnetic Swipe Reader קורא פס מגנטי
PAN	Personal Account Number. The full credit card number (4580….) מספר כרטיס אשראי
PCL	Payment Communication Layer A software/hardware layer that creates the channel that enables the ECR to communicate with the Pinpad and enables the Pinpad to use the ECR machine as a tunnel to the outside world. The PCL component is developed by Ingenico
PIN	Personal Identification Number. The secret code of the credit card. קוד סודי
Pinpad	A device that accepts all kind of cards: magnetic, contact and contactless, and then performs the transaction.
Router app	An application on the Pinpad that routes the request packets, that comes from the ECR, to the right applications. And then routes back the response to the ECR.
Shva	ABS – Automatic Banking Systems. The main payment switch of Israel. שב"א – שירותי בנק אוטומטיים
SmartRetail app	An application on the Pinpad that gives the ECR - the API to control the Pinpad
Switch	A server that will "stand" between the Pinpad and Shva and will be responsible for passing all the communication going between the Pinpad and Shva. The Switch will also accumulate all the transactions by himself, instead of the Pinpad. מתג אשראי
Telium	A family of Pinpads. It includes the following Pinpad models: iPP320, iPP350, iSC250, iCMP, iSMP

Term	Description
Tetra	A family of Pinpads. It includes the following Pinpad models: Lane5000, Lane7000, Lane8000.
TMS	Terminal Management System. A system that takes care of software downloads and updates for the Pinpads. The Pinpad will call the TMS every night to check for updates. שרת טעינות תוכנה

Appendix E – SmartRetail Tester

The SmartRetail Tester is a Windows application written in VB.NET.

The Tester is using the CaspitPayment.dll

This tester gives you the option to test all the features of the Pinpad SmartRetail application and with the help of the tester, you can learn the correct XML that is needed to perform EMV transactions and how to perform other commands and use all the features of the SmartRetail application.

Caspit can provide the full source code of the tester.

TODO: add screenshot of Tester

Appendix F – Caspit Payment Windows Service

If your ECR is Windows based, then you will be using the Caspit Payment Windows Service. This Windows service is encapsulating the PCL component that enables the communication between Windows and the Pinpad. The Windows service is another "link" in the communication chain between the ECR software and the SmartRetail application on the Pinpad.

If you are using Android OS or iOS then you will need to handle the PCL by yourself.

This appendix is relevant only to Windows based ECR that uses the Caspit Payment Windows Service.

The Caspit Payment Service is supported on Windows 7 and Windows 10 (XP is not supported)

The Caspit Payment Service has a TCP listener that waits for data coming from the ECR. Once the Service got the data, the Service will send it to the Pinpad.

The ECR can work in two ways:

Using a DLL

The DLL is developed by Caspit and it is written in C#.

What the DLL does is simply handling the TCP communication for the ECR. In addition, beside communication, the DLL gives you the option to work with a .NET Object and properties instead of writing and parsing XMLs.

Using a DLL, the flow of transaction will look like this:

ECR → DLL → Windows Service → Pinpad → Windows Service → DLL → ECR

Without a DLL

If you prefer to work without the DLL, it is your choice.

Without the DLL, the ECR will be responsible to handle the communication between the ECR and the Caspit Payment Service. If you are not using the DLL, you must use the PTL header before any XML string that you send to the Caspit Payment Service (see Appendix B about the PTL header).

ECR → Windows Service → Pinpad → Pinpad → Windows Service → ECR

Appendix F – Communication Problems

1. No matter if the ECR is using a DLL or not using a DLL, if the Caspit Payment Service cannot connect to the Pinpad then the Service will return the following XML (this XML will be created by the Service)

<Response>

<ResultCode>10041</ResultCode> (RETVAL_ERROR_CONNECTING_TO_PINPAD)

</Response>

2. If the ECR is using the Caspit DLL and the **Caspit DLL** cannot connect to the Caspit Payment Service, then the DLL will return the following XML (this XML will be created by the DLL)

<Response>

<ResultCode>10043</ResultCode> (RETVAL_CASPIT_WINDOWS_SERVICE_NOT_RESPONDING)

</Response>

3. If the ECR is NOT using the Caspit DLL, then it is the responsibility of the ECR to handle the TCP connection to the Caspit Payment Service. And if the Service is not running or the Service doesn't answer then the ECR will have to handle these problems. Just to make things clear, the part that is in the responsibility of the ECR is just the part that is relevant to the TCP socket creation and connection. If the TCP connection has been established and there is a problem between the Service and the Pinpad then the Service will return the XML (with result code 10041) as described above.

4. If, for some reason, the Pinpad returned an empty response, the Service will return the following XML

<Response>

<ResultCode>10053</ResultCode> (RETVAL_PINPAD_RETURNED_EMPTY_RESPONSE)

</Response>

5. If a timeout event happended in the Caspit DLL, the DLL will return the following XML:

<Response>

<ResultCode>10032</ResultCode> (RETVAL_TIMEOUT_EVENT_FROM_CASPIT_DLL)

</Response>

Appendix F – Getting Windows Service Information

If the Windows Service is working, the ECR can get some information about the Service.

The ECR needs to send an XML that is targeted at the Windows Service, **the AppCode in the PTL header will be 03**.

<Request>

<Command>GetInfo</Command>

</Request>

The response will be:

<Response>

<Command>GetInfo</Command>

<ServiceVersion>1.0.5</ServiceVersion>

<PCLVersion>2.10</PCLVersion>

<SerialPortName>COM5</SerialPortName>

<TimeoutSocket>30</TimeoutSocket>

```xml
<PCLStatus>1</PCLStatus>
<IPPort>6656</IPPort>
<LogEnabled>true</LogEnabled>
<ArchiveDays>10</ArchiveDays>
</Response>
```

Tag name	**Description**	**Matching TLV tag (Caspit internal)**
Response	The opening tag of the Response XML	N/A
Command	GetInfo	001
ServiceVersion	The version of the Caspit Payment Windows Service software	006
PCLVersion	The version of the PCL component from Ingenico	040
SerialPortName	Like COM1	051
TimeoutSocket		
PCLStatus	0 – PCL is not connected 1 – PCL is connected.	010
IPPort		
LogEnabled		
ArchiveDays		
Response	Closing tag of the response XML	N/A

Appendix F – Windows Service Config.xml

```xml
<?xml version="1.0" encoding="utf-8" ?>
<Configuration>
  <Settings id="Service">
    <SerialPortName>COM5</SerialPortName>
    <!--Possible Communication Types: 0 - Bluetooth, 1 - USB, 2 - RS232 -->
    <CommunicationType>1</CommunicationType>
    <IPAddress></IPAddress>
    <IPPort>6656</IPPort>
    <TimeoutSocket>30</TimeoutSocket>
```

```
  </Settings>
  <Settings id="Logger">
    <IsEnabled>true</IsEnabled>
    <IPAddress></IPAddress>
    <IPPort>0</IPPort>
    <SendLogByTCP>false</SendLogByTCP>
    <ArchiveDays>10</ArchiveDays>
  </Settings>
</Configuration>
```

Appendix G – Idle Swipe

There are three "Swipe" features in the Pinpad, some may find it a bit confusing, and so I will summarize them here:

- The **Swipe Command** is described in chapter 21. It is a command that is initiated by the PC.
- The **Smart Swipe** is a feature that changes a transaction request into a Swipe response. See chapter 9.5.
- The **Idle Swipe** is a feature that allows swiping cards while the Pinpad is in idle mode. It is described here.

Idle Swipe refers to the feature of the Pinpad to read magnetic cards **not during the transaction**, but while the Pinpad is in idle state (some people refer to Idle Swipe as "passive swipe"). Although we have the Swipe Command (chapter 21) that enables the ECR to ask for a magnetic swipe, the Swipe Command is initiated by the ECR. But some retailers require the ability to read magnetic cards without the initiation of the ECR. They need it because their ECR business logic is based on a simple MSR (magnetic stripe reader) and they don't wish to change all their normal business logic.

While the Pinpad is not doing anything (i.e. not in the middle of a transaction), we say that it is in idle state. A cardholder can swipe his magnetic card and the card track2 data will be sent to the ECR. The ECR can do whatever it needs to do with that data. Note that the Pinpad will not send out track2 data that is identified as a credit card. Only non-credit card data will be sent out to the ECR. This is the same behavior as in the Swipe Command, and it is mandatory because of PCI regulations.

TCP Listener

Because the Pinpad and the ECR are using TCP/IP to communicate with each other (even with USB connection, there is a layer of TCP/IP on top of the USB later), the only way to send the track2 data is by using TCP. Therefore, the ECR must keep a TCP listener opened all the time. Once a card is swipe, the Pinpad will open a TCP socket to the IP of the ECR and then the Pinpad will send the track2 data to the ECR. After sending the data, the Pinpad will close the TCP connection. I would like to emphasize this point: the TCP connection between the Pinpad and the ECR will NOT stay opened all the time. It will be opened and then closed for each idle swipe that is done. The USB connection has a limited bandwidth for the TCP layer and we cannot leave the TCP connection all the time. Otherwise, it will interrupt the other Pinpad communication needs. Pay attention that the TCP connection is always done **from** the Pinpad to the ECR and not the other way around. The Pinpad does not have a TCP listener on his side.

One possible way of using the Idle Swipe is by developing the TCP listener as a small application, running on the PC, and that listener can catch the Idle Swipe event (XML), parse the XML and then send the track2 data as a keyboard input. This will enable the ECR software to keep its normal way of getting magnetic card data, for example, loyalty cards or employee cards.

You need to use the Pinpad Configuration command to enable/disable the Idle Swipe.

Following is an example of a Pinpad Configuration command with only the relevant tags to enable the Idle Swipe.

<Request>

<Command>013</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<EnableIdleSwipe>1</EnableIdleSwipe>

<IdleSwipeAddress>192.168.0.1/1200</IdleSwipeAddress>

<RequestId>20160216155540</RequestId>

</Request>

(I don't show here the response of the Pinpad Configuration command)

The data sent by the Pinpad, on an Idle Swipe event, will be the same XML that is sent in the Swipe Command response. The ECR does not respond to the Idle Swipe XML that is coming from the Pinpad.
The Idle Swipe TCP communication is a one way direction (from the Pinpad to the ECR).

<Response>

<Command>023</Command>

<ResultCode>0</ResultCode>

<Track1>23093809348503</Track1> **(if exists)**

<Track2>4580458045804580230298304294809З</Track2>

<Track3>3095830958309530459384O</Track3> **(if exists)**

</Response>

If the swiped card is identified as a credit card, the response will be like this:

<Response>

<Command>023</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<ResultCode>11000</ResultCode>

<RequestId>20160216155540</RequestId>

<CardHash>2F9A8B7C2F9A8B7C2F9A8B7C2F9A8B7C</CardHash>

</Response>

Appendix H – Shva <Status> codes

Copied from Appendix 1 of Shva mefizim EMV document.

Status	Hebrew Description	English Description
0000	מאושר	Approved
1001	העבר\הכנס\הצג כרטיס	Swipe/Insert/Present card
1002	העבר\הצג כרטיס	Swipe/Present card
1003	העבר\הכנס כרטיס	Swipe/Insert card
1004	הכנס כרטיס לקורא חכם	Insert card into card reader
1005	הכנס\הצג כרטיס	Insert/Present card
1006	הצג כרטיס	Present card
1007	נא הוצא כרטיס	Please remove card
1008	העבר שנית	Swipe again
1009	השתמש בקורא פס מגנטי	Use magnetic reader
1010	המסוף לא רשאי לבצע עיסקאות בכרטיס	The terminal is not allowed to perform transactions in this card
1011	אנא המתן השאר כרטיס	Please wait. Leave card.
1012	כרטיס לא תקין	Card invalid
1013	כרטיס לא נתמך	Card not supported
1014	קיימות עסקאות ישנות. בצע שידור	Old transactions. Call Shva
1015	קובץ חסום לא מעודכן. בצע שידור	Old black list. Call Shva
1016	כרטיס לא מורשה לסוג אשראי	Card not allowed for this credit term
1017	אין הרשאה לסוג מטבע	No permission for this currency
1018	מספר כרטיס שגוי	Invalid card number
1019	הכרטיס לא מורשה במסוף	Card is not allowed in this terminal
1020	כרטיס לא מורשה לסוג העסקה	Card is not allowed for this transaction type
1021	הכרטיס מחוץ לתחום	Card out of limit
1022	קוד מועדון לא בתחום	Club code out of limit
1023	מנפיק לא קיים	Issuer doesn't exist
1024	מותג לא קיים	Brand doesn't exist
1025	מספר רכב לא תקין	Invalid card license plate
1026	כרטיס לא בתוקף	Card expired
1027	נא הקלד נתונים נוספים	Please enter additional data
1028	הקש 3 ספרות אחרונות בגב הכרטיס	Enter 3 digits from back of the card
1029	הקש 4 ספרות מודפסות בכרטיס	Enter 4 digits from back of the card

Status	Hebrew Description	English Description
1030	נא הקש ת.ז. כולל ספרת ביקורת	Enter ID
1031	ת.ז. שגויה	Invalid ID
1032	ת.ז. שגויה נסה שנית	Invalid ID. Try again.
1033	נא הקש כתובת לקוח	Enter customer address
1034	לאלץ או להתקשר	Force or call
1035	הקשה F1 לבקשת אישור	Press F1 to authorize
1036	התקשר לחברת אשראי	Call the credit company
1037	מאושר, הוצא כרטיס	Approved. Remove card.
1038	העסקה לא אושרה	Declined
1039	דחייה. תקלה בתקשורת	Declined. Communication problem.
1040	דחה עסקה cvv2 או ת.ז. שגוי	Declined. Wrong CVV2 or ID
1041	דחייה. כרטיס לא משוייך לרשת	Declined. Card doesn't belong to network
1042	דחה עסקה. cavv/ucaf שגוי	Declined. Invalid cavv/ucaf
1043	דחה עסקה. כתובת שגויה	Declined. Invalid address
1044	כרטיס גנוב. החרם	Stolen card
1045	כרטיס חסום	Blocked card
1046	כרטיס מזוייף. החרם	Forged card
1047	דחה בקשה. קוד יתרה שגוי	Declined. Invalid balance code
1048	דחה עסקה. חוסר בנק'/כוכבים/מיילים/הטבה אחרת	Declined. Missing benefit
1049	ממתין לאישור. השאר כרטיס	Please wait. Leave card
1050	העסקה אושרה. הוצא כרטיס	Approved. Remove card
1051	נא הקלד מס' אישור קולי שהתקבל מחברת האשראי	Please enter authorization number
1052	מהדורת תוכנה שגויה	Invalid software version
1053	נתוני עסקה שגויים	Invalid transaction details
1054	מסוף לא ביצע הקמה	Terminal was not setup
1055	כרטיס לא ניתמך, קוד שרות	Card not supported because of service code
1056	סולק לא קיים	Acquirer doesn't exist
1057	סט פרמטרים שגוי	Invalid parameters set
1059	דחייה, כרטיס נטען	Declined, gift card
1060	נתוני עסקת הטבה שגויים	Invalid benefit details
1061	אין הרשאה לביצוע עסקה	No permission for transaction
1062	נא הקש מיקוד הלקוח	Please enter zip code
1063	מזהה שגוי	Invalid identification

Status	**Hebrew Description**	**English Description**
2000	שגיאה אפליקטיבית	Applicative error

Appendix I – Shva <AshStatus> codes

Copied from Appendix 1 of Shva mefizim EMV document.

Status	Hebrew Description	English Description
	קודי שגיאה המתקבלים בתקשורת	**Error codes that come during communication**
000	מאושר	Approved
001	כרטיס חסום	Blocked card
002	גנוב החרם כרטיס	Stolen card
003	התקשר לחברת האשראי	Call credit company
004	העסקה לא אושרה	Declined
005	כרטיס מזוייף החרם	Forged card
006	דחה עסקה cvv2/id שגוי	Declined. Invalid cvv2/id
007	דחה עסקה cavv/ucaf שגוי	Declined. Invalid cavv/ucaf
008	דחה עסקה avs שגוי	Declined. Invalid AVS
009	דחייה – נתק בתקשורת	Declined. Communication problem.
010	אישור חלקי	Partial approval
011	דחה עסקה: חוסר בנקודות/כוכבים/מיילים/הטבה אחרת	Declined. Missing benefit.
012	הכרטיס לא מורשה במסוף	Card not allowed
013	דחה בקשה. קוד יתרה שגוי	Declined. Invalid balance code
014	דחייה. כרטיס לא משוייך לרשת	Declined.
015	דחה עסקה: הכרטיס אינו בתוקף	Declined. Card expired
016	דחייה – אין הרשאה לסוג מטבע	Declined. Currency not permitted
017	דחייה – אין הרשאה לסוג אשראי בעסקה	Declined. Credit term not permitted
026	דחה עסקה id שגוי	Declined. Invalid Id
041	ישנה חובת יציאה לשאילתא בגין תקרה בלבד לעסקה עם פרמטר j2	It is mandatory to go online because of ceiling-only for J2
042	ישנה חובת יציאה לשאילתא לא רק בגין תקרה, לעסקה עם פרמטר J2	It is mandatory to go online not only because of ceiling
	חוסר בקבצי מערכת	**System files missing**
051	חסר קובץ ווקטור 1	File vector 1 is missing
052	חסר קובץ ווקטור 4	File vector 4 is missing
053	חסר קובץ ווקטור 6	File vector 6 is missing
055	חסר קובץ ווקטור 11	File vector 11 is missing

Status	Hebrew Description	English Description
056	חסר קובץ ווקטור 12	File vector 12 is missing
057	חסר קובץ ווקטור 15	File vector 15 is missing
058	חסר קובץ ווקטור 18	File vector 18 is missin
059	חסר קובץ ווקטור 31	File vector 31 is missing
060	חסר קובץ ווקטור 34	File vector 34 is missing
061	חסר קובץ ווקטור 41	File vector 41 is missing
062	חסר קובץ ווקטור 44	File vector 44 is missing
063	חסר קובץ ווקטור 64	File vector 64 is missing
064	חסר קובץ ווקטור 80	File vector 80 is missing
065	חסר קובץ ווקטור 81	File vector 81 is missing
066	חסר קובץ ווקטור 82	File vector 82 is missing
067	חסר קובץ ווקטור 83	File vector 83 is missing
068	חסר קובץ ווקטור 90	File vector 90 is missing
069	חסר קובץ ווקטור 91	File vector 91 is missing
070	חסר קובץ ווקטור 92	File vector 92 is missing
071	חסר קובץ ווקטור 93	File vector 93 is missing
073	חסר קובץ PARAM_3_1	File PARAM_3_1 is missing
074	חסר קובץ PARAM_3_2	File PARAM_3_2 is missing
075	חסר קובץ PARAM_3_3	File PARAM_3_3 is missing
076	חסר קובץ PARAM_3_4	File PARAM _3_4 is missing
077	חסר קובץ PARAM_361	File PARAM_361 is missing
078	חסר קובץ PARAM_363	File PARAM_363 is missing
079	חסר קובץ PARAM_364	File PARAM_364 is missing
080	חסר קובץ PARAM_61	File PARAM_61 is missing
081	חסר קובץ PARAM_62	File PARAM_62 is missing
082	חסר קובץ PARAM_63	File PARAM_63 is missing
083	חסר קובץ CEIL_41	File CEIL_41 is missing
084	חסר קובץ CEIL_42	File CEIL_42 is missing
085	חסר קובץ CEIL_43	File CEIL_43 is missing
086	חסר קובץ CEIL_44	File CEIL_44 is missing
087	חסר קובץ DATA	File DATA is missing
088	חסר קובץ JENR	File JENR is missing

Status	Hebrew Description	English Description
089	חסר קובץ Start	File Start is missing
	חוסר כניסה מתאימה בקבצי הוקטורים	**Entry missing in vector files**
101	חסרה כניסה בוקטור 1	Entry is missing in vector 1
103	חסרה כניסה בוקטור 4	Entry is missing in vector 4
104	חסרה כניסה בוקטור 6	Entry is missing in vector 6
106	חסרה כניסה בוקטור 11	Entry is missing in vector 11
107	חסרה כניסה בוקטור 12	Entry is missing in vector 12
108	חסרה כניסה בוקטור 15	Entry is missing in vector 15
110	חסרה כניסה בוקטור 18	Entry is missing in vector 18
111	חסרה כניסה בוקטור 31	Entry is missing in vector 31
112	חסרה כניסה בוקטור 34	Entry is missing in vector 34
113	חסרה כניסה בוקטור 41	Entry is missing in vector 41
114	חסרה כניסה בוקטור 44	Entry is missing in vector 44
116	חסרה כניסה בוקטור 64	Entry is missing in vector 64
117	חסרה כניסה בוקטור 81	Entry is missing in vector 81
118	חסרה כניסה בוקטור 82	Entry is missing in vector 82
119	חסרה כניסה בוקטור 83	Entry is missing in vector 83
120	חסרה כניסה בוקטור 90	Entry is missing in vector 90
121	חסרה כניסה בוקטור 91	Entry is missing in vector 91
122	חסרה כניסה בוקטור 92	Entry is missing in vector 92
123	חסרה כניסה בוקטור 93	Entry is missing in vector 93
	חוסר כניסה מתאימה בקבצי הפרמטרים	**Entry missing in parameter files**
141	חסרה כניסה מתאימה בקובץ פרמטרים 3.2	Entry is missing in parameter file 3.2
142	חסרה כניסה מתאימה בקובץ פרמטרים 3.3	Entry is missing in parameter file 3.3
143	חסרה כניסה בקובץ תחומי מועדון 3.6.1	Entry is missing in club range file 3.6.1
144	חסרה כניסה בקובץ תחומי מועדון 3.6.3	Entry is missing in club range file 3.6.3
145	חסרה כניסה בקובץ תחומי מועדון 3.6.4	Entry is missing in club range file 3.6.4
146	חסרה כניסה בקובץ תקרות לכרטיסי PL 4.1	Entry is missing in ceiling file for PL cards 4.1
147	חסרה כניסה בקובץ תקרות לכרטיסים ישראלים שאינם PL שיטה 0 4.2	Entry is missing in ceiling file for Israeli cards which are not PL. Method 0 4.2
148	חסרה כניסה בקובץ תקרות לכרטיסים ישראלים שאינם PL שיטה 1 4.3	Entry is missing in ceiling file for Israeli cards which are not PL. Method 1 4.3

Status	Hebrew Description	English Description
149	חסרה כניסה בקובץ תקרות לכרטיסי תייר 4.4	Entry is missing in Ceiling file for tourist cards 4.4
150	חסרה כניסה בקובץ כרטיסים תקפים – ישראכרט	Entry is missing in valid cards file – Isracard
151	חסרה כניסה בקובץ כרטיסים תקפים – כאל	Entry is missing in valid cards file – Cal
152	חסרה כניסה בקובץ כרטיסים תקפים – מנפיק עתידי	Entry is missing in valid cards file – Future issuer
	שגיאות בערכי נתוני קבצי מערכת	**Errors in system files**
182	שגיאה בערכי וקטור 4	Error in vector 4
183	שגיאה בערכי וקטור 6/12	Error in vector 6/12
186	שגיאה בערכי וקטור 18	Error in vector 18
187	שגיאה בערכי וקטור 34	Error in vector 34
188	שגיאה בערכי וקטור 64	Error in vector 64
190	שגיאה בערכי וקטור 90	Error in vector 90
191	נתונים לא תקינים בוקטור הרשאות מנפיק	Invalid data in issuer permissions vector
192	נתונים לא ולידים בסט הפרמטרים	Invalid data in parameters set
193	נתונים לא ולידים בקובץ פרמטרים ברמת מסוף	Invalid data in terminal parameters file
	הרשאות סולק	**Acquirer permissions**
300	אין הרשאה לסוג עסקה – הרשאת סולק	No permission for transaction type
301	אין הרשאה למטבע – הרשאת סולק	No permission for currency
303	אין הרשאת סולק לביצוע עסקה כאשר הכרטיס לא נוכח	No permission for card not present transaction
304	אין הרשאה לאשראי – הרשאת סולק	No permission for credit
308	אין הרשאה להצמדה – הרשאת סולק	No permission for index
309	אין הרשאת סולק לאשראי במועד קבוע	No permission for recurring payment
310	אין הרשאה להקלדת מספר אישור מראש	No permission for authorized transaction
311	אין הרשאה לבצע עסקאות לקוד שרות 587	No permission for service code 587
312	אין הרשאת סולק לאשראי דחוי	No permission for delayed credit
313	אין הרשאת סולק להטבות	No permission for benefits
314	אין הרשאת סולק למבצעים	No permission for sales
315	אין הרשאת סולק לקוד מבצע ספציפי	No permission for specific sale code
316	אין הרשאת סולק לעסקת טעינה	No permission for charge transaction
317	אין הרשאת סולק לטעינה/פריקה בקוד אמצעי התשלום בשילוב קוד מטבע	No permission for to charge/discharge with that currency
318	אין הרשאת סולק למטבע בסוג אשראי זה	No permission for that currency with that credit term

Status	**Hebrew Description**	**English Description**
319	אין הרשאת סולק לטיפ	No permission for tip
	הרשאות מנפיק	**Issuer permissions**
341	אין הרשאה לעסקה – הרשאת מנפיק	No permission for transaction
342	אין הרשאה למטבע – הרשאת מנפיק	No permission for currency
343	אין הרשאת מנפיק לביצוע עסקה כאשר הכרטיס לא נוכח	No permission for card not present transaction
344	אין הרשאה לאשראי – הרשאת מנפיק	No permission for credit
348	אין הרשאה לביצוע אישור בקשה יזומה ע"י קמעונאי	No permission for must authorized transaction
349	אין הרשאה מתאימה לביצוע בקשה לאישור ללא עסקה J5	No permission for pre authorization
350	אין הרשאת מנפיק להטבות	No permission for benefits
351	אין הרשאת מנפיק לאשראי דחוי	No permission for delayed credit
352	אין הרשאת מנפיק לעסקת טעינה	No permission for charge transaction
353	אין הרשאת מנפיק לטעינה/פריקה בקוד אמצעי התשלום	No permission for charge/discharge with that credit term
354	אין הרשאת מנפיק למטבע בסוג אשראי זה	No permission for that currency with that credit term
	הרשאות מסוף	**Terminal permissions**
381	אין הרשאה לבצע עסקת contactless מעל סכום מרבי	No permission for contactless above this amount
382	במסוף המוגדר כשרות עצמי ניתן לבצע רק עסקאות בשירות עצמי	Only self service transaction can be done in a self service terminal
384	מסוף מוגדר כרב-ספק/מוטב – חסר מספר ספק/מוטב	Missing rav-sapak/mutav number
385	במסוף המוגדר כמסוף סחר אלקטרוני חובה להעביר eci	Missing ECI
	שגיאות אפליקטיביות	**Application errors**
401	מספר התשלומים גדול מערך שדה מספר תשלומים מקסימלי	Number of installments is above the maximum
402	מספר התשלומים קטן מערך שדה מספר תשלומים מינימלי	Number of installments is below the minimum
403	סכום העסקה קטן מערך שדה סכום מינימלי לתשלום	The amount is below the minimum amount
404	לא הוזן שדה מספר תשלומים	Missing number of installments
405	חסר נתון סכום תשלום ראשון/קבוע	Missing first installment amount or other installments amount
406	סה"כ סכום העסקה שונה מסכום תשלום ראשון + סכום תשלום קבוע * מספר תשלומים	The total number of transaction doesn't match the installments details

Status	Hebrew Description	English Description
408	ערוץ 2 קצר מ-37 תווים	Track2 is shorter than 37 characters
410	דחיה מסיבת dcode	Decline because of dcode
414	בעסקה עם חיוב בתאריך קבוע הוכנס תאריך מאוחר משנה מבצוע העיסקה	Invalid date in recurring transaction
415	הוזנו נתונים לא תקינים	Invalid data
416	תאריך תוקף לא במבנה תקין	Expiration date format is invalid
417	מספר מסוף אינו תקין	Invalid terminal id
418	חסרים פרמטרים חיוניים	Missing essential parameters
419	שגיאה בעברת מאפיין clientInputPan	Invalid clientInputPan
420	מספר כרטיס לא ולידי – במצב של הזנת ערוץ 2 בעסקה ללא כרטיס נוכח	Invalid card number
421	שגיאה כללי – נתונים לא ולידים	Invalid data
422	שגיאה בבנית מסר ISO	Error building ISO message
424	שדה לא נומרי	Field is not numeric
425	רשומה כפולה	Double record
426	הסכום הוגדל לאחר ביצוע בדיקות אשראית	Amount was incremented after performing payment application logic
428	חסר קוד שרות בכרטיס	Missing service code in card
429	כרטיס אינו תקף לפי קובץ כרטיסים תקפים	Card expired according to valid cards file
431	שגיאה כללית	General error
432	אין הרשאה להעברת כרטיס דרך קורא מגנטי	MSR is not allowed
433	חיוב להעביר ב- PinPad	Must use Pinpad
434	אסור להעביר כרטיס במכשיר ה -PinPad	Pinpad is not allowed
435	המכשיר לא מוגדר להעברת כרטיס מגנטי CTL	The device is not configured for Magnetic Contactless
436	המכשיר לא מוגדר להעברת כרטיס EMV CTL	The device is not configured for EMV Contactless
439	אין הרשאה לסוג אשראי לפי סוג עסקה	Credit term is not allowed for that transaction type
440	כרטיס תייר אינו מורשה לסוג אשראי זה	Tourist card is not allowed for this credit term
441	אין הרשאה לביצוע סוג עסקה – כרטיס קיים בוקטור 80	No permission for transaction type – Card exists in vector 80
442	אין לבצע Stand-in לאימות אישור לסולק זה	Can't perform Stand-in
443	לא ניתן לבצע עסקת ביטול – כרטיס לא נמצא בקובץ תנועות הקיים במסוף	Can't void transaction – Card doesn't exist in TRAN file
445	בכרטיס חיוב מיידי ניתן לבצע אשראי חיוב מיידי בלבד	Immediate card can be used only with credit term immediate
447	מספר כרטיס שגוי	Invalid card number

Status	Hebrew Description	English Description
448	חיוב להקליד כתובת לקוח (מיקוד, מספר בית ועיר)	Must enter AVS
449	חיוב להקליד מיקוד	Must enter ZIP
450	קוד מבצע מחוץ לתחום, צ"ל בתחום 12 – 1	Sale code out of range. Must be between 1 – 12
451	שגיאה במהלך בנית רשומת עסקה	Error while building transaction record
452	בעסקת טעינה/פריקה/בירור יתרה חיוב להזין שדה קוד אמצעי תשלום	Must enter payment method code
453	אין אפשרות לבטל עסקה פריקה 7.9.3	It is not possible to void a discharge transaction
455	לא ניתן לבצע עסקת חיוב מאולצת כאשר נדרשת בקשה לאישור (למעט תקרות)	Cannot force transaction when authorization is needed
456	כרטיס נמצא בקובץ תנועות עם קוד תשובה 'החרם כרטיס'	Card is insideTRAN file with response code "Confiscate card"
457	בכרטיס חיוב מיידי מותרת עסקת חיוב רגילה/זיכוי/ביטול	Immediate card – Only debit/refund and void are possible
458	קוד מועדון לא בתחום	Club code out of range
472	בעסקת חיוב עם מזומן חיוב להזין סכום במזומן	Must enter cash amount in a cashback transaction
473	בעסקה חיוב עם מזומן סכום המזומן צריך להיות קטן מסכום העסקה	In a cashback transaction, the cash amount must be lower than transaction amount
474	עסקת איתחול בהוראת קבע מחייבת פרמטר J5	Must use J5 for initializing a recurring transaction
475	עסקת ה"ק מחייבת אחד מהשדות: מספר תשלומים או סכום כולל	Must input number of installments or total amount for recurring transaction
476	עסקת תורן בהוראת קבע מחייבת שדה מספר תשלום	Current installment in recurring transaction must have field current installment number
477	עסקת תורן בהוראת קבע מחייבת מספר מזהה של עסקת איתחול	Current installment in recurring transaction must have id number of the initializiation transaction
478	עסקת תורן בהוראת קבע מחייבת מספר אישור של עסקת איתחול	Current installment in recurring transaction must have authorization number of initialization transaction
479	עסקת תורן בהוראת קבע מחייבת שדות תאריך וזמן עסקת איתחול	Current installment in recurring transaction must have date and time of initialization transaction
480	חסר שדה מאשר עסקת מקור	Missing field original authorization
481	חסר שדה מספר יחידות כאשר העסקה מתבצעת בקוד אמצעי תשלום השונה ממטבע	Missing field number of units when transaction is done without currency
482	בכרטיס נטען מותרת עסקת חיוב רגילה/זיכוי/ביטול/פריקה/טעינה/בירור יתרה	A giftcard can be used with the following transaction types: regular, refund, discharge, topup and balance
483	עסקה עם כרטיס דלק במסוף דלק חיוב להזין מספר רכב	Must enter card license plate number when doing transaction in a petrol terminal
484	מספר רכב המוקלד שונה ממספר הרכב הצרוב ע"ג הפס המגנטי/מספר בנק שונה מ-012/ספרות שמאליות של מהפסר הסניף שונה מ- 44	Invalid license plate number

Status	Hebrew Description	English Description
485	מספר רכב קצר מ – 6 ספרות/שונה ממספר הרכב המופיע ע"ג ערוץ 2 (פוזיציה 34 בערוץ 2) כרטיס מאפיין דלק של לאומי קארד	Invalid license plate number
486	ישנה חובת הקלדת קריאת מונה (פוזיציה 30 בערוץ 2) כרטיס מאפיין דלק של לאומי קארד	Must enter speedometer
487	רק במסוף המוגדר כדלק דו שלבי ניתן להשתמש בעדכון אובליגו	You can perform an obligo update only in a two-phase petrol terminal
489	בכרטיס דלקן מותרת עסקת חיוב רגילה בלבד (עסקת ביטול אסורה)	Only regular debit is allowed with petrol card
490	בכרטיסי דלק/דלקן/דלק מועדון ניתן לבצע עסקאות רק במסופי דלק	Petrol cards can be used only with petrol terminals
491	עסקה הכוללת המרה חייבת להכיל את הכל השדות conversion_amount_06, conversion_currency_51, conversion_rate_09	A transaction with conversion must include the following fields
492	אין המרה על עסקאות שקל/דולר	No covfersion on NIS/Dollar transaction
493	בעסקה הכוללת הטבה חיוב שיהיו רק אחד מהשדות הבאים: סכום הנחה/מספר יחידות/ % ההנחה	A transaction with benefit must include one of the following fields: discount amount, number of units or discount percentage
494	מספר מסוף שונה	Different terminal id
495	אין הרשאת fallback	Fallback is not allowed
496	לא ניתן להצמיד אשראי השונה מאשראי קרדיט/תשלום	Cannot link credit-term which is different than "Credit with installments"
497	לא ניתן להצמיד לדולר/מדד במטבע השונה משקל	Cannot link to dollar/index in a currency which is different than Nis
498	כרטיס ישראכרט מקומי הספרטור צ"ל בפוזיציה 18	Local Isracard card. The separator must be in position 18
500	העסקה הופסקה ע"י המשתמש	Transaction aborted by user
504	חוסר התאמה בין שדה מקור נתוני הכרטיס לשדה מספר כרטיס	Pan entry mode and card number are incompatible
505	ערך לא חוקי בשדה סוג עסקה	Invalid transaction type
506	ערך לא חוקי בשדה eci	Invalid ECI
507	סכום העסקה בפועל גבוה מהסכום המאושר	Transaction amount is bigger than authorized amount
512	לא ניתן להכניס אישור שהתקבל ממענה קולי לעסקה זו	You can't use void authorization number with this transaction
	שגיאות אפליקטיביות בתשובה המתקבלת מהטנדם	
551	מסר תשובה אינו מתאים למסר הבקשה	Response message is incompatible with request message
552	שגיאה בשדה 55	Error in field 55

Status	Hebrew Description	English Description
553	התקבלה שגיאה מהטנדם	Error from Tandem
554	במסר התשובה חסר שדה mmc_18	Response is missing field mmc_18
555	במסר התשובה חסר שדה response_code_25	Response is missing field response_code_25
556	במסר התשובה חסר שדה rrn_37	Response is missing field rrn_37
557	במסר התשובה חסר שדה comp_retailer_num_42	Response is missing field comp_retailer_num_42
558	במסר התשובה חסר שדה auth_code_43	Response is missing field auth_code_43
559	במסר התשובה חסר שדה f39_response_39	Response is missing field f39_response_39
560	במסר התשובה חסר שדה authorization_no_38	Response is missing field authorization_no_38
561	במסר התשובה חסר/ריק שדה additional_data_48.solek_auth_no	Response is missing field additional_data_48.solek_auth_no
562	במסר התשובה חסר אחד מהשדות conversion_rate_09, conversion_amount_08, conversion_currency_51	Response is missing one of these fields: conversion_rate_09, conversion_amount_08, conversion_currency_51
563	ערך השדה אינו מתאים למספרי האישור שהתקבלו auth_code_43	Field value is incompatible with field auth_code_43
564	במסר התשובה חסר/ריק שדה additional_amounts54.cashback_amount	Response is missing field additional_amounts54.cashback_amount
565	אי-התאמה בין שדה 25 לשדה 43	Incompatibility between fields 25 and 43
566	במסוף המוגדר כתומך בדלק דו-שלבי יש חובה להחזיר שדות 90,119	Field 90 and 119 must return in a two-phases petrol terminal
567	שדות 25,127 לא תקינים במסר עידכון אובליגו במסוף המוגדר כדלק דו-שלבי	Fields 25,127 are invalid in a two-phases petrol terminal
-98	ERROR_IN_NEG_FILE	ERROR_IN_NEG_FILE
-99	שגיאה כללית	General error
	שגיאות המתקבלות מה- PinPad	
700	עסקה נדחתה ע"י מכשיר PinPad	Transaction rejected by Pinpad
701	שגיאה במכשיר PinPad	Error in Pinpad
702	יציאת com לא תקינה	Invalid COM port
703	עסקה נכשלה	Transaction failed
704	עסקה בוטלה	Transaction cancelled
705	עסקה הופסקה ע"י המשתמש	Transaction aborted by user
706	זמן המתנה ארוך מדי	Wait time is too long
707	משתמש הוציא כרטיס לפני סיום ביצוע העסקה	Cardholder removed his card before transaction end
708	PINPAD_UserRetriesExceeded	PINPAD_UserRetriesExceeded

Status	**Hebrew Description**	**English Description**
709	PINPAD_PINPadTimeout	PINPAD_PINPadTimeout
710	תקלת תקשורת עם מכשיר	Communication error with Pinpad
711	PINPAD_PINPadMessageError	PINPAD_PINPadMessageError
712	PINPAD_PINPadNotInitialized	PINPAD_PINPadNotInitialized
713	PINPAD_PINPadCardReaderError	PINPAD_PINPadCardReaderError
714	PINPAD_ReaderTimeout	PINPAD_ReaderTimeout
715	PINPAD_ReaderCommsError	PINPAD_ReaderCommsError
716	PINPAD_ReaderMessageError	PINPAD_ReaderMessageError
717	PINPAD_HostMessageError	PINPAD_HostMessageError
718	PINPAD_HostConfigError	PINPAD_HostConfigError
719	PINPAD_HostKeyError	PINPAD_HostKeyError
720	PINPAD_HostConnectError	PINPAD_HostConnectError
721	PINPAD_HostTransmitError	PINPAD_HostTransmitError
722	PINPAD_HostReceiveError	PINPAD_HostReceiveError
723	PINPAD_HostTimeout	PINPAD_HostTimeout
724	PINVerificationNotSupportedByCard	PINVerificationNotSupportedByCard
725	PINVerificationFailed	PINVerificationFailed
726	שגיאה בקליטת קובץ config.xml	Error parsing config.xml file
730	מכשיר אישר עסקה בניגוד להחלטת אשראית	Pinpad authorized transaction in contrary to payment application decision
731	כרטיס לא הוכנס	Card was not inserted

Appendix J – Transaction Steps

Explanation

The Pinpad is facing the cardholder and most of the time the Pinpad screen will not be visible from the cashier point of view. Because of that, the cashier will not know what is happening on the Pinpad and what the delay is (if there is a delay). For example, the Pinpad may ask the cardholder to enter his PIN, but if the cardholder doesn't pay attention to the Pinpad or if he doesn't understand what to do, a delay will happen. For that purpose, we provide the Transaction Steps feature. This feature will allow the cash register to get the current step (or state) of the Pinpad. The cash register will know if the Pinpad is waiting for a card or waiting for Pin entry and the cashier can decide whether to abort the transaction or simply instruct the cardholder what he needs to do.

There are tree options to get Transaction Steps and to send Abort from ECR. The esiest way is to use http to get the step and to send abort. Second option is using Windows Event Watcher and the thired way is to implement TCP Listener.

To use Http method STEP application should be download to the pinpad. In such case TCP method will be disabled.

Enable Transaction Steps with Http

The **HTTP GET** shell be used as follows

http://pinpadIPaddress:8080/step

http://pinpadIPaddress:8080/abort

Enable Transaction Steps vs FileSystemWatcher

The **FileSystemWatcher** class in the **System.IO** namespace can be used to monitor changes to the transaction steps. It watches a two files "TransactionStep.event" and "TransactionAbort.event" in a working directory for changes and triggers events when changes occur. The FileSystemWatcher raises the following events when changes occur to a directory that it is monitoring..

- Changed: This event is triggered when a file“TransactionStep.event” being monitored is changed. It used for getting step information from file “TransactionStep.event” and/or setting RedButtonAbort to anabled or disabled state.

- Deleted: This event is triggered when a file “TransactionStep.event” being monitored is deleted. It used for settind RemoteButtonAbort to hide and close. On starting monitoring we recommended create an empty file "TransactionStep.event".

- Creted: This event is triggered when a file “TransactionAbort.event” being monitored is created. It used for emulation of pressed RedButtonAbort on the PinPad. On starting monitoring we recommended to remove the “TransactionAbort.event”.

For example, the default monitoring process name using in “**Caspit SmartRetail Tester**” is “**Rba**”. The process “**Rba**” communicates with the CaspitPayment.dll using the above events.

The process “**Rba**” starts in “**Caspit SmartRetail Tester**” after the transaction (Command>**001**</Command> <!-- `Performing Transaction` -->) begins or other Ashrait commands (Command>**0xx**</Command>)

The main functions of the “Rba” process after it is started are:

- Display RedButtonAbort, and display step information. (Changed event)
- Display transaction estimated time remaining. (Changed event)
- Control of RedButtonAbort Enable/Disable states. (Changed event)
- Send “TransactionAbort.event” event after user hit RedButtonAbort on the screen or after timeout expired. (Creted event - “TransactionAbort.event”)

The “Rba” process should be closed after the **`Performing Transaction`** ends (Deleted event).

Enable Transaction Steps vs TCP Listener

The ECR can use the Pinpad Configuration command to configure the Pinpad to send the transaction steps. Use the tags <EnableTranSteps> and

<TranStepsAddress> - the IP address and the port number of the listener for the transaction steps.

Here you can see an example of a Pinpad Configuration command that enables the transaction steps, sets the IP address and port of the listener and tells the Pinpad that the transaction steps are mandatory.

```xml
<Request>
  <Command>013</Command>                          <!-- PinPad Configuration -->
  <TerminalId>0880302</TerminalId>
  <TermNo>1</TermNo>                              <!-- Station number 1 (Not multiple pinpads).  -->
  <RequestId>20180628115807</RequestId> <!-- Request ID (yyyyMMddHHmmss)  -->
  <EnableTranSteps>1</EnableTranSteps>     <!-- Transaction steps are enabled  -->
  <TranStepsAddress>192.168.1.17:24321</TranStepsAddress>
</Request>
```

Disable Transaction Steps

Here you can see an example of a Pinpad Configuration command that disables the transaction steps

```xml
<Request>
  <Command>013</Command>                          <!-- PinPad Configuration -->
  <TerminalId>0880302</TerminalId>
  <TermNo>1</TermNo>                              <!-- Station number 1 (Not multiple pinpads).  -->
  <RequestId>20180628115808</RequestId> <!-- Request ID (yyyyMMddHHmmss)  -->
  <EnableTranSteps>0</EnableTranSteps>     <!-- Transaction steps are enabled  -->
  <MustTranSteps>1</MustTranSteps>
</Request>
```

If transaction steps are enabled, the Pinpad will make a TCP connection to the configured address and it will update the ECR about the current step of the transaction.

The ECR software is responsible for creating a TCP listener for the Pinpad. Caspit doesn't provide this listener.

Beware that if the Pinpad is configured to send transaction steps but the ECR TCP listener is not working, this may cause a delay at the start of the transaction because the Pinpad is trying to establish the TCP connection **at the start of the transaction**. The Pinpad will **close the TCP connection at the end of the transaction**.

If the ECR is using also the Idle Swipe feature (see Appendix G), then the same listener of the Idle Swipe can be used for the transaction steps. But it still required to configure the Transaction steps using the tags <EnableTranSteps> and <TranStepsAddress>. See Appendix G for Idle Swipe

With the <MustTranSteps> tag you can tell the Pinpad that the transaction steps are mandatory for the transaction. If you configure the Pinpad with <MustTranSteps>1</MustTranSteps> and the Pinpad is not able to open the TCP connection, the Pinpad will abort the transaction and return with Result Code 10026 (RETVAL_TRAN_STEPS_LISTENER_NOT_WORKING).

The last Pinpad step is also available in the transaction response XML (<PinpadStep> tag). So if the transaction was not completed successfully you will see on which step the transaction failed.

The following commands will support the steps sending, i.e. during these commands the ECR can expect to get step notifications:

- Transaction
- Transmit transactions to Shva
- Swipe command
- UI command

Basically, the idea is that every command that takes time to finish, and the ECR "doesn't know" the current status of the command, that command will send notifications to tell the ECR what is going on.

The Pinpad will send the following XML for each step:

```xml
<Step>
    <StepCode>0010</StepCode>
    <AP>1</AP>
    <Ver>1</Ver>
</Step>
```

The <StepCode> is a number from the possible transaction steps table that you can see below.

The <AP> tag is a short for "Abort Possible".

If <AP> is 1, it means the ECR can send an abort command to the Pinpad.

If the <AP> is 0 or if <AP> tag is missing then the ECR cannot send an abort command.

The ECR does not need to send response back to the Pinpad.

The ECR software can do what ever it likes with the steps information. It can ignore the steps or it may use them to display the process of the transaction on the ECR screen for the comfort of the merchant. For example, during PIN entry (secret code entry) the Pinpad will send steps describing every key the cardholder is pressing on thePinpad. The ECR may show a nice screen which shows the cardholder key presses (of course, the specific PIN digits are not sent from the Pinpad).

Note that during the transaction, the step numbers don't need to be consecutive. It is possible that a high step code will come before a low step code.

The CaspitPayment.dll will send to Rba(Red button abort) Event the following XML for each step:

```xml
<Step>
    <StepCode>0010</StepCode>
    <AP>1</AP>
```

```
        <Ver>1</Ver>
        <RT>60</RT>
</Step>
```

Were **RT** tag value may be used for show estimated time remaining in seconds.

Note: the remaining time will be starting from Request tag “**TimeoutInSeconds**” value.

The CaspitPayment.dll may send the following XML also:

```
<Remaining>
        <RT>60</RT>
</Remaining>
```

Abort

If the last step XML the ECR got included the <AP>1</AP> then the ECR may send an abort command to the Pinpad. The ECR must use the same TCP connection to send the abort command.

The abort command is this short XML: **<Abort>**

Table of steps

Step	Definition and explanation	Application (Caspit Internal)
0010	STEP_REQUEST_ARRIVED_TO_ASHRAIT_EMV Transaction request has arrived to Ashrait EMV application	Ashrait EMV
0020	STEP_USER_MUST_PRESENT_CARD Cardholder is required to present his card Abort is possible in this step	Ashrait EMV
0030	STEP_USER_PRESENTED_MAGNETIC_CARD The cardholder swiped the card in the magnetic card reader	Ashrait EMV
0031	STEP_USER_MUST_USE_CHIP The cardholder swiped a magnetic card but he must use the chip Abort is possible in this step	Ashrait EMV
0040	STEP_USER_PRESENTED_CONTACT_CARD The cardholder inserted the card into the smart card reader	Ashrait EMV
0050	STEP_USER_PRESENTED_CONTACTLESS_CARD The cardholder tapped the card on the contactless reader	Ashrait EMV
0060	STEP_USER_PRESENTED_MOBILE_PHONE The cardholder tapped his mobile phone on the contactless reader	Ashrait EMV

Step	Definition and explanation	Application (Caspit Internal)
0070	STEP_USER_MUST_ENTER_PIN The cardholder is required to enter the PIN (secret code) Abort is possible in this step	EPA
0071	STEP_USER_MUST_RETRY_PIN The cardholder is required to enter the PIN (secret code) another time (the previous PIN was wrong) Abort is possible in this step	EPA
0072	STEP_USER_MUST_RETRY_PIN_LAST_TRY The cardholder is required to enter the PIN (secret code) another time. **But Watch out!** This is the last try. Another wrong PIN will cause his credit card to be blocked. Abort is possible in this step	EPA
0073	STEP_USER_MUST_ENTER_PIN_LAST_TRY The cardholder is required to enter the PIN (secret code). This is the first try but it is **also the last try**!. A wrong PIN will cause his credit card to be blocked. Abort is possible in this step	EPA
0074	STEP_USER_ENTERED_A_PIN_DIGIT The cardholder entered a PIN digit (don't worry, there is no way to know which digit was pressed) Abort is possible in this step	EPA
0075	STEP_USER_ERASED_A_PIN_DIGIT The cardholder used the backspace to erase one digit of the PIN Abort is possible in this step	EPA
0076	STEP_USER_CONFIRMED_THE_PIN The cardholder pressed on the green button to confirm the PIN. That **doesn't** mean that the PIN is correct or incorrect.	EPA
0077	STEP_USER_ABORTED_PIN_ENTRY The cardholder pressed on the red button to abort the PIN entry. The transaction cannot be completed.	EPA
0078	STEP_PIN_CORRECT The PIN the cardholder entered was found to be **correct**	EPA
0079	STEP_PIN_INCORRECT The PIN the cardholder entered was found to be **incorrect**	EPA
0085	STEP_USER_REMOVED_CARD_DURING_PIN_ENTRY The cardholder removed his card in the middle of PIN entry. The transaction cannot be completed.	EPA
0086	STEP_PINPAD_WAIT_FOR_CARD_REMOVAL The Pinpad is waiting for card removal	EPA

Step	Definition and explanation	Application (Caspit Internal)
0087	STEP_WAITING_FOR_CARDHOLDER_INPUT The cardholder needs to input something using the Pinpad keys Abort is possible in this step	EPA
0100	STEP_USER_MUST_ENTER_ID The cardholder is required to enter his ID number Abort is possible in this step	Ashrait EMV
0110	STEP_USER_MUST_ENTER_CVV The cardholder is required to enter the card CVV Abort is possible in this step	Ashrait EMV
0120	STEP_USER_MUST_ENTER_CARD_NUMBER The cardholder is required to enter the card number. Most likely, this will be done by the cashier and not by the cardholder. Abort is possible in this step	Ashrait EMV
0130	STEP_USER_MUST_ENTER_EXPIRATION_DATE The cardholder is required to enter the expiration date. Most likely, this will be done by the cashier and not by the cardholder. Abort is possible in this step	Ashrait EMV
0200	STEP_BEFORE_WRITING_TRANSACTION_TO_BATCH The Pinpad is going to write the transaction into the transaction batch (the TRAN file)	Ashrait EMV
0210	STEP_AFTER_WRITING_TRANSACTION_TO_BATCH The Pinpad has finished writing the transaction into the transaction batch (the TRAN file)	Ashrait EMV
0250	STEP_CALLING_SHVA The Pinpad is calling Shva to get authorization for the transaction	Ashrait EMV
0260	STEP_SHVA_COMMUNICATION_ENDED Shva communication has ended	Ashrait EMV
0220	STEP_CHECKING_CARD_PERMISSIONS Ashrait EMV is checking the permissions of the card	Ashrait EMV
0300	STEP_STARTING_SHVA_FULL_COMMUNICATION Ashrait EMV is starting the full communication to Shva. In the full communication, transactions will be send and the Pinpad will update its parameters and black list.	Ashrait EMV
0310	STEP_RECEIVING_PARAMETERS While Ashrait EMV is doing the full communication to Shva, this is the step that Ashrait EMV is getting the parameters.	Ashrait EMV

Step	Definition and explanation	Application (Caspit Internal)
0320	STEP_SENDING_TRANSACTIONS While Ashrait EMV is doing the full communication to Shva, this is the step that Ashrait EMV is sending the transactions.	Ashrait EMV
0330	STEP_RECEIVING_BLACK_LIST While Ashrait EMV is doing the full communication to Shva, this is the step that Ashrait EMV is getting the black list.	Ashrait EMV
0340	STEP_RECEIVING_PINPAD_PARAMETERS Whie Ashrait EMV is doing the full communication to Shva, this is the step that Ashrait EMV is getting the Pinpad parameters.	Ashrait EMV
0350	STEP_ENDING_SHVA_FULL_COMMUNICATION End of full communication to Shva	Ashrait EMV
0400	STEP_USER_MUST_SWIPE_CARD This step is for the **Swipe Command**. The cardholder must swipe his card. Abort is possible in this step	
0500	STEP_CONTACT_TRAN_APPROVED_REMOVE_CARD The contact transaction has been approved. The cardholder must remove his card	EPA
0505	STEP_CONTACT_TRAN_DECLINED_REMOVE_CARD The contact transaction has been declined. The cardholder must remove his card.	EPA
0510	STEP_CONTACT_TRAN_PROCESSING_ERROR_REMOVE_CARD There is a processing error in the contact transaction. The cardholder must remove his card.	EPA
0515	STEP_CONTACT_TRAN_ABORTED_REMOVE_CARD The contact transaction was aborted. The cardholder must remove his card.	EPA
0520	STEP_CONTACT_TRAN_INVALID_CARD_REMOVE_CARD The contact transaction failed because of invalid card. The cardholder must remove his card.	EPA
0525	STEP_USER_REMOVED_THE_CARD The cardholder removed his card	EPA
0530	STEP_CONTACT_TRAN_AID_BLOCKED_REMOVE_CARD The contact transaction failed because the card AID is blocked. The cardholder must remove his card.	EPA
0535	STEP_USER_CANCELLED_PIN_ENTRY_REMOVE_CARD The contact transaction failed because the cardholder cancelled the PIN entry. The cardholder must remove his card.	EPA
0540	STEP_CARD_BLOCKED_REMOVE_CARD The contact transaction failed because the card is blocked. The cardholder must remove his card.	EPA

Step	Definition and explanation	Application (Caspit Internal)
0545	STEP_NO_MATCHING_AID_REMOVE_CARD The contact transaction failed because there is no matching AID. The cardholder must remove his card.	EPA
0600	STEP_USER_SHOULD_TRY_TAPPING_AGAIN	CPA
0605	STEP_USER_TAPPED_TOO_MANY_CARDS_TRY_AGAIN	CPA
9000	STEP_END_OF_TRANSACTION	Ashrait EMV

Example

During a standard Pin entry process, the Pinpad may send the following XMLs

<Step>

<StepCode>0070</StepCode> **(the cardholder must enter PIN)**

<AP>1</AP>

</Step>

<Step>

<StepCode>0074</StepCode> **(the cardholder entered a digit)**

<AP>1</AP>

</Step>

<Step>

<StepCode>0074</StepCode> **(the cardholder entered a digit)**

<AP>1</AP>

</Step>

<Step>

<StepCode>0074</StepCode> **(the cardholder entered a digit)**

<AP>1</AP>

</Step>

<Step>

<StepCode>0074</StepCode> **(the cardholder entered a digit)**

<AP>1</AP>

</Step>

<Step>

<StepCode>0076</StepCode> (**the cardholder pressed on the green button)**

<AP>0</AP>

</Step>

<Step>

<StepCode>0078</StepCode> **(PIN is correct)**

</Step>

Appendix K – Various Tables

Caspit Internal Errors Table

The transaction response XML may include the tag <CaspitInternalError>. This tag gives the internal error number of Ashrait EMV application on the Pinpad. The explanation of each error number is out of the scope of this document. But I will put here the definition of all the internal errors, it may help sometimes.

```
#define _PARAMETER_ERROR                           _ERROR_START       + 100
#define _TRANSACTION_ERROR                         _ERROR_START       + 200
#define _BLACKLIST_ERROR                           _ERROR_START       + 300
#define _GENERAL_ERROR                             _ERROR_START       + 500
#define _CREDIT_ERROR                              _ERROR_START       + 900
#define _IDX_ERROR                                 _ERROR_START       + 1000
#define _CARD_IDENTIFICATION_ERROR                 _ERROR_START       + 1100
#define _REPORT_GENERATOR_ERROR                    _ERROR_START       + 1200
#define _COMM_ERROR                                _ERROR_START       + 1300
#define _AUTHORIZATION_ERROR                       _ERROR_START       + 1400
#define _DATA_PERMISSION_ERROR                     _ERROR_START       + 1500
#define _PIN_PAD_ERROR                             _ERROR_START       + 1600

#define _ECR_ERROR                                _ERROR_START       + 3000

/*************************************************************************************/

#define ERR_TOO_MANY_NAKS                          _PARAMETER_ERROR   + 0
#define ERR_CANT_UPDATE_CLOCK                      _PARAMETER_ERROR   + 1
#define ERR_WRONG_SITE_ID                          _PARAMETER_ERROR   + 2
#define ERR_BAD_PARAM_BLOCK_SEQUENCE               _PARAMETER_ERROR   + 3
#define ERR_BAD_PARAM_BLOCK_TYPE                   _PARAMETER_ERROR   + 4
#define ERR_BAD_PARAM_VECTOR_TYPE                  _PARAMETER_ERROR   + 5
#define ERR_CANT_OPEN_VECTOR_FILE                  _PARAMETER_ERROR   + 6
#define ERR_VECTOR_WRITE_TRUNCATED                 _PARAMETER_ERROR   + 7
#define ERR_VECTOR_SEEK_ERROR                      _PARAMETER_ERROR   + 8
#define ERR_VECTOR_READ_ERROR                      _PARAMETER_ERROR   + 9
#define ERR_PRINT_PARAM_ILLEGAL_BLOCK_ID            _PARAMETER_ERROR   + 10
#define ERR_PRINT_PARAM_OPEN_PARAM                  _PARAMETER_ERROR   + 11
#define ERR_PRINT_PARAM_SEEK_PARAM                  _PARAMETER_ERROR   + 12
#define ERR_PRINT_PARAM_READ_PARAM                  _PARAMETER_ERROR   + 13
#define ERR_PRINT_PARAM_WRITE_OUT                   _PARAMETER_ERROR   + 14
#define ERR_PRINT_PARAM_OPEN_OUTPUT                 _PARAMETER_ERROR   + 15
#define ERR_PRINT_PARAM_OPEN_PARAM_INDEX            _PARAMETER_ERROR   + 16
#define ERR_PRINT_PARAM_SEEK_PARAM_INDEX            _PARAMETER_ERROR   + 17
#define ERR_PRINT_PARAM_READ_PARAM_INDEX            _PARAMETER_ERROR   + 18
#define ERR_CALL_PARAM                              _PARAMETER_ERROR   + 19
#define ERR_SEEK_PARAM                              _PARAMETER_ERROR   + 20


/*************************************************************************************/

#define ERR_BAD_OK_BLOCK                           _TRANSACTION_ERROR + 0
#define ERR_TRN_COPY_CREATE                         _TRANSACTION_ERROR + 1
#define ERR_TRN_COPY_OPEN                           _TRANSACTION_ERROR + 2
#define ERR_TRN_COPY_WRITE                          _TRANSACTION_ERROR + 3
#define ERR_TRN_COPY_READ                           _TRANSACTION_ERROR + 4
#define ERR_BATCH_OPEN                              _TRANSACTION_ERROR + 5
```

```
#define ERR_BATCH_CREATE                          _TRANSACTION_ERROR + 6
#define ERR_BATCH_SEEK                            _TRANSACTION_ERROR + 7
#define ERR_BATCH_READ                            _TRANSACTION_ERROR + 8
#define ERR_BATCH_WRITE                            _TRANSACTION_ERROR + 9
#define ERR_BATCH_TOO_MANY_TRANSACTIONS            _TRANSACTION_ERROR + 10
#define ERR_BATCH_WRITE_COUNTER                   _TRANSACTION_ERROR + 11
#define ERR_CANT_DISPLAY_BLOCK_NUM                _TRANSACTION_ERROR + 12
#define ERR_EMPTY_BATCH_FILE                      _TRANSACTION_ERROR + 13
#define ERR_ARRAY_CREATE                          _TRANSACTION_ERROR + 14
#define ERR_ARRAY_READ                            _TRANSACTION_ERROR + 15
#define ERR_ARRAY_SEEK                            _TRANSACTION_ERROR + 16
#define ERR_ARRAY_WRITE                           _TRANSACTION_ERROR + 17
#define ERR_NO_TRANSACTION_IN_SHIFT               _TRANSACTION_ERROR + 18
#define ERR_EMPTY_DEPOSIT_FILE                    _TRANSACTION_ERROR + 19
#define ERR_INVALID_AMOUNT                        _TRANSACTION_ERROR + 20

/***************************************************************************************/

#define ERR_CANT_OPEN_ADD_LIST_FILE               _BLACKLIST_ERROR   + 0
#define ERR_CANT_OPEN_DEL_LIST_FILE               _BLACKLIST_ERROR   + 1
#define ERR_BAD_UPDATE_TYPE                       _BLACKLIST_ERROR   + 2
#define ERR_BAD_BLACK_LIST_BLOCK_SEQUENCE         _BLACKLIST_ERROR   + 3
#define ERR_WRITE_TO_BLACKLIST_FILE_FAILED        _BLACKLIST_ERROR   + 4
#define ERR_SMALL_HOT_WRITE                       _BLACKLIST_ERROR   + 5
#define ERR_SMALL_HOT_CREATE                      _BLACKLIST_ERROR   + 6
#define ERR_BL_OFF_LINE                           _BLACKLIST_ERROR   + 7
#define ERR_SHORT_CARD_LEN                        _BLACKLIST_ERROR   + 8
#define ERR_SHORT_ISRACARD_CARD_NUM               _BLACKLIST_ERROR   + 9
#define ERR_SHORT_AMEX_CARD_NUM                   _BLACKLIST_ERROR   + 10
#define ERR_SHORT_DINERS_CARD_NUM                 _BLACKLIST_ERROR   + 11
#define ERR_SHORT_VISA_CARD_NUM                   _BLACKLIST_ERROR   + 12

/***************************************************************************************/

#define ERR_EXPECTED_DELIMITER_MISSING            _GENERAL_ERROR     + 0
#define ERR_MEMORY_ALOCATION_FAILED               _GENERAL_ERROR     + 1
#define ERR_CALL_SHVA                             _GENERAL_ERROR     + 2
#define ERR_TRANSMIT_TRANZACTIONS                 _GENERAL_ERROR     + 3
#define ERR_TERMINAL_BLOCKED                      _GENERAL_ERROR     + 4

#define ERR_KEYED_FILE_OPEN                       _GENERAL_ERROR     + 5
#define ERR_EDIT_DISABLE                          _GENERAL_ERROR     + 6
#define ERR_LAST_REPORT_DISABLE                   _GENERAL_ERROR     + 7

#define ERR_HIGH_MEM_BOUNDARY                     _GENERAL_ERROR     + 8
#define ERR_LOW_MEM_BOUNDARY                      _GENERAL_ERROR     + 9
#define ERR_TEST_VERSION                          _GENERAL_ERROR     + 10
#define ERR_TRANSMIT_DISABLE                      _GENERAL_ERROR     + 11
#define ERR_LOW_TRN_MEM_BOUNDARY                  _GENERAL_ERROR     + 12
#define ERR_LOW_FILE_BOUNDARY                     _GENERAL_ERROR     + 13

#define ERR_KUPA_HIGH_MEM_BOUNDARY                _GENERAL_ERROR     + 14
#define ERR_KUPA_LOW_MEM_BOUNDARY                  _GENERAL_ERROR     + 15

/***************************************************************************************/
```

```
#define ERR_CARD_REJECTED                          _CREDIT_ERROR      + 0
#define ERR_CARD_REJECTED_BLOCKED                  _CREDIT_ERROR      + 1
#define ERR_CARD_REJECTED_STOLEN                   _CREDIT_ERROR      + 2
#define ERR_CARD_REJECTED_CALL_CREDIT_COMPANY      _CREDIT_ERROR      + 3
#define ERR_CARD_REJECTED_DECLINED                 _CREDIT_ERROR      + 4
#define ERR_CARD_REJECTED_FORGED                   _CREDIT_ERROR      + 5

#define ERR_CARD_REJECTED_TZ                       _CREDIT_ERROR      + 6
#define ERR_CARD_REJECTED_CVV                      _CREDIT_ERROR      + 7
#define ERR_CARD_BLOCKED                           _CREDIT_ERROR      + 8
#define ERR_BAD_CHECK_DIGIT                        _CREDIT_ERROR      + 9
#define ERR_WRONG_MASTER_NUMBER                    _CREDIT_ERROR      + 10
#define ERR_CANT_UNDO_LAST_TRANSACTION             _CREDIT_ERROR      + 11
#define ERR_NO_TRANSACTIONS_TO_UNDO                _CREDIT_ERROR      + 12
#define ERR_BAD_TRANSACTION_RECORD                 _CREDIT_ERROR      + 13
#define ERR_NO_TRANSACTIONS                        _CREDIT_ERROR      + 14
#define ERR_NO_TRANSACTIONS_IN_SHIFT               _CREDIT_ERROR      + 15
#define ERR_CARD_NOT_ACCEPTED                      _CREDIT_ERROR      + 16
#define ERR_NO_PHONE_NUMBER                        _CREDIT_ERROR      + 17
#define ERR_PHONE_TRN_BLOCKED                      _CREDIT_ERROR      + 18
#define ERR_LIMIT_NO_OPTIONS                       _CREDIT_ERROR      + 19
#define ERR_NO_AUTHORIZATION                       _CREDIT_ERROR      + 20
#define ERR_READ_READER                            _CREDIT_ERROR      + 21
#define ERR_ISSUER_NOT_OPEN                        _CREDIT_ERROR      + 22
#define ERR_PRE_AUTHOR_NOT_ALLOWED                 _CREDIT_ERROR      + 23
#define ERR_CARD_EXPIRED                           _CREDIT_ERROR      + 24
#define ERR_BAD_TRN_TYPE                           _CREDIT_ERROR      + 25
#define ERR_CREDIT_NOT_ALLOWED                     _CREDIT_ERROR      + 26
#define ERR_DEBIT_NOT_ALLOWED                      _CREDIT_ERROR      + 27
#define ERR_TELEPHONE_TRANSACTION_NOT_PERMITTED    _CREDIT_ERROR      + 28
#define ERR_SIGNATURE_TRANSACTION_NOT_PERMITTED    _CREDIT_ERROR      + 29
#define ERR_INVALID_CARD_DATE                      _CREDIT_ERROR      + 30
#define ERR_PREAUTHOR_NO_UNDO                      _CREDIT_ERROR      + 31
#define ERR_PREAUTHOR_NO_COPY                      _CREDIT_ERROR      + 32
#define ERR_CREDIT36_NOT_ALLOWED                   _CREDIT_ERROR      + 33
#define ERROR_CANT_UNDO_GIFT_TRANSACTION           _CREDIT_ERROR      + 34
#define ERROR_CEIL0                                _CREDIT_ERROR      + 35
#define ERROR_CEIL0_PHONE_SIGN                     _CREDIT_ERROR      + 36
#define ERROR_CEIL0_ZIKUI                          _CREDIT_ERROR      + 37
#define ERROR_CREDIT_BLOCKED                       _CREDIT_ERROR      + 38
#define ERROR_INVALID_PAYM_NUM                     _CREDIT_ERROR      + 39
#define ERROR_BRAND_NOT_OPEN                       _CREDIT_ERROR      + 40 //emv
#define ERROR_INVALID_PARAM_SET                    _CREDIT_ERROR      + 41 //emv
#define ERROR_ACQUIRE_NOT_OPEN                     _CREDIT_ERROR      + 42 //emv
#define ERROR_INVALID_ID_NUMBER                    _CREDIT_ERROR      + 43
#define ERROR_CARD_PERMISSION                      _CREDIT_ERROR      + 44 //emv
#define ERROR_TRANS_PERMISSION                     _CREDIT_ERROR      + 45 //emv
#define ERROR_LOAD_CARD_PERMISSION                 _CREDIT_ERROR      + 46 //emv
#define ERR_CARD_REJECTED_AVS                      _CREDIT_ERROR      + 47
#define ERROR_INVALID_PARAM_SET_SOLEK              _CREDIT_ERROR      + 48

/*************************************************************************************/

#define ERR_MEMORY_ALLOCATION_ERROR                _IDX_ERROR         + 0
#define ERR_FILE_OPEN_ERROR                        _IDX_ERROR         + 1
#define ERR_FILE_NAME_TOO_LONG                     _IDX_ERROR         + 2
```

```
#define ERR_READ_ERROR                                      _IDX_ERROR          + 3
#define ERR_BAD_KEY_TYPE                                    _IDX_ERROR          + 4
#define ERR_SEEK_ERROR                                      _IDX_ERROR          + 5
#define ERR_BAD_OFFSET_SIZE                                 _IDX_ERROR          + 6
#define ERR_BAD_LENGTH_SIZE                                 _IDX_ERROR          + 7
#define ERR_WRITE_ERROR                                     _IDX_ERROR          + 8
#define ERR_DUPLICATE_KEY                                   _IDX_ERROR          + 9
#define KEY_OUT_OF_RANGE                                    _IDX_ERROR          + 10
#define KEY_NOT_FOUND                                       _IDX_ERROR          + 11

/*************************************************************************************/

#define ERR_ISSUER_14_DIGIT                           _CARD_IDENTIFICATION_ERROR        + 0
#define ERR_BAD_CARD_TYPE                             _CARD_IDENTIFICATION_ERROR        + 1
#define ERR_ISSUER_15_DIGIT                           _CARD_IDENTIFICATION_ERROR        + 2
#define ERR_ISSUER_16_DIGIT                           _CARD_IDENTIFICATION_ERROR        + 3
#define ERR_ISSUER_4_NOT_FIRST_DIGIT                  _CARD_IDENTIFICATION_ERROR        + 4
#define ERR_CARD_NUM_LENGTH                           _CARD_IDENTIFICATION_ERROR        + 5
#define ERR_BAD_CHECK_DIGIT_LOCAL_ISRACARD            _CARD_IDENTIFICATION_ERROR        + 6
#define ERR_BAD_CHECK_DIGIT_OTHER_CARDS               _CARD_IDENTIFICATION_ERROR        + 7
#define ERR_ISSUER_NUMBER                             _CARD_IDENTIFICATION_ERROR        + 8
#define ERR_SEPARATOR_NOT_FOUND                       _CARD_IDENTIFICATION_ERROR        + 9
#define ERR_DINERS_PRE_6_NOT_IN_VECTOR_21             _CARD_IDENTIFICATION_ERROR        + 10
#define ERR_AMEX_WRONG_GROUP_NUMBER                   _CARD_IDENTIFICATION_ERROR        + 11
#define ERR_YOUTH_CARD_GROUP_NOT_VALID                _CARD_IDENTIFICATION_ERROR        + 12
#define ERR_ISRACARD_GROUP_NUMBER_BLOCKED             _CARD_IDENTIFICATION_ERROR        + 13
#define ERR_ISRACARD_DELEK_CARD                       _CARD_IDENTIFICATION_ERROR        + 14
#define ERR_ISRACARD_VECTOR_7                         _CARD_IDENTIFICATION_ERROR        + 15
#define ERR_ISRACARD_VECTOR_5                         _CARD_IDENTIFICATION_ERROR        + 16
#define ERR_ISRACARD_SERVICE_CODE                     _CARD_IDENTIFICATION_ERROR        + 17
#define ERR_VISA_DELEK_CARD                           _CARD_IDENTIFICATION_ERROR        + 18
#define ERR_VISA_SERVICE_CODE                         _CARD_IDENTIFICATION_ERROR        + 19
#define ERR_VISA_NEED_COMM_REJECT_CODE                _CARD_IDENTIFICATION_ERROR        + 20
#define ERR_VISA_PRE_3090                             _CARD_IDENTIFICATION_ERROR        + 21
#define ERR_WRONG_SERVICE_CODE                        _CARD_IDENTIFICATION_ERROR        + 22
#define ERR_SEC_SEPARATOR_NOT_FOUND                   _CARD_IDENTIFICATION_ERROR        + 23
#define ERR_CARD_OUT_OF_RANGE                         _CARD_IDENTIFICATION_ERROR        + 24
#define ERR_COMP_LAST_DIGITS                          _CARD_IDENTIFICATION_ERROR        + 25
#define ERR_ILLEGAL_CLUB_RANGE                        _CARD_IDENTIFICATION_ERROR        + 26
#define ERR_CARD_NOT_DEFINED                          _CARD_IDENTIFICATION_ERROR          + 27
#define ERR_VECTOR_57                                  _CARD_IDENTIFICATION_ERROR         + 28
#define ERR_VECTOR_58                                 _CARD_IDENTIFICATION_ERROR + 29
#define ERR_BAD_TECH_CARD                            _CARD_IDENTIFICATION_ERROR + 30

/***********************************************************************************/

#define REPORT_RETURN_CAN_NOT_OPEN_TEMPLATE                _REPORT_GENERATOR_ERROR + 0
#define REPORT_RETURN_CAN_NOT_OPEN_DEVICE                  _REPORT_GENERATOR_ERROR + 1
#define REPORT_RETURN_READ_TEMPLATE_ERROR                  _REPORT_GENERATOR_ERROR + 2
#define REPORT_RETURN_WRITE_DEVICE_ERROR                   _REPORT_GENERATOR_ERROR + 3
#define REPORT_RETURN_ILLEGAL_ARGUMENT                     _REPORT_GENERATOR_ERROR + 4
#define REPORT_RETURN_CONVERT_ARGUMENT_ERROR                _REPORT_GENERATOR_ERROR + 5
#define REPORT_RETURN_VARIABLE_NOT_FOUND                   _REPORT_GENERATOR_ERROR + 6
#define REPORT_RETURN_ILLEGAL_DIRECTIVE                    _REPORT_GENERATOR_ERROR + 7
#define REPORT_RETURN_GET_DIRECTIVE_ERROR                  _REPORT_GENERATOR_ERROR + 8
#define REPORT_RETURN_ILLEGAL_GROUP_NUMBER                  _REPORT_GENERATOR_ERROR + 9
```

```
#define REPORT_RETURN_SKIP_CURRENT_GROUP               _REPORT_GENERATOR_ERROR + 10
#define REPORT_RETURN_HANDLE_CURRENT_GROUP             _REPORT_GENERATOR_ERROR + 11
#define REPORT_RETURN_CONTINUE_IN_TEMPLATE             _REPORT_GENERATOR_ERROR + 12
#define REPORT_RETURN_REPEAT_CURRENT_GROUP             _REPORT_GENERATOR_ERROR + 13
#define REPORT_RETURN_END_GROUP_NOT_FOUND              _REPORT_GENERATOR_ERROR + 14
#define REPORT_RETURN_COMMAND_NOT_FOUND                _REPORT_GENERATOR_ERROR + 15
#define REPORT_RETURN_ESC_SEQ_TOO_BIG                  _REPORT_GENERATOR_ERROR + 16
#define REPORT_RETURN_NOT_ENOUGH_MEMORY_FOR_TEMPLATE _REPORT_GENERATOR_ERROR + 17
#define REPORT_RETURN_TEMPLATE_SEEK_ERROR              _REPORT_GENERATOR_ERROR + 18
#define REPORT_RETURN_GROUP_ENTRY_ALLOCATION_ERROR     _REPORT_GENERATOR_ERROR + 19

/*****************************************************************************/

#define ERR_CE_COMM_INIT                          _COMM_ERROR         + 0
#define ERR_CE_OPEN_DEVICE                        _COMM_ERROR         + 1
#define ERR_CE_BUILD_DCB                          _COMM_ERROR         + 2
#define ERR_CE_COMM_STATE                         _COMM_ERROR         + 3
#define ERR_CE_MEM_ALLOC                          _COMM_ERROR         + 4
#define ERR_CE_COMM_DIAL                          _COMM_ERROR         + 5
#define ERR_CE_CONNECT_TYPE                       _COMM_ERROR         + 6
#define ERR_CE_CHECK_RESPONSE                     _COMM_ERROR         + 7
#define ERR_CE_NOT_SUPPORTED_OPTION               _COMM_ERROR         + 8
#define ERR_CE_WRITE_TO_DEVICE                    _COMM_ERROR         + 9
#define ERR_CE_OVERFLOW                           _COMM_ERROR         + 10
#define ERR_CE_NO_RESPONSE                        _COMM_ERROR         + 11
#define ERR_CE_RECEIVE_HOST_STRING                _COMM_ERROR         + 12
#define ERR_CE_SEND_HOST_STRING                   _COMM_ERROR         + 13
#define ERR_CE_GET_HOST_ANSWER                    _COMM_ERROR         + 14
#define ERR_CE_WRONG_CHECK_SUM_TYPE               _COMM_ERROR         + 15
#define ERR_CE_SEND                               _COMM_ERROR         + 16
#define ERR_CE_RECEIVE                            _COMM_ERROR         + 17
#define ERR_CE_DATA_LEN                           _COMM_ERROR         + 18
#define ERR_CE_START_CHAR                         _COMM_ERROR         + 19
#define ERR_CE_NAK_RECEIVE                        _COMM_ERROR         + 20
#define ERR_CE_TOO_LARGE_PACKET                   _COMM_ERROR         + 21
#define ERR_CE_END_BLOCK_NOT_RECEIVED             _COMM_ERROR         + 22
#define ERR_CE_WRONG_CHECK_SUM                    _COMM_ERROR         + 23
#define ERR_CE_BLOCK_SEQ                          _COMM_ERROR         + 24
#define ERR_CE_ENVELOPE_NAK_PACKET                _COMM_ERROR         + 25
#define ERR_CE_SEND_NAK_PACKET                    _COMM_ERROR         + 26
#define ERR_CE_NO_CARRIER                         _COMM_ERROR         + 27
#define ERR_DOWNLOAD                              _COMM_ERROR         + 28
#define ERR_SSL_ERROR                             _COMM_ERROR         + 29
#define ERR_FTP_NOT_CONNECTED                     _COMM_ERROR         + 30
#define ERR_FTP_DIR_NOT_CHANGED                   _COMM_ERROR         + 31
#define ERR_CE_DECLINE                            _COMM_ERROR         + 32

/*********************************************************************************************/

#define ERR_STOP_ACTIVITY_AND_IMMEDIATE_CARD      _AUTHORIZATION_ERROR    + 0
#define ERR_MUST_AUTHOR_NOT_ALLOWED               _AUTHORIZATION_ERROR    + 1
#define ERR_MUST_CALL_AND_SHOULD_NOT_CALL         _AUTHORIZATION_ERROR    + 2
#define ERR_NEGATIVE_RESPONSE_FROM_CRED_CO        _AUTHORIZATION_ERROR    + 3
#define ERR_VOICE_AUTHOR_NOT_ALLOWED_FOR_CARD     _AUTHORIZATION_ERROR    + 4
#define ERR_VOICE_NOT_ALLOWED_FOR_TRANSACTION     _AUTHORIZATION_ERROR    + 5
#define ERR_INQ_CREDIT_ABOVE_LIMIT_NOT_ALLOWED    _AUTHORIZATION_ERROR    + 6
```

```
#define ERR_DEBIT_IMMED_INQUERY_NOT_ALLOWED      _AUTHORIZATION_ERROR    + 7
#define ERR_AUTHOR_UNKNOWN_RESPONSE              _AUTHORIZATION_ERROR    + 8
#define ERR_RANDOM_CREATE                         _AUTHORIZATION_ERROR    + 9
#define ERR_RANDOM_OPEN                           _AUTHORIZATION_ERROR    + 10
#define ERR_RANDOM_SEEK                           _AUTHORIZATION_ERROR    + 11
#define ERR_RANDOM_READ                           _AUTHORIZATION_ERROR    + 12
#define ERR_RANDOM_WRITE                          _AUTHORIZATION_ERROR    + 13
#define ERR_AUTHOR_PINPAD_REQUIRED               _AUTHORIZATION_ERROR    + 14
#define ERR_AUTHOR_UNKNOWN_BLOCK_TYPE            _AUTHORIZATION_ERROR    + 15

/************************************************************************/
#define ERR_DATA_PERMISSION_FILE_OPEN            _DATA_PERMISSION_ERROR  + 0
#define ERR_DATA_PERMISSION_SEEK                 _DATA_PERMISSION_ERROR  + 1
#define ERR_DATA_PERMISSION_READ                 _DATA_PERMISSION_ERROR  + 2
#define ERR_PERMISSION_CARD_NUMBER               _DATA_PERMISSION_ERROR  + 3
#define ERR_DATA_NO_OPTIONS                      _DATA_PERMISSION_ERROR  + 4
#define ERR_PERMISSION_CURRENCY                   _DATA_PERMISSION_ERROR  + 5
/************************************************************************/
#define ERR_PP_TOO_LARGE_PACKET                  _PIN_PAD_ERROR          + 0
#define ERR_PP_TRANSMIT_PACKET                   _PIN_PAD_ERROR          + 1
#define ERR_PP_READ_PINPAD_DEVICE                _PIN_PAD_ERROR          + 2
#define ERR_PP_GET_PACKET_WRONG_LRC              _PIN_PAD_ERROR          + 3
#define ERR_PP_GET_PACKET_SIZE                   _PIN_PAD_ERROR          + 4
#define ERR_PP_GET_PACKET_DIRECTION              _PIN_PAD_ERROR          + 5
#define ERR_PP_REMOVE_CARD                       _PIN_PAD_ERROR          + 6
#define ERR_PP_TRANSACTION_CANCELED              _PIN_PAD_ERROR          + 7
#define ERR_PP_VISA_SERVICE_CODE                 _PIN_PAD_ERROR          + 8
#define ERR_LOCKED_PIN_TRANS_CANCELED            _PIN_PAD_ERROR          + 9
#define ERR_PP_TIMEOUT_ON_RECEIVE                _PIN_PAD_ERROR          + 10
#define ERR_PP_NO_ACK                            _PIN_PAD_ERROR          + 11
#define ERR_PP_NO_MATCH_TRACK2                   _PIN_PAD_ERROR          + 12
#define ERR_PP_GOT_NAK                           _PIN_PAD_ERROR          + 13
#define ERR_PP_MAG_STRIPE                        _PIN_PAD_ERROR          + 14
#define ERR_TIMEOUT_TRANS_CANCELED               _PIN_PAD_ERROR          + 15
#define ERR_PP_CTLS                                  _PIN_PAD_ERROR          + 16
/************************************************************************/
#define _CASPIT_CENTER_ERROR                        _ERROR_START       + 1800
/************************************************************************/
#define ERR_WRONG_OPERATOR_NUMBER                _CASPIT_CENTER_ERROR + 5
#define ERR_WRONG_TERMINAL_ID                    _CASPIT_CENTER_ERROR + 6
#define ERR_TOO_MANY_SHIFTS                      _CASPIT_CENTER_ERROR + 7
#define ERR_CASPIT_CENTER_FS_ERROR               _CASPIT_CENTER_ERROR + 8
#define ERR_CASPIT_CENTER_NEG_RESPONSE           _CASPIT_CENTER_ERROR + 9
#define ERR_CASPIT_CENTER_ENQ                    _CASPIT_CENTER_ERROR + 10
#define ERR_CASPIT_CENTER_ACK                    _CASPIT_CENTER_ERROR + 11

/************************************************************************/
#define ERR_ECR_PAYMENTS_NOT_EQUAL_TOTAL_AMOUNT        _ECR_ERROR + 37
```

Appendix L – Beginner's Guide

I wrote this Beginner's Guide to ease the integration phase for new integrators of the Caspit SmartRetail Solution. However, this guide is **not** an excuse not to read the complete description of each command you need to use. Each command described in this guide has many more options and tags than described here. You should refer to the relevant chapters and learn about all the possible feature of the commands. Also, don't neglect chapter 9.2, where you can find XML examples for most of the transaction types.

Before you begin, you need to have to following items:

- A Pinpad device with applications that are ready to be used.
- A USB cable for the Pinpad (unless it is a BlueTooth Pinpad)
- Ingenico Driver for Windows (needed only for USB Pinpad)
- A power supply for the Pinpad (depending on the Pinpad model)
- Caspit Payment Windows Service software
- Caspit Payment DLL
- SmartRetail Tester
- This document about the Caspit SmartRetail Solution (read it!)
- Test credit cards

Step 1 – Installation

Install the following components:

- Install the Ingenico driver.
- Connect a Pinpad to the PC and make sure the Pinpad is found in the Device Manager
- Or, pair the BlueTooth Pinpad to Windows (or to the tablet/smartphone)
- Install the Caspit Payment Windows Service (only for Windows of course. Not relevant for iOS/Android)
- In some cases, you may need to install the Visual C++ Redistributable package. You can download it from the following link:
  - https://drive.google.com/open?id=0B3lyoKvcD8chRUQ3SkFQTFdSTW8
- Install the SmartRetail Tester (again, it is only for Windows)

Step 2 – Use the SmartRetail Tester

Now that the installation is behind us, you are going to use the tester to perform three essential commands.

Even if you are integrating the Pinpad with iOS or Android, I strongly recommend that you install the Pinpad on Windows and use the SmartRetail tester to learn how to write the correct XML. If something is not working, always go back to Windows and try it with the tester.

Learn to use the SmartRetail Tester and perform the following three commands using the Tester:

- Communication Test (command 003)
- Transmit Transactions (command 006)
- Perform a Transaction (command 001)

You must have success with all three commands before starting your own implementation.

You can get the full source code of the Tester and learn from it how to implement these three commands.

Step 3 – Implement Communication Test

Great, tester is working, now let's get down to business and start the real implementation.

The first thing you need to have is a successful Communication Test. A successful Communication Test will give you the following indications:

- That the Windows Service is installed correctly and working properly.
- That the CaspitPayment.dll is working and that you know how to use the Dll.
- That the Pinpad is connected and have a good setup.

During production, it is always a good practice to perform Communication Test every X minutes.

So to send the Communication Test command, just send the following XML to the Pinpad (either directly or using the DLL).

This is a minimal format of the Communication Test command. As you can see in chapter 16, the Communication Test has more options.


```
<Request>
<Command>003</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<TimeoutInSeconds>10</TimeoutInSeconds>
<RequestId>20160216155540</RequestId>
</Request>
```


If all goes well, you should get the following response XML:


```
<Response>
<Command>003</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<SerialNumber>30303PP3039393</SerialNumber>
<SmartRetailVersion>4872830001</SoftwareVersion>
<AshraitEMVVersion>5959590040</AshraitEMVVersion>
<EPAVersion>33939392122</EPAVersion>
<CPAVersion>20202022020</CPAVersion>
<RouterVersion>30303030303</RouterVersion>
<SDKVersion>9.22.0</SDKVersion>
<ResultCode>0</ResultCode>
<ShvaCommStatus>9</ShvaCommStatus>
<SwitchCommStatus>9</SwitchCommStatus>
<CaspitCommStatus>9</CaspitCommStatus>
<TMSCommStatus>9</TMSCommStatus>
<CertServerStatus>9</CertServerStatus>
```

<ReportServerStatus>9</ReportServerStatus>

<FTPServerStatus>9</CheckFTPStatus>

<ContactStatus>1</ContactStatus>

<ContactlessStatus>1</ContactlessStatus>

<CardInside>0</CardInside>

<RequestId>20160216155540</RequestId>

</Response>

Some of the tags you get back may have different values than the example, but the <ResultCode> must be zero, to indicate success.

Step 4 – Implement Communication to Shva

Wonderful. Now that you have mastered the Communication Test command, let's move on to make a communication to Shva.

A successful communication to Shva is a must. Otherwise, transactions will stay inside the Pinpad and money will not be transferred. During a transaction, the Pinpad may do a communication to Shva, but this is only to get authorization for the transaction. A transaction that got online authorization is not a complete transaction. Only the transmission of transactions to Shva, at the end of day, will complete the transactions.

A successful communication to Shva will give the following indications:

- That all the indications from Step 3 are OK.
- That the Pinpad is able to use the USB layer to connect to Shva

Note that if you are using USB connection, all the communication from the Pinpad to the outside world, like Shva and the Terminal Management System (TMS), all this communication is done through the USB. The Pinpad doesn't need a separate Ethernet cable, because thanks to Ingenico PCL technology all TCP based communication can be done over the USB cable.

So to make a communication to Shva, send the Transmit Transactions command. Just send the following XML to the Pinpad (either directly or using the DLL).

<Request>

<Command>006</Command>

<TerminalId>0880264</TerminalId>

<TermNo>001</TermNo>

<RequestId>20160216155540</RequestId>

</Request>

A successful response should look like this:

<Response>

<Command>006</Command>

<TerminalId>0880264</TerminalId>

<ResultCode>0</ResultCode>

<RequestId>20160216155540</RequestId>

<DATA> here will be the text of the DATA file. It is 47 characters long </DATA>

</Response>

If there were transactions in the Pinpad, you may get additional tags in the response. Again, the <ResultCode> must be zero to indicate a successful response.

Step 5 – Perform a Transaction

Well done! You really are something special. I think you are ready for your first transaction. So fasten you seat belt and get ready for lift off.

Here is the XML of a simple transaction (this example is taken from chapter 9.2. You can find more examples there).

Send it to the Pinpad

<Request>
<Command>001</Command>

<RequestId>23049820984</RequestId>
<TerminalId>0880264</Terminalid>
<TermNo>001</TermNo>

<TimeoutInSeconds>90</TimeoutInSeconds>
<Mti>100</Mti>
<CreditTerms>1</CreditTerms>
<TranType>1</TranType>
<Amount>10000</Amount> **(Amount is in agorot)**
<Currency>376</Currency> **(Currency is in ISO currency code)**
<PanEntryMode>PinPad</PanEntryMode>

<Xfield>20398049823</Xfield>

</Request>

A successful response will look like that:

<EMV_Output>

<Command>1</Command>

<TerminalId>0880264</TerminalId>

<ResultCode>0</ResultCode>

<Status>0</Status>

<AshStatus>0</AshStatus>

<Mti>100</Mti>

<Pan>0004580289999999709</Pan>

<PanEntryMode>40</PanEntryMode>

<CardName>ויזה זהב </CardName>

<Manpik>2</Manpik>

<Brand>2</Brand>

Bla Bla Bla Bla. More tags will come here.

<RequestId>20170109130703</RequestId>

</EMV_Output>

The important thing is that the <ResultCode>, <Status> and <AshStatus>, they all should be zero. If they are, you have successfully done you first transaction. Congratulations.

Appendix M – Troubleshooting

1. **Pinpad is not shown in the Device Manager of Windows**
   - Is the Pinpad connected to the PC using a USB cable? (if you connection type is RS232, you will not see the Pinpad in the device manager)
   - Is the Pinpad turned on? (there is no **ON** switch, if there is power, the Pinpad should be on). The iPP350 needs only the USB cable; it doesn't need any other external power supply. If you are using the iSC250 then you will need an external power supply.
   - Did you install the Ingenico driver?
   - Did you restart your PC after installing the Ingenico driver?

2. **Pinpad is blinking with an "Unauthorized" message**
   - The Pinpad is broken. You must replace the Pinpad.

3. **Troubleshooting Result codes**

Code	Define
0	RETVAL_SUCCESS
10000	RETVAL_TIMEOUT
10001	
10002	
10003	RETVAL_INCORRECT_TERMINAL_ID Check the tag <TerminalId> in the request XML. The <TerminalId> value must be identical to the Terminal Id of the Pinpad.
10004	
10005	RETVAL_NO_LAST_TRAN_FILE
10006	RETVAL_CANT_FIND_REQUESTED_TRAN_FILE
10007	
10008	
10009	
10010	
10011	
10012	
10013	
10014	
10015	
10016	
10017	

10018	
10019	RETVAL_SWITCH_TRAN_FILE_NOTHING_TO_SEND
10020	RETVAL_NO_PAPER_IN_PRINTER
10021	RETVAL_SWITCH_TRAN_FILE_IS_EMPTY
10022	RETVAL_INCORRECT_ECR_NUMBER The ECR number aka Station number or Kupa number Check the tag <TermNo> in the request XML. The <TermNo> value must be identical to the <TermNo> defined in the Pinpad.
10023	RETVAL_NO_STATIS_FILE_ON_POS
10024	RETVAL_SWITCH_TRAN_HAS_UNSENT_TRANSACTIONS
10025	RETVAL_SWITCH_TRANSACTIONS_SENT_FAILED
10026	RETVAL_TRAN_STEPS_LISTENER_NOT_WORKING
10027	RETVAL_INVALID_MESSAGE_CODE
10028	RETVAL_CANT_CONTINUE_J2
10029	RETVAL_INVALID_REQUEST_ID
10030	RETVAL_SHVA_PARAMETERS_DELETION_FAILED
10031	RETVAL_SHVA_BLACKLIST_DELETION_FAILED
10032	RETVAL_TIMEOUT_EVENT_FROM_CASPIT_DLL
10033	RETVAL_SHVA_CALL_FAILED
10034	RETVAL_REPORTS_SERVER_CALL_FAILED
10035	RETVAL_XML_INVALID_METHOD
10036	RETVAL_XML_MISSING_TERMINAL_ID_TAG
10037	RETVAL_CERTIFICATES_SERVER_CALL_FAILED
10038	RETVAL_SWITCH_CALL_FAILED
10039	RETVAL_INVALID_TRAN_RECORD_NUMBER
10040	RETVAL_FILE_CANNOT_BE_RETRIEVED
10041	RETVAL_ERROR_CONNECTING_TO_PINPAD
10042	RETVAL_XML_MISSING_REQUEST_ID_TAG
10043	RETVAL_CASPIT_WINDOWS_SERVICE_NOT_RESPONDING
10044	RETVAL_USER_ABORTED_TRANSACTION
10045	RETVAL_TIMEOUT_CHECK_TRAN *There was a timeout during the transaction and the status of the transaction is not clear. The PC should query the POS to know if the transaction failed or succeeded.*

10046	RETVAL_XML_MISSING_TIMEOUT_TAG
10047	RETVAL_XML_MISSING_INT_IN_TAG
10048	RETVAL_CHECK_OTHER_STATUSES It means you have to check the other statuses in the response. Check <Status> and <AshStatus>
10049	RETVAL_TRANSACTION_NOT_FOUND
10050	RETVAL_DOUBLE_TRANSACTION
10051	RETVAL_BATCH_NUMBER_DOESNT_EXIST
10052	RETVAL_INVALID_BLOCK_NUMBER_TO_RESUME
10053	RETVAL_PINPAD_RETURNED_EMPTY_RESPONSE
10054	RETVAL_ASHRAIT_INVALID_SETUP
10055	RETVAL_NO_PING
10056	RETVAL_UNSUPPORTED_GENERIC_COMMAND_CODE
10057	RETVAL_TRANSMIT_LOGS_FAILED
10058	RETVAL_NO_CONNECTION_TO_SHVA
10059	RETVAL_DELETE_LOGS_FAILED
10060	RETVAL_CASPIT_TECH_CENTER_FAILED
10061	RETVAL_TERMINAL_REBOOT_FAILED
10062	RETVAL_STARTING_TMS_CALL_FAILED
10063	RETVAL_CLEAN_COMMUNICATION_ENGINE_FAILED
10064	RETVAL_SETTING_DATE_TIME_FAILED
10065	RETVAL_TRANSACTIONS_CENTER_CALL_FAILED
10066	RETVAL_TRANSACTIONS_CENTER_TOGGLE_FAILED
10099	RETVAL_UNKNOWN_ERROR
10100	RETVAL_INVALID_PAN_ENTRY_MODE
10101	RETVAL_INVALID_MTI
10102	RETVAL_INVALID_X_FIELD
10103	RETVAL_CANT_FIND_DEPOSIT_REPORT
10104	RETVAL_INVALID_SESSION_NUMBER_XML_TAG
10105	RETVAL_INVALID_DATE_XML_TAG
10106	RETVAL_NO_TOTAL_FILE
10501	RETVAL_TIMEOUT_WHILE_WAITING_FOR_SWIPE *There was a timeout while the POS was waiting for a swipe. The customer didn't swipe his card and eventually the operation was terminated because of a*

	timeout.
10502	RETVAL_TIMEOUT_AFTER_FAILED_SWIPE_ATTEMPT *There was a timeout while the POS was waiting for a swipe. There is an indication that the customer swiped his card at least one time, but the card was not read properly so the POS continued to wait until the operation was terminated because of a timeout.*
10503	RETVAL_CANCEL_WITHOUT_SWIPING *The operation was cancelled by the user, he pressed on the red button.*
10504	RETVAL_CANCEL_AFTER_FAILED_SWIPE_ATTEMPT *The operation was cancelled by the user, he pressed on the red button. But before pressing on the red button, the customer tried to swipe his card at least one time, but the card was not read properly so the POS continued to wait until the operation was terminated by the customer.*
11000	RETVAL_CANNOT_SEND_OUT_CREDIT_CARDS
11001	RETVAL_INVALID_CARD_SWIPE
11002	RETVAL_FILE_BAD_RECORD *The operation was cancelled by terminal program,* because bad record found in file. This error stops terminal activity to avoid transaction collection without their transmission .

Appendix N – FAQ

Q: How can the ECR check if there is a card inside the smart card reader?

A: The simplest way to check it is to run a minimal communication test command

```
<Request>
<Command>003</Command>
<TerminalId>0880264</TerminalId>
<TermNo>001</TermNo>
<RequestId>20160216155540</RequestId>
</Request>
```

The response will be:

```
<Response>
<Command>003</Command>
…
<CardInside>1</CardInside>
</Request>
```

The important tag from the response is the <CardInside> tag, which tells you if there is a card inside or not.

Note that a card is considered to be inside only if it is inserted all the way, as needed for performing a transaction. If the card is not all the way in, the <CardInside> tag will return with zero.

Q: What is the fastest way to check if the Pinpad has transactions that need to be transmitted to Shva

A: You can do it using the Pinpad Configuration command. The response will include the tag <UnsentTransactions> that holds the number of transactions currently inside the Pinpad. Actually, I recommend using this method also after calling Transmit Transation command. That way you can be sure that the transactions were transmitted successfully to Shva.

Appendix O – CaspitPayment.dll

דוגמאות לביצוע עסקאות בתצורות התקנה שונות

כל הדוגמאות מבוססות על פיתוח ב- Visual Studio 2013 C# FW 4.5.

לתוכנית המפיץ יש להוסיף "Reference" לספריה CaspitPayment.dll (c:\Caspit\CaspitPaymentService\ CaspitPayment.dll) המסופקת לכם.

להלן דוגמה להגדרת פרמטרי קלט, קריאה לפונקציה deal.PerformingTransactionObj

וקריאת נתוני פלט:

```
CaspitPayment.RequestTreatment _deal      = new CaspitPayment.RequestTreatment ();
CaspitPayment.Request          _inputObj  = new CaspitPayment.Request ();
CaspitPayment.GlobalObject     _outputObj= new CaspitPayment.GlobalObject();

// כתובת IP של מחשב בו מותקן CaspitPaymentService – תוכנת Windows Service המאפשרת העבורה של נתונים בין מחשב של מפיץ קופות לבין מכשירי Pinpad של כספיט. התוכנה מסופקת ע"י כספיט.
_deal.IpAddress = "127.0.0.1";
_deal.Port      = 6900; // Config.xml פורט – ראה קובץ
_deal.Timeout   = 60;   // (in seconds) זמן המתנה ליצירת חיבור בין המחשב ל-סרוויס
_deal.Target     = 52;   // יעד – קבוע
```

מכיוון ששני האובייקטים **inputObj** ו **outputOjb** מכילים עשרות פרמטרים , בדוגמאות נוכל להתמקד רק בפרמטרים העיקריים.

דוגמה לעסקה J2 :

```
_inputObj.TerminalId            = "0880302";
_inputObj.Command               = "001";
_inputObj.TimeoutInSeconds      = "90";
_inputObj.Mti                   = "100;          // 100 - Regular; 400 - Cancel
_inputObj.TranType              = "1";
_inputObj.Amount                = "1000";        // Amount is in agorot
_inputObj.TermNo                = "1";
_inputObj.Currency              = "376";         // NIS (שקל) New Israeli Shekel
_inputObj.XField                = "20398049823";
_inputObj.CreditTerms           = "1";
_inputObj.IndexPayment          = "0";
_inputObj.ParameterJ            = "2";
_inputObj.RequestId             = DateTime.Now.ToString ("yyyyMMddHHmmss");
_inputObj.SkipDoubleTranCheck   = "0";
_inputObj.AllowCardInsideBefore = "1";
_inputObj.ContinueJ2            = "0";
_inputObj.AllowCardInsideAfter  = "1";
_inputObj.Pan                   = "";
_inputObj.CreditTerms           = "1";           // Credit type
_inputObj.TranType              = "1";           // Charge
_inputObj.PanEntryMode          = "PinPad";

/*
המילה "PinPad" מעידה על כך שנתוני הכרטיס צפויים להגיע מה-PinPad
ברוב המקרים ל PinPad -יש גם קורא מגנטי ולעתים גם קורא קירבה (contactless) ובכל המקרים האלה
PanEntryMode = "PinPad"
*/
```

```
int sts = _deal.PerformingTransactionObj(_inputObj, ref _outputObj);
if (sts != 0) {MessageBox.Show("Error"); return;}
/*                        Created string _outputObj.Xml and
 Response EMV_Output Objector _outputObj.Output
                                   Or/and _outputObj.Response

If Command = "001"  ("Performing Transaction")
The outputObj object will contain the Emv_Output object.
In any other command - you must access the Response object

  Note: Here we would to check: ResultCode!= 0 && Status != 0 && AshStatus !=0
*/
```

C# EMV_Output Class Example:

```
using System.Runtime.InteropServices;
using System.Xml.Serialization;

namespace CaspitPayment
{
    // [Serializable]
    [ClassInterface(ClassInterfaceType.AutoDual)]
    public class EMV_Output
    {
        public string Command   { get; set; }// <Command>1</Command>
        public string RequestId { get; set; }// <RequestId>20180108130813</RequestId>
        public string TerminalId{ get; set; }// <TerminalId>0880302</TerminalId>
//......................

        [XmlElementAttribute("PossibleCreditTerms", Form =
System.Xml.Schema.XmlSchemaForm.Unqualified)]
        public EmvOutputPossibleCreditTerms[] PossibleCreditTerms { get; set; }

        public string PossibleCreditTermsString { get; set; }
// Code,Avail,MinPayments,MaxPayments|Code,Avail,MinPayments,MaxPayments|...
// 1,True,2,5|2,False,0,0|...

//......................

        public EMV_Output()
        {
            Command    = string.Empty;
            RequestId  = string.Empty;
            TerminalId = string.Empty;
//......................
        }
//......................
        [System.ComponentModel.DesignerCategoryAttribute("code")]
        [XmlTypeAttribute(AnonymousType = true)]
        public class EmvOutputPossibleCreditTerms
        {
            private EmvOutputPossibleCreditTermsCreditTerm[] _creditTermField;
            /// <remarks/>
            [XmlElementAttribute("CreditTerm", Form =
System.Xml.Schema.XmlSchemaForm.Unqualified)]
            public EmvOutputPossibleCreditTermsCreditTerm[] CreditTerm
            {
                get { return _creditTermField; }
                set {_creditTermField = value; }
            }
        }
        [System.SerializableAttribute()]
        [System.Diagnostics.DebuggerStepThroughAttribute()]
        [System.ComponentModel.DesignerCategoryAttribute("code")]
        [XmlTypeAttribute(AnonymousType = true)]
        public class EmvOutputPossibleCreditTermsCreditTerm
        {
            private string _codeField;
            private string _availField;
            private string _minPaymentsField;
```

```
            private string _maxPaymentsField;
            [XmlAttributeAttribute()]
            public string Code
            {
                get { return _codeField; }
                set { _codeField = value; }
            }
            [XmlAttributeAttribute()]
            public string Avail
            {
                get { return _availField; }
                set { _availField = value; }
            }
            [XmlAttributeAttribute()]
            public string MinPayments
            {
                get { return _minPaymentsField; }
                set { _minPaymentsField = value; }
            }
            [XmlAttributeAttribute()]
            public string MaxPayments
            {
                get { return _maxPaymentsField; }
                set { _maxPaymentsField = value; }
            }
        }
    }


//...................... Using for get values from PossibleCreditTerms.
//...................... The PossibleCreditTerms is an Array of Objects
   CaspitPayment.GlobalObject _outputObj = new CaspitPayment.GlobalObject();
   void Get()
      {
             int sts = _deal.PerformingTransactionObj(_inputObj, ref _outputObj);
//......................

            var possibleCreditTerms = _outputObj.Output.PossibleCreditTerms;
            if (possibleCreditTerms != null && possibleCreditTerms.Length > 0)
            {
                var creditTerm = possibleCreditTerms[0].CreditTerm;
                for (int i = 0; i < creditTerm.Length; i++)
                {
                    string code        = creditTerm[i].Code;
                    string avail       = creditTerm[i].Avail;
                    string minPayments  = creditTerm[i].MinPayments;
                    string maxPayments  = creditTerm[i].MaxPayments;
                }
            }


//...................... Also possible get values from PossibleCreditTermsString
//                       where data like "1,True,2,5|2,False,0,0|..."
//(Code,Avail,MinPayments,MaxPayments| Code,Avail,MinPayments,MaxPayments|...)
      }
}
```

Sending request in XML string

```
string ResponseInternal = "";
int Result = locDeal.PerformingTransactionXml(eArgument as string, out ResponseInternal);
```

where eArgument is

```
<Request>
  <Command>001</Command>
- <!--  Performing Transaction  -->
  <TerminalId>0880302</TerminalId>
- <!--  Terminal ID  -->
  <TimeoutInSeconds>60</TimeoutInSeconds>
  <Mti>100</Mti>
  <TranType>1</TranType>
- <!--  Regular  -->
  <Amount>1000</Amount>
  <TermNo>1</TermNo>
  <PanEntryMode>PinPad</PanEntryMode>
  <Currency>376</Currency>
- <!--  NIS (שקל) New Israeli Shekel  -->
  <XField>1234567890</XField>
  <CreditTerms>1</CreditTerms>
  <ParameterJ>4</ParameterJ>
  <RequestId>20180205153858</RequestId>
</Request>
```

Response EMV_Output XML Example


```
<?xml version="1.0" ?>
- <EMV_Output>
  <Command>1</Command> - <!-- Performing Transaction -->
  <TerminalId>0880302</TerminalId> - <!-- Terminal ID -->
  <ResultCode>0</ResultCode> - <!-- SUCCESS -->
  <Status>0</Status> - <!-- מאושר (Successful/approved transaction) -->
  <AshStatus>0</AshStatus> - <!-- מאושר -->
  <Mti>100</Mti>
  <Pan>00045574499999994812</Pan> - <!-- Same value as it came in the J2 transaction -->
  <PanEntryMode>5</PanEntryMode> - <!-- Contactless EMV -->
  <CardName>ויזה זהב</CardName> - <!-- שם כרטיס -->
  <CardHash>21D6F4983DC741FB6C831F3F86578402</CardHash> - <!-- Hash result on the card number (the PAN). MD5.-->
  <Manpik>1</Manpik>
  <Brand>2</Brand>
  <Solek>2</Solek> - <!-- כ.א.ל (CAL) -->
  <CardType>041</CardType> - <!-- Not defined yet -->
  <spType>0</spType>
  <Amount>1</Amount>
  <TranType>1</TranType> - <!-- Regular -->
  <CreditTerms>1</CreditTerms>
  <Currency>376</Currency> - <!-- NIS (שקל) New Israeli Shekel -->
  <FileNo>07</FileNo>
  <TermNo>001</TermNo> - <!-- Station number 001 (Not multiple pinpads). -->
  <TermSeq>001</TermSeq>
  <TerminalName>טיפכס תוכנה</TerminalName>
  <Retailer>0880302015</Retailer> - <!-- The long terminal ID - מספר מסוף ארוך -->
  <ComRetailerNum>7007245</ComRetailerNum>-<!--Business number -מספר ספק בחברה הסולקת-->
  <DateTime>0205154447</DateTime>
  <ResponseId>0</ResponseId> - <!-- Not entered – לא הוכנס -->
  <ResponseCvv2>0</ResponseCvv2> - <!-- Not entered – לא הוכנס -->
  <ResponseAvs>0</ResponseAvs> - <!-- Not entered – לא הוכנס -->
  <AuthCodeManpik>0</AuthCodeManpik>- <!-- Transaction without authorization number- עסקה ללא מספר אישור מנפיק -->
  <AuthCodeSolek>0</AuthCodeSolek> - <!-- Transaction has no authorization number from acquirer -->
  <Uid>18020515444708803020013</Uid> - <!-- YYMMDDHHMMSS_TTTTTTT_RRR_C (TimeStamp_TerminalId_ECR_Checksum -->
  <SkipDoubleTranCheck>0</SkipDoubleTranCheck>
  <appVersion>CST000128T</appVersion> - <!-- Application version of the Ashrait EMV -->
  <AID>A0000000031010</AID> - <!-- Chip Application Id - מזהה אפליקציה על כרטיס חכם -->
```

```
<ATC>0502</ATC>
<TVR>0000000000</TVR> - <!-- Terminal Verification results - תוצאת בדיקות המסוף -->
<AddDspBalance>0</AddDspBalance>
<TelNoCom>0035726333</TelNoCom> - <!-- Phone number of the acquirer - מספר טלפון של החברה הסולקת -->
<Xfield>1234567890</Xfield> - <!-- Transaction Query -->
<RequestId>20180205154428</RequestId> - <!-- מזהה כלשהו של הבקשה -->
<ParameterJ>4</ParameterJ>
<PinpadStep>9000</PinpadStep>
- <ReceiptMerchant>
<Line name="Merchant">טיפסכ תוכמסה</Line>
<Line name="Empty" />
<Line name="TerminalId">טיפסכ .סמ: 0880302</Line>
<Line name="SWVersion">הנכות: CST000128T</Line>
<Line name="BusinessNumber">קסע רפסמ: 7007245</Line>
<Line name="Empty" />
<Line name="TranDateTime">05/02/18 15:44</Line>
<Line name="Empty" />
<Line name="CardName">בהז הזיו</Line>
<Line name="CardNumber">XXXXXXXXXXXX4812</Line>
<Line name="Voucher">רבוש 'סמ: 07001001</Line>
<Line name="TranType">הבוח :הקסע</Line>
<Line name="PanEntryMode">עוציב: Contactless EMV</Line>
<Line name="CreditTerms">ליגר: יארשא</Line>
<Line name="Amount">הקסיע םוכס:</Line>
<Line name="Empty" />
<Line name="Amount">10.00</Line>
<Line name="Empty" />
<Line name="Currency">ח"ש :עבטמ</Line>
<Line name="Empty" />
<Line name="Uid">UID:18020515444708803020013</Line>
<Line name="EMV">ATC:0502,TVR:0000000000</Line>
<Line name="EMV">AID:A0000000031010</Line>
<Line name="Empty" />
<Line name="Signature">המיתח____________________</Line>
<Line name="Empty" />
<Line name="Phone">ןופלט____________________</Line>
<Line name="Empty" />
</ReceiptMerchant>
```

```
- <ReceiptCustomer>
  <Line name="Merchant">טיפסכ תוכמסה</Line>
  <Line name="Empty" />
  <Line name="TerminalId">0880302 :מס. טיפסכ</Line>
  <Line name="SWVersion">CST000128T :הנכות</Line>
  <Line name="BusinessNumber">7007245 :קסע רפסמ</Line>
  <Line name="Empty" />
  <Line name="TranDateTime">05/02/18 15:44</Line>
  <Line name="Empty" />
  <Line name="CardName">בהז הזיו</Line>
  <Line name="CardNumber">XXXXXXXXXXXX4812</Line>
  <Line name="Voucher">07001001 :רבוש 'מס</Line>
  <Line name="TranType">הבוח :הקסע</Line>
  <Line name="PanEntryMode">Contactless EMV :עוציב</Line>
  <Line name="CreditTerms">ליגר :יארשא</Line>
  <Line name="Amount">הקסיע םוכס:</Line>
  <Line name="Empty" />
  <Line name="Amount">0.01</Line>
  <Line name="Empty" />
  <Line name="Currency">עבטמ: ח"ש</Line>
  <Line name="Empty" />
  <Line name="Uid">UID:18020515444708803020013</Line>
  <Line name="EMV">ATC:0502,TVR:0000000000</Line>
  <Line name="EMV">AID:A0000000031010</Line>
  <Line name="Empty" />
  <Line name="Copy">*** חוקלל קתוע ***</Line>
  <Line name="Empty" />
  <Line name="Goodbye">תוארתהל הדות!</Line>
  <Line name="Empty" />
  </ReceiptCustomer>
  </EMV_Output>
```

Appendix P – PINPad connection over Broker

- Broker API is a REST api works with POST XML requests.
- The API Domain is 212.235.022.050
- The API connection must be established through SSL , port 5000
- The API is synchronous, Broker sends to PINPad to execute request and returns the response received from PINPad.
- The api URI = **https://[Api Domain]:5000/cashregister/request/[terminalId]/[terminalNo]**
- The api body = Caspit Request Text in XML format